

# Bankers' Outside Options and Financial Risk<sup>\*</sup>

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## Abstract

This paper documents that bankers' labor market outside options shape risk-taking in the financial industry. We construct a bank-level measure based on hiring dynamics at connected financial institutions using granular employment histories and predetermined coworker links. Improved outside options predict faster growth in non-investment-grade lending, bank risk, and systemic risk. Individual loans shift toward riskier borrowers without compensating spreads and experience worse subsequent outcomes. Evidence from individually matched bankers shows that the risk response is concentrated among younger, more mobile bankers and that outside options reduce the separation penalty following poor loan performance. We also provide complementary evidence from investment advisers, where better outside options translate into more misconduct cases and weaker ensuing employment penalties. The results are consistent with outside options diluting the disciplining force of dismissals and provide a labor-market channel connecting procyclical outside options to credit supply.

**Keywords:** Financial Stability, Risk, Labor Markets, Outside Options

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# 1 Introduction

In imperfect labor markets, workers' outside options play a central role in determining wages and job mobility. Improvements in the quality or quantity of alternative employment opportunities strengthen workers' bargaining positions, translating into higher pay and more frequent transitions across employers (Caldwell and Harmon, 2019; Caldwell and Danieli, 2024; Johnson et al., 2025). However, outside options may not only affect workers' matching to firms but also their actions within firms. We investigate the link between outside options and on-the-job behavior in the financial industry, a setting well suited to studying this question for two reasons. First, job-to-job transitions represent a relatively high share of new hires in the financial sector (Azzopardi et al., 2020). Bankers should consequently attend closely to the arrival of outside offers, making their conduct inside the firm sensitive to the state of the external labor market. Second, employees have considerable discretion to make risky, high-stakes choices on behalf of firms that have limited capacity to monitor these decisions.

We present empirical evidence that bankers with better labor market outside options make riskier lending decisions and argue that an implicit insurance channel drives these results. In the presence of an agency problem between the bank and the banker, improved outside options reduce the effectiveness of the bank's main disciplining tool, dismissal. Because employment opportunities are procyclical, this effect varies systematically over the business cycle and can contribute to fluctuations in credit supply. Given the large negative spillovers of financial crises, these dynamics in a narrow segment of the labor market can have consequences well beyond the firms involved.

To formalize the mechanism, we present a conceptual framework featuring optimal disciplining by banks in the presence of outside options. A banker chooses their screening intensity, trading off the private gains from laxer screening against the increased dismissal risk that follows loan defaults. The employing bank, in turn, sets the dismissal rate applied after a default, weighing lower default losses against the replacement costs of dismissing bankers. When bankers are more likely to receive attractive offers that the incumbent does not match, the dismissal threat is diluted and risk taking increases. This response is strongest for young bankers, for whom outside offers are more likely to dominate their current match. Since the marginal gain from enforcing discipline

is reduced by outside options, banks respond optimally by reducing dismissal rates, amplifying the effect. Outside offers thus insure bankers against the consequences of poor performance, a mechanism we term the implicit insurance channel.

The empirical challenge is to measure outside options without using the focal bank's own contemporaneous mobility or financial performance, as these factors are plausibly correlated with lending behavior. We construct Inflows-at-Ties, a network-based measure from employment biographies of senior US bankers. For each bank and quarter, it combines job-to-job hiring at other institutions with predetermined coworker links to those institutions, while mechanically excluding contemporaneous transitions from the focal bank. The measure therefore captures exposure to hiring news from attainable employers using information generated outside the focal bank. The causal interpretation of our results requires a conditional independence assumption, which we examine through balance tests, shift-share diagnostics, fixed-exposure specifications, and controls for correlated product-market activity.

We link employment histories drawn from nearly 500 million public profiles to manually verified bank parent companies, syndicated loans, borrower ratings, lender and borrower accounts, compensation records, and bank and systemic risk measures. We further extract banker signatures from SEC loan contracts and match the signers to their employment histories. In a separate exercise, we link our measure of outside options to annual records for approximately 140,000 financial advisers. This allows us to follow the same mechanism from bank outcomes to individual credit decisions and subsequent career outcomes.

The measure passes several relevance and design checks. Stronger inflows at ties predict job-to-job mobility and higher salary growth, while transitions out of the financial industry, bonuses, and equity at risk show little response. Past outflows and observed bank fundamentals do not predict subsequent inflows at ties. The identifying events are dispersed across roughly 164,000 movers and 367 connected hiring institutions, serial switching is rare, and residual hiring co-movement is small within the observed industry and geographic groupings after absorbing year-quarter trends through fixed effects. We first examine outcomes at the level of the lending bank. Better outside options increase lending growth, with the most precisely estimated response in non-investment-grade credit. A one standard deviation increase raises non-investment-grade volume growth by 1.6 to 2.8 percentage points across the reported specifications. Measures of the

lending bank's tail risk and the financial sector's risk conditional on that bank being in distress both increase, with larger and more precise estimates at the 99th percentile. In the preferred specification, the 99th-percentile VaR and CoVaR responses each equal about 4% of the standard deviation of quarterly risk growth. The parallel VaR and CoVaR responses indicate that the lending risk associated with outside options passes through to standard measures of bank and systemic risk.

Estimates for volume and risk growth remain elevated in the quarters after a shock, although statistical significance is concentrated in the first quarter. The results also survive alternative lender controls and fixed effects, origination-only volume, allocation across co-leads, exclusion of the great financial crisis, winsorization, alternative tie windows, controls for realized outflows and inflows at co-syndicators, and measures built from different seniority groups.

We then examine the loan-level choices underlying these bank outcomes. A one standard deviation improvement in outside options raises the probability of lending to a borrower below investment grade by between 0.5 and 1.3 percentage points across fixed-effects specifications. The shift also appears in an alternative ex ante ratings measure and in borrower book leverage at origination. We find no evidence that banks receive a compensating increase in spreads. Borrowers subsequently experience more severe rating downgrades and pay more when refinancing their next loans. Additional analyses under highly restrictive lender-by-year-quarter fixed effects, which are defined at the same level as our proxy, suggest a stronger response for first-time lender-borrower relationships, which require greater screening effort. The loan-level effect is also larger when state-level changes reduce non-compete enforceability, as outside options influence worker decisions more when mobility is less restricted.

The loan-level findings are robust to controlling for lender inflows, outflows and syndicate size, limiting the sample to originations and non-GFC years, winsorizing the outside options measure, and using alternative tie durations. Reconstructing the measure from transitions of very senior bankers roughly doubles the coefficient for borrower risk relative to the coefficient obtained for less senior hires. This result suggests that while mobility is lower for senior bankers, it is more strongly influenced by personal networks. We provide two checks that directly address correlated lending cycles at connected institutions. First, the estimates remain stable when we control for total and leveraged lending at connected banks and when we add hiring at banks connected

through prior co-syndication. Second, the estimates also survive segment-by-time controls,<sup>1</sup> first differences, and a fixed-exposure measure that removes changes in the network weights.

We next use the linked banker-loan sample to study individual bankers. Younger signers are substantially more mobile, and shocks to outside options predict realized loan deterioration primarily for these younger bankers, even when comparing signers at the same bank in the same quarter. The corresponding age interaction for risk measured at origination has the same sign but is less precise. Poor loan performance raises subsequent separation rates when outside options are limited, yet a one standard deviation improvement in outside options offsets the estimated separation penalty. These findings suggest that outside options affect bankers who are interested in exercising them by reducing the penalty associated with adverse lending outcomes.

Finally, we provide complementary evidence from financial advisers. Better outside options predict more misconduct disclosures, customer complaints, and settled complaints. These results are robust to adviser-firm fixed effects, state-by-year fixed effects, alternative timing assumptions, controls for lagged outflows, and binary outcomes. Outside options also reduce the likelihood of job separation and industry exit after misconduct and raise within-sector reemployment conditional on misconduct-related separation. This evidence shows that the mechanism operates in a distinct but related labor market. Better outside options induce choices that are misaligned with the firm and reduce the effective career penalty from dismissal, leading firms to rely on it less.

Our results also speak to aggregate dynamics. Figure A3 shows that labor mobility in the financial industry co-moves with lending standards over time. Our evidence supplies a worker-level mechanism consistent with this aggregate relationship. Because outside options are procyclical in theoretical and empirical accounts (Mortensen and Pissarides, 1999; Nakamura et al., 2019; Figueiredo, 2022), shocks that expand credit demand and competition for banking talent may reinforce one another and contribute to risky lending during expansions. Since credit growth predicts the likelihood and severity of financial crises (Schularick and Taylor, 2012; Jordà et al., 2013), the results suggest that labor-market conditions in the financial industry may contain information relevant for macroprudential monitoring. Deferred compensation and clawback provisions strengthen discipline while bankers remain at their firms but become less effective when

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<sup>1</sup>We add fixed effects for the lender's dominant connected-bank industry segment interacted with the year, at the two- and three-digit NAICS levels. We also use an exposure-weighted control that captures the segment-time hiring to which the lender's network is exposed.

outside options improve. Making such compensation partially portable across employers could represent a unilateral solution to the externality generated by interbank hiring.

**Literature.** Our main contribution is to provide empirical evidence on a novel mechanism through which labor markets affect credit supply, adding to four strands of literature. First, a small theoretical literature has studied potential interactions between agency problems within banks, labor market dynamics in the financial sector, and the quality and quantity of credit (Myerson, 2012, 2014; Thanassoulis, 2012; Axelson and Bond, 2015; Acharya et al., 2016; Boxtel, 2017). These papers analyze different channels through which the labor market for bankers can impact credit supply, including frictions in the supply of bankers, externalities generated by competition for talent, and bankers' job opportunities outside the financial sector. We add to this literature by providing empirical evidence on this interaction, and in particular on a channel whereby hiring at one bank, by improving the outside options of bankers elsewhere, weakens their alignment with their own employer.

Second, we inform the discussion about the role of career concerns in mitigating risky behavior in the financial sector, which can be traced back to the seminal work of Holmström (1999). The evidence on whether bankers are punished for adverse choices is mixed: some studies report no effect in response to corporate misconduct (Griffin et al., 2019; Hamdi et al., 2023), while others detect such effects in the context of syndicated loans and hedge funds (Ellul et al., 2020; Gao et al., 2020). Our study complements these findings by showing that the career consequences of risky choices vary with outside options, and that bankers internalize this state-dependence.

Third, a related empirical literature has examined different aspects of the labor market in the financial sector, such as the high labor mobility of investment bankers (Gao et al., 2024a,b) and the effect of labor mobility restrictions on financial intermediation (Cici et al., 2021; Agarwal et al., 2024; Bonelli, 2019; Norden et al., 2025). Other papers have highlighted the discretionary role of individual bankers in setting lending terms for syndicated loans (Herpfer, 2021; Bushman et al., 2021; Carvalho et al., 2023). We contribute to this literature by showing that labor mobility across firms can lead to the origination of more credit, more risky credit, and more financial risk in the context of corporate lending.

Finally, our paper speaks to a recent literature examining the role of outside options in labor markets more generally (Caldwell and Danieli, 2024; Caldwell et al., 2024; Schubert et al., 2022). In particular, Jäger et al. (2024) show that information from coworkers shifts workers’ beliefs about outside options, which in turn affect their search and bargaining behavior. Caldwell and Harmon (2019) investigate the role of personal networks in transmitting such information about outside options across firms, showing that news transmitted through these networks promotes both wage growth and labor mobility. Ahammer et al. (2023) document a negative relationship between outside options and worker effort by exploiting variation in unemployment insurance. We add to this work by documenting that outside options shape the job-related decisions of highly skilled workers through a risk-taking channel, with potentially adverse implications for their employers.

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework and its predictions. Section 3 describes the linked data, and Section 4 constructs the outside options measure and states the identifying assumptions. Section 5 presents lender-level volume and risk results before turning to loan-level borrower outcomes. Section 6 uses individually matched corporate bankers to study age heterogeneity and the dilution of discipline, then presents complementary evidence from financial advisers. Section 7 concludes.

## 2 Conceptual Framework

To interpret our results, we lay out a simple framework capturing how banks govern banker risk-taking in practice. In the presence of information frictions, a bank disciplines a banker by imposing a penalty after adverse outcomes, but better outside options erode the effect of that penalty. We focus on the penalty of discontinued employment. A banker who performs poorly can be dismissed and lose the value of the ongoing relationship. An accepted outside offer protects the banker by allowing them to depart before the loan outcome is observed. Two features affect how this discipline dilution channel operates. First, firing is costly to the bank, so the intensity of the dismissal threat is an equilibrium object. We show that banks optimally reduce discipline as offers become more likely, even though they correctly anticipate the resulting increase in risk-taking. Second, an outside offer affects discipline only when the banker is able and willing to

exercise it through departure. We embed the lending game in a minimal on-the-job search environment in the spirit of Burdett and Mortensen (1998). Over a career, a banker samples offers and moves only to better matches. An offer is attractive only if it is better than any previous draw, an event whose probability declines mechanically over time. Because outside options are more likely to be attractive for young bankers than for older bankers, fluctuations in labor demand at other banks dilute discipline more for the former group.

**Setup.** There are three dates,  $t = 0, 1, 2$ , and two risk-neutral players, a bank and a banker. The banker has no wealth and is protected by limited liability. The disciplinary instrument at the bank's disposal is the dismissal threat. The bank commits at  $t = 0$  to a dismissal probability  $\delta \in [0, 1]$  applied after a default at  $t = 2$  to a banker who remains employed by the bank. A dismissed banker forfeits the value  $D \geq 0$  of continued employment, which includes any unvested deferred pay (Edmans et al., 2017; Hoffmann et al., 2022). This disciplinary technology is the one Gao et al. (2020) document, where corporate bankers face career penalties, such as dismissal and demotion, for risk-taking that ends in poor loan outcomes. Building on their findings, we show how the effectiveness of dismissal moves with the labor market. The banker chooses screening laxity  $r \in [0, a]$ , where  $r = 0$  is the bank's preferred intensity and  $r = a$  is no screening at all. As this choice is unobserved, the firing decision can only be conditioned on the default outcome (Holmström, 1979). We also assume non-observability of defaults across banks. Laxity yields the banker a private benefit  $g(r) = ar - r^2/2$ , which is increasing and concave on  $[0, a]$ , and captures saved screening effort, volume-based bonuses and resume-building. This represents the tension between originating loans and screening them studied in the literature on loan officers (Heider and Inderst, 2012; Behr et al., 2020). Higher screening laxity increases the default probability  $p(r) = \gamma r$ , with  $\gamma > 0$  and  $\gamma a \leq 1$ , and imposes an expected loss  $\ell(r) = hr$  on the bank.<sup>2</sup>

We model the dismissal costs for the bank as increasing and convex in dismissal strictness,

$$C(\delta) = \frac{k}{2} \delta^2, \quad k > 0. \tag{1}$$

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<sup>2</sup>We abstract from the bank's choice of which risk target to set for bankers and focus on the banker's unobservable deviation from it. We adopt the linear loss function for tractability. For the superlinear case, the interiority conditions change and additional assumptions are required for the dismissal response and the age gradient prediction to carry the same signs as below.

This choice is motivated by the observation that experienced bankers' client relationships are difficult to replace in a sector that hires little from outside, requiring the remaining staff to absorb the extra workload when a banker departs.

**Job ladder.** Before the lending game, the banker has spent a career in the sector, which we summarize by the number  $n \geq 1$  of match values the banker has sampled, drawn i.i.d. from a continuous distribution  $F$ . This count grows with time in the labor market and can be interpreted as career age. Because a searching worker moves only to matches better than the current one, the banker enters the lending game occupying the best match sampled so far,

$$v_n = \max\{x_1, \dots, x_n\}. \quad (2)$$

This is the selection mechanism at the heart of on-the-job search models (Burdett and Mortensen, 1998; Jovanovic, 1979). Seniority and match quality accumulate together because bankers who have searched longer have had more draws.

At  $t = 1$ , before loan outcomes are observed, the banker receives an offer  $x \sim F$  with probability  $\lambda$ . Denote the new total number of draws by  $\eta \equiv n + 1$ . The offer is an attractive option only if it dominates the incumbent match,  $x \geq v_n$ . The probability of this event does not depend on  $F$  and is given by  $1/\eta$ . The arrival rate of dominating offers is

$$\lambda_d(n) = \frac{\lambda}{\eta}, \quad \eta \equiv n + 1. \quad (3)$$

Conditional on receiving a dominating offer, the incumbent does not match the offer with probability  $\mu \in (0, 1)$ , in which case the banker departs. Otherwise, the banker accepts a matched contract from the incumbent. Job-to-job departures therefore occur at rate

$$\lambda_I(n) = \frac{\lambda\mu}{\eta}.$$

The effective insurance rate is therefore  $\lambda\mu/\eta$ . Protection arises only when a dominating offer arrives and the incumbent does not match it, because departure places the banker beyond the incumbent bank's reach before dismissal at  $t = 2$ . A banker who accepts the incumbent's matched

contract receives the higher bargained pay associated with a credible outside offer but remains subject to dismissal if the loan subsequently defaults.<sup>3</sup> Once the banker accepts or declines an offer, the offer expires. A dominated offer is declined without affecting either bargaining or dismissal exposure.

**The banker's choice.** The banker retains employment value  $D$  unless the banker remains with the incumbent, the loan defaults, and dismissal occurs. Expected utility is

$$U(r) = w + g(r) + \left[ 1 - \left( 1 - \frac{\lambda\mu}{\eta} \right) p(r) \delta \right] D, \quad (4)$$

where  $w$  is the banker's expected bargained pay. The first-order condition

$$g'(r) = \left( 1 - \frac{\lambda\mu}{\eta} \right) p'(r) \delta D$$

yields

$$r^* = a - \gamma \left( 1 - \frac{\lambda\mu}{\eta} \right) \delta D. \quad (5)$$

That is, risk-taking increases with the arrival rate of outside offers and decreases with career age.

**The bank's problem.** With additive base pay, the bank chooses the firing intensity by solving

$$\max_{0 \leq \delta \leq 1} -hr - \frac{k}{2} \delta^2, \quad r = a - \gamma \left( 1 - \frac{\lambda\mu}{\eta} \right) \delta D. \quad (6)$$

Writing  $B(n) \equiv \gamma \left( 1 - \frac{\lambda\mu}{\eta} \right) D$  for the marginal disciplinary power of the firing threat, risk-taking follows  $r = a - B\delta$ . With a linear loss function, the first-order condition  $hB = k\delta$  yields

$$\delta^*(n) = \frac{hB(n)}{k}, \quad r^*(n) = a - \frac{hB(n)^2}{k}. \quad (7)$$

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<sup>3</sup>In a matched state, the incumbent and the banker divide the surplus of the continued match  $V$  by Nash bargaining. Given a dominating offer  $x$  and banker bargaining weight  $\beta \in [0, 1]$ , the retained wage solves  $\max_{w_m} (w_m - x)^\beta (V - w_m)^{1-\beta}$ , giving  $w_m = \beta V + (1 - \beta)x$ . The banker occupies a matched state with probability  $\lambda(1 - \mu)/\eta$  and is otherwise paid a base wage  $w_0 = \beta V + (1 - \beta)v_0$ , with  $v_0$  the banker's outside value absent an offer, so  $w_0 < w_m$ . Both expected pay and departures increase with outside options.

Discipline is interior,  $\delta^* < 1$ , whenever

$$h\gamma D < k, \quad (8)$$

a single condition that holds uniformly across career ages since  $B(n) \leq \gamma D$ . The firing cost is large enough that the bank does not dismiss every uninsured defaulter. We maintain this assumption based on the finding of Gao et al. (2020) that banks do not dismiss every banker with non-performing loans.

**Comparative statics.** The bank's first-order condition equates the marginal value of discipline to its marginal cost. As  $\lambda$  rises, unmatched dominating offers become more likely, the dismissal threat is diluted,  $B$  falls, each unit of intensity buys less discipline, and the bank optimally scales back  $\delta$ . Differentiating equilibrium risk in Equation (7),

$$\frac{\partial r^*}{\partial \lambda} = \frac{2h\gamma^2 D^2}{k} \frac{\mu(\eta - \mu\lambda)}{\eta^2} > 0, \quad (9)$$

so risk rises with the offer rate even though firing is costly and optimally chosen.

The distortion is not resolved by the bank's optimization around dismissal strictness because its source lies outside the contract. The bank cannot contract with the rival employers whose unmatched offers allow departure and neutralize the dismissal threat. The residual distortion is an externality across employers competing for the same talent, complementing accounts in which competition for bankers shapes pay and risk taking in equilibrium (Thanassoulis, 2012; Acharya et al., 2016).

The model also speaks to the firing decision itself. Differentiating the chosen intensity in Equation (7),

$$\frac{\partial \delta^*}{\partial \lambda} = -\frac{h\gamma D}{k} \frac{\mu}{\eta} < 0, \quad (10)$$

that is, the bank fires less intensely when  $\lambda$  rises. This qualifies the findings of Gao et al. (2020), as the career penalty itself depends on labor market outside options. The effect is proportional to  $1/\eta$ , with larger reductions in separation penalties for bankers who are more likely to act on offers.

The age heterogeneity result follows from the ladder. Differentiating the risk response (9),

$$\frac{\partial^2 r^*}{\partial \lambda \partial \eta} = -\frac{2h\gamma^2 D^2}{k} \frac{\mu(\eta - 2\mu\lambda)}{\eta^3} < 0 \quad \text{for all } \eta \geq 2, \quad (11)$$

since  $\lambda \leq 1$ . The response of risk-taking to improved outside options is more pronounced among young bankers, for whom offers are more likely to be acceptable. For senior bankers, who occupy the best match of a long career, the same improvement in  $\lambda$  barely moves the effective insurance rate  $\lambda/\eta$  and hence barely moves behavior. The same margin governs mobility itself. Departures occur at rate  $m(n) = \lambda\mu/\eta$ , which declines with career age, consistent with the decline in job-to-job transitions documented since Topel and Ward (1992), and rises with outside options. The mobility response to outside options,  $\partial m/\partial \lambda = \mu/\eta$ , is likewise concentrated among the young.

**Predictions.** The framework yields three testable predictions.

1. *Pay and mobility.* Improvements in outside options raise both the expected pay of retained bankers and the rate of departures  $\lambda\mu/\eta$ , as banks match and raise pay for a share of dominating offers and bankers depart when those offers are not matched (Thanassoulis, 2012; Acharya et al., 2016). Because the probability that an offer dominates falls with career age, mobility and its response to outside options are concentrated among young bankers.
2. *Risk taking.* As better outside options dilute the threat of dismissal, banker screening relaxes, inducing an increase in lending volumes and a shift toward riskier borrowers. The level result operates through actionable outside offers and is weaker if labor mobility is restricted. With better outside options, the bank also reduces firing conditional on bad realizations.
3. *Heterogeneity and misconduct.* The risk response to outside options declines with banker age. Among young bankers, offers are more likely to dominate their current match and provide effective insurance, making them more responsive to fluctuations. The erosion of the separation penalty as shown in Equation (10) is concentrated among the young for the same reason.

**Discussion.** The framework treats  $\lambda$  as exogenous to the bank's own situation and the banker's career age  $n$  as predetermined at the time of the lending decision. Our empirical strategy uses

hiring variation generated at other institutions and excludes the focal bank’s contemporaneous employment dynamics by construction.

Three implicit assumptions deserve comment. First, changes in other banks’ hiring behavior move bankers’ outside options, which is reasonable given limited entry into the sector and common job-to-job transitions (Philippon and Reshef, 2012; Bell and Van Reenen, 2014; Böhm et al., 2023). Second, individual bankers influence loan decisions, for which there is extensive evidence in the syndicated loan market (Herpfer, 2021; Bushman et al., 2021; Carvalho et al., 2023) and more broadly that loan officers’ incentives shape credit provision (Hertzberg et al., 2010). Third, we model the banker as risk neutral to isolate the contractual channel. A risk-averse banker would weigh dominating offers as insurance against the income loss following dismissal, reinforcing the prediction.

Other channels could link outside options to risk-taking. Better outside options raise turnover, and shorter expected tenure mechanically lowers the probability of being employed at the bank when a default materializes. In our framework the risk choice responds to the anticipated probability of escaping the penalty, so the prediction holds for bankers who never move, and our main results are robust to controlling for realized outflows. Behavioral channels could operate alongside the contractual one. Favorable labor market news could make bankers optimistic about their own portfolios and lead to riskier lending (Bordalo et al., 2024). The misconduct results in Section 6.2 are difficult to reconcile with optimism alone, as they show that the labor market penalties for lax screening and misconduct themselves fall when outside options improve. Finally, bankers may use a stronger labor market position to bargain for more generous severance or greater lending discretion, channels that operate through the same position and reinforce the relationship we document.

### **3 Data**

We construct a linked dataset that connects labor-market conditions to bank-level outcomes, individual loans, and banker careers. We focus on the syndicated lending market for three main reasons. First, both the size of individual loans and the size of the overall loan market make syndicated loans important for financial stability. With a median loan size in our sample of USD

400 million, these contracts represent high-stakes decisions. In 2018, the USD 2.8 trillion syndicated lending market exceeded the USD 1.3 trillion corporate bond market. Second, default risk can spread through syndication, secondary-market trading, and securitization (Mugasha, 2004). Finally, lending choices can be attributed to arranging banks and, for the matched subsample, to individual signers, while prior work shows that lead bankers influence loan design (Herpfer, 2021; Bushman et al., 2021).

**Dealscan.** We obtain data on bank lending activity from *Dealscan*. From the universe of loans recorded in the database, we identify the 1,500 largest lead-arranging banks by number of loans issued and include all tranches arranged by these banks in our sample. For each loan, we include the identities of the borrowing and lending firms, the tranche amount and maturity, and indicators for whether the loan is secured and whether covenants are attached. Table A1 reports summary statistics for the tranche-level data. To measure credit quantity, we also aggregate the data to the lender-quarter level to obtain the number of loans and total lending volume.

**Revelio Labs.** We draw on labor market data from Revelio Labs, which collects employment biographies reported on LinkedIn. The dataset includes employment details for the universe of public profiles, covering nearly 500 million workers across approximately 1.8 billion employment spells. The linked employer-employee data contain employment spell start and end dates, as well as self-reported job titles. From this information, Revelio Labs generates a task-based classification of job titles, which maps activities to each title by analyzing descriptions from resumes and online profiles, alongside responsibilities listed in job postings.<sup>4</sup> Initially, the algorithm identifies 1,500 job roles, which are successively aggregated using a clustering algorithm. The data also include a seniority ranking derived from job titles, ranging from 1 (lowest) to 7 (highest).

To construct our sample of senior bankers, we proceed as follows. First, we exclude all employment spells outside the United States. Second, we use the job taxonomy at the hierarchical level that defines 300 distinct roles, allowing us to identify the individuals employed in the relevant roles.<sup>5</sup> Third, to identify senior bankers, we retain spells with a seniority level of 4 or above. Fi-

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<sup>4</sup>For additional details, see [www.reveliolabs.com](http://www.reveliolabs.com).

<sup>5</sup>The included roles are “Investment Specialist”, “Investment Banking”, “Banker”, “Credit Specialist”, “Financial Advisor”, “Loan Officer”, and “Financial Analyst”. The extensive set of included roles is deliberate. Revelio’s 300 role taxonomy remains coarse within finance, so a narrow label rule would exclude workers who perform similar credit and

nally, we retain full-time bankers by excluding overlapping spells. We use all employment spells between 1994 and 2020.<sup>6</sup>

Panel A of Table A3 reports summary statistics for this sample. The sample predominantly consists of male, white, and highly educated employees with average annual incomes of around USD 150,000. Because the employment histories are backfilled, the mean employee age is relatively young at 37 years. Table A4 compares key demographic variables between the LinkedIn dataset and administrative data sources. The sample captures the main characteristics of individuals employed in the financial industry reported in the CPS and ACS.<sup>7</sup> These similarities are stronger when we remove the seniority restriction.

For our main analysis, we combine the Dealscan lending data with an outside options proxy calculated from employment histories at the parent-company level. We sequentially merge on exact matches of parent companies' GVKEYs, tickers, and company names, which we clean and normalize beforehand, retaining only unique matches. For unmatched observations, we repeat the procedure at the subsidiary level, linking parent companies through their subsidiaries when matches are identified. When multiple subsidiaries match, we prioritize the match associated with the highest number of spells for each lender parent. We manually verify all matches and identify additional ones. In this way, we merge 1,268 parent companies, comprising two-thirds of the subsidiaries in Dealscan. Among the 1,000 largest lenders, the matched sample accounts for 99% of individual loans and 91% of total lending volume.

For a limited sample of loans, we identify individual lead-arranging bankers by name and affiliation and match them to their respective employment histories. We follow Herpfer (2021) in extracting banker signatures from syndicated loan contracts contained in SEC filings and match these signers to their employment histories using the name, affiliation, and filing date. Panel B of Table A3 reports summary statistics for this sample.

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client functions under different titles. We verified this issue manually in the sample of individual loan signers. Signers who can be identified from loan contracts appear across the broader set of classifications, which makes broad coverage necessary to capture the relevant banker network. The wider scope also reflects lateral moves into related financial markets. Hiring for these roles can therefore affect the attainable outside options of corporate bankers even when the destination title differs from the current one. We apply the same role definition to the banker and financial adviser samples so that the outside options measure has a consistent labor market boundary across the two analyses.

<sup>6</sup>The data have been collected since 2019 but contain all self-reported employment spells from earlier periods.

<sup>7</sup>We obtain both sources through IPUMS. For details on the ACS data, see Ruggles et al. (2025).

**Compustat.** We obtain data on firm-level fundamentals from Compustat. For both borrowing and lending firms, we include ratings, assets, liabilities, net income, EBIT, short-term debt, long-term debt, employees, shareholders' equity, and capital expenditure. Table A5 reports summary statistics. We also use borrowers' S&P credit ratings as a measure of credit risk and repayment probability. We use the linking tables of Chava and Roberts (2008) and Schwert (2020) to match borrower and lender firms in Dealscan and Compustat. Panels A and B of Table A4 report the results. To ensure a consistent sample of firms throughout the analysis, we use only banks for which we observe syndicated loan activity, employment histories, and balance sheet information.

**Mergent FISD.** We use Mergent's Fixed Income Securities Database (FISD) to obtain bond credit ratings, rating actions, and defaults from Moody's, S&P, and Fitch. For each loan tranche, we construct a contemporaneous issuer rating as the median of the most recent active ratings, on or before origination, across the borrower's bonds and rating agencies. We map ratings to a 22-point scale ranging from 1 (D) to 22 (AAA/Aaa). Table A2 reports summary statistics for the rated-loan subsample.

**(Conditional) Value at Risk.** To measure financial risk, we use the quarterly data of Adrian and Brunnermeier (2016), which span 1994 to 2009. The first measure is *Value at Risk* (VaR), which captures the maximum loss for institution  $i$  at the 95% or 99% confidence level and thus the financial risk of the lending bank. The second variable is *Conditional Value at Risk* (CoVaR), defined as the VaR of the financial sector conditional on bank  $i$  being in distress. CoVaR quantifies the expected loss to the financial system conditional on a 95th- or 99th-percentile loss at bank  $i$ , capturing the noncausal contribution of bank  $i$  to systemic risk. Adrian and Brunnermeier (2016) estimate these measures using quantile regressions. We transform the data into symmetric growth rates and report summary statistics in Table A6.

**BoardEx.** We use information on banker pay from BoardEx, a database provided by *Delinian*. These data include detailed information on compensation for top executives of large, listed companies in the United States since 1997. We construct year-over-year growth rates in the firm-level median of salaries, bonuses, and equity at risk. To better capture wage bargaining, we restrict

the sample to roles with operational responsibilities rather than executive oversight.<sup>8</sup> Summary statistics are shown in Panel D of Table A4.

**Investment adviser data.** We use data on misconduct among U.S. investment advisers, covering individuals registered with the Securities and Exchange Commission (SEC), a state authority, or both. The dataset is a panel spanning 2007 to 2015 and includes detailed employment histories alongside annual counts of misconduct disclosures and customer complaints. We combine this information with our measure of outside options at the parent-company level, matching sequentially on CIK, web domain, and bank name. The final sample includes approximately 140,000 distinct individuals.<sup>9</sup>

## 4 Measuring Outside Options

This section describes our network-based measure of outside options, followed by validation tests and a discussion of the identifying assumptions. The measure generates differential exposure to hiring at other banks using predetermined coworker connections to those institutions and is constructed independently of the focal bank’s contemporaneous employment dynamics. The role of social networks in shaping labor market outcomes has been extensively documented both theoretically (Montgomery, 1991; Dustmann et al., 2016) and empirically (Bramoullé and Saint-Paul, 2010; Shue, 2013; Hensvik and Skans, 2016; Glitz, 2017; Gee et al., 2017; Saygin et al., 2021; Cingano and Rosolia, 2012). Our measure is built from LinkedIn data on individual career histories drawn from public profiles.

**Construction.** Our outside options measure counts new hires at other firms, weighted by the share of former coworkers now employed there, capturing bank-time variation in exposure to signals about labor market dynamics at broadly attainable firms. More formally, for bankers at a

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<sup>8</sup>This includes all role names containing “Director”, “President”, “VP”, or “MD”, excluding those that contain “CEO”, “Chair”, “Trustee”, or “Partner”.

<sup>9</sup>Egan, Matvos, and Seru construct the underlying data using publicly available records from the SEC’s Investment Adviser Public Disclosure website. For additional details, see Egan, Matvos, and Seru’s homepage, Egan et al. (2019, 2022), and the recent literature review by Egan et al. (2024).

given bank  $i$  in quarter  $t$ , our measure of outside options is defined as follows:

$$InflowsAtTies_{it} = \sum_{j \neq i} w_{ijt} \times Inflows_{-i,jt} \quad (12)$$

Thus,  $InflowsAtTies_{it}$  is the weighted sum of inflows at other banks  $j$ , denoted by  $Inflows_{-i,jt}$ . The latter term is defined as job-to-job transitions from all other firms  $k \neq i$  to firm  $j$  in quarter  $t$ :

$$Inflows_{-i,jt} = \sum_{k \neq i} Transitions_{kjt}. \quad (13)$$

We exclude transitions from firm  $i$  to  $j$  in  $t$  from the hiring component. This mechanically isolates the measure from the focal bank's contemporaneous outflows that might be related to factors also influencing lending decisions.

The weights measure the share of former coworkers now employed at a given bank. More specifically, the weight bank  $i$  attributes to  $j$  at time  $t$  counts the number of past transitions from the former to the latter, normalized by the number of past transitions to all other banks:

$$w_{ijt} = \frac{\sum_{h=4}^{12} Transitions_{ij,t-h}}{\sum_{m \neq i} \sum_{h=4}^{12} Transitions_{im,t-h}}. \quad (14)$$

The weights thus capture the social ties a bank's workers have to other firms and hence their exposure to hiring at those firms.<sup>10</sup> In our baseline specification, we focus on ties that are between 4 and 12 quarters old. We limit the tie duration to 3 years because the median spell length in our sample is around 13 quarters. We begin including ties only one year after the transition to ensure that outflows from a firm do not contemporaneously affect the tie-weight component of our measure. A bank's contemporaneous outflows therefore enter neither the hiring nor the weight component of the measure. We demonstrate the robustness of our main results to alternative assumptions in the appendix.

The measure  $InflowsAtTies_{it}$  was initially proposed by Caldwell and Harmon (2019) and has several properties that make it well-suited to our setting. First, it captures firm-level variation

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<sup>10</sup>The LinkedIn dataset does not include information about connections between profiles. We are therefore unable to base our network measure on social ties indicated by these connections. Instead, we assume that ties form between senior investment bankers working at the same firm at the same time and persist for a limited period after one banker moves to another firm.

in signals about labor demand, shifting workers’ beliefs about their outside options. Prior work shows that such beliefs affect not only the job search and bargaining intentions of workers (Jäger et al., 2024) but also their search and bargaining outcomes (Caldwell and Harmon, 2019). We expect this information transmission channel to be relevant, given the importance of professional (Godechot, 2014) and alumni networks (McNamara and McLoughlin, 2009) and the widespread use of LinkedIn in the financial industry (Fracassi et al., 2016; Jiang et al., 2018). In many cases, senior positions remain unadvertised, increasing banks’ reliance on established network connections and employee referrals. Second, the weighting approach provides an intuitive method of accounting for a bank  $i$ ’s worker placement record in the recent past. Workers should place more weight on labor demand at firms that are achievable for them, which our proxy captures. Because we weight inflows by the share rather than the number of ties, firm size and the corresponding size of the coworker network do not mechanically inflate the proxy. Third, the measure relies exclusively on variation in employment flows across firms. Alternative approaches (e.g. as in Caldwell et al., 2024) based on worker concentration across industries or occupations are unsuitable in the context of a single occupational group. We also caution against relying solely on spatial variation, given the concentration of senior investment bankers in a small number of financial hubs as well as potential demand-side effects (Ee et al., 2023; Ee and Huang, 2024).

To construct the measure, we identify senior investment bankers as those with at least one employment spell that meets the exclusion criteria outlined in Section 3. Next, we retrieve the employment histories of these individuals to trace their connections to other institutions. We also extract all senior investment banking spells at those other firms. For the hiring dynamics, we focus on job-to-job transitions, which we define as those with employment gaps of one month or less. In a given quarter, we observe on average 317 bankers per institution, with 24 connected banks, each of which hires 7 bankers.<sup>11</sup> Based on this sample of spells, we compute  $InflowsAtTies_{it}$  in the way described above and standardize the resulting measure to have a mean of zero and a standard deviation of one. As we plot in Figure A4, the outside options measure exhibits a procyclical pattern over time, closely co-moving with the credit-to-GDP ratio and the number of employees in credit intermediation.

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<sup>11</sup>We refer to Table A8 for summary statistics.

**Validation.** We validate the relevance of the measure by replicating the findings of Caldwell and Harmon (2019) on the effects of outside options on mobility and wages. To examine the relationship between outside options and employee mobility, we regress employee outflows from a given institution on our proxy for outside options, including bank and year-quarter fixed effects. We estimate separate models for each lead and lag of the outside options measure. Figure A1 reports the results for moves to other banking roles in Panel a and for all employee outflows in Panel b. Outside options significantly increase both measures of outflows on impact and in subsequent periods. This persistence is consistent with news shocks shifting beliefs and influencing search behavior over the medium term. The coefficients on the leads are insignificant, indicating that elevated past outflows do not predict our outside options proxy. The pattern also remains unchanged for alternative timing assumptions for the construction of  $InflowsAtTies_{it}$  (Figure A2).

At the spell level, the same shock in outside options predicts subsequent worker mobility: Appendix Table A10 shows that a one standard deviation increase in  $InflowsAtTies$  raises the probability of a job-to-job move over the next two years, while leaving moves out of finance essentially unchanged.

Second, we examine whether better outside options translate into pay increases for employees who remain with their firms. To this end, we regress symmetric growth rates of firm-level compensation components on our outside options proxy, controlling for lender and year-quarter fixed effects. Table A9 presents the results. Columns (1) and (2) show that improved outside options are associated with substantial increases in median salary growth, with and without lender controls. Columns (3) to (6) consider other key pay components, bonuses and equity at risk, and find no significant response to variation in outside options. This contrast indicates that, conditional on controls, the proxy does not capture cyclical variation in banks' financial position, as we would otherwise expect more volatile pay components to respond more strongly. It also aligns with prior evidence that workers primarily bargain over wages rather than bonuses or amenities (Caldwell et al., 2024). Overall, the results indicate that banks raise fixed pay to retain talent when outside options improve, supporting the interpretation that our proxy reflects meaningful variation in outside options.

**Identification.** Our estimand is the effect of a change in the focal bank’s exposure to hiring at connected institutions on its lending and risk outcomes. We can give the coefficient a causal interpretation under the assumption that, after conditioning on the stated controls and fixed effects, the hiring shifts transmitted through the network are unrelated to unobserved determinants of the focal bank’s outcomes.

The design is a panel shift-share design. The shares are the tie weights  $w_{ijt}$ , and the shifts are job-to-job inflows  $Inflows_{-i,jt}$  at connected banks. We follow the exogenous-shifts approach of Borusyak et al. (2022) and Borusyak et al. (2025). Identification requires conditionally quasi-random hiring shifts and sufficiently dispersed variation in the underlying shifts to prevent a small number of connected institutions from determining the estimate. The network can therefore reflect persistent bank characteristics, provided that residual hiring at the institutions reached through that network is unrelated to residual changes in the focal bank’s outcomes.

We eliminate two potential sources of reverse causality by construction, ensuring that the measure is not affected by the focal bank’s contemporaneous outflows. First, the hiring shifts exclude moves from focal bank  $i$  to connected bank  $j$  in quarter  $t$ . Second, the shares are based on transitions formed four to twelve quarters earlier.

Hiring at a connected institution can arise for a number of reasons, including retirements, leadership changes, team staffing needs, or departures from banking. To support our reliance on such idiosyncratic variation, we present several observable diagnostics. Current bank balance sheet variables, in levels and growth rates, do not predict subsequent inflows at ties after lender and year-quarter fixed effects (Tables A11 and A12). The mobility validation also shows no elevated outflow coefficients before the measured outside options shock in either panel of Figure A1. These tests provide evidence against reverse causality operating through observed bank conditions or prior mobility.

The shocks are also dispersed. The average bank is connected to 24 institutions in a quarter, while 367 connected banks record positive hiring over the sample. The largest average shock exposure is 2.7%,<sup>12</sup> and the average bank is connected to about 7% of the hiring institutions. Year-quarter fixed effects account for about 2% of the variation in hiring behavior. After residualizing

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<sup>12</sup>Randomly sampling a bank-quarter and then sampling a connected bank according to the tie weights in that quarter yields the most connected bank in 2.7% of cases.

for them, correlations within three-digit NAICS industries and metro areas range from 0.02 to 0.04 (Table A13), providing evidence that hiring behavior is dispersed even within narrow sectors and regions.

A remaining concern is bank-specific exposure to a common or industry-level shock that affects both hiring at connected banks and the focal bank’s lending (Guren et al., 2021). We address this concern in Section 5, using several robustness checks.

Consistent with this, serial switching is rare. Among bankers who move, the large majority transition only once over the entire sample period, and moving four or more times is exceptional (Figure A6). The inflow events underlying the measure are correspondingly dispersed: the most frequently mobile 1% of movers account for only about 4% of all transitions, and the Gini coefficient of events across movers is just 0.18 (Figure A7).

## 5 Main Results

This section presents the main results. We begin with the effects of outside options on lender-level outcomes and study the timing of these responses. We then turn to loan-level choices and test whether the composition and subsequent performance of borrowers move in the direction implied by laxer screening.

### 5.1 Lender-Level Outcomes

We ask whether improved outside options are associated with a risky credit expansion and faster growth in bank and systemic risk. As discussed in Section 2, the model predicts more lending and risk, as more potential borrowers pass a lowered screening threshold when the dismissal threat weakens. To test this, we estimate the following specification for bank  $i$  in year-quarter  $t$ :

$$y_{it} = \beta \text{InflowsAtTies}_{it} + \Gamma V_{it} + \alpha_i + \lambda_t + \epsilon_{it} \quad (15)$$

where  $y_{it}$  is the outcome of interest and  $V_{it}$  is a vector of time-varying controls, comprising the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. The terms  $\alpha_i$  and  $\lambda_t$

are fixed effects at the bank and year-quarter level. Standard errors are double-clustered at the lender and year-quarter level.

**Loan Volume.** We assess the effects of outside options on lending volumes constructed from the syndicated loan data, with results presented in Table 1. We measure a bank’s quarterly lending volume as the total amount of the facilities in which it is a lead arranger.<sup>13</sup> The dependent variable is the symmetric (midpoint) growth rate of this volume in percentage points, and we decompose total lending (Columns 1 to 3) into its non-investment-grade (Columns 4 to 6) and investment-grade (Columns 7 to 9) components. Within each block we report three specifications that absorb time-varying lender heterogeneity in complementary ways: bank and year-quarter fixed effects (Columns 1, 4, 7); year-quarter and bank-by-year fixed effects (Columns 2, 5, 8), and bank and year-quarter fixed effects augmented with time-varying lender controls (Columns 3, 6, 9). Because the lender controls are drawn from Compustat and are available only for the subset of matched, listed lenders, they shrink the sample. We therefore treat the bank-by-year fixed effects and the explicit lender controls as interchangeable strategies for netting out lender-level confounders, and report both.

Improved outside options predict higher lending-volume growth, with the most precise response in risky credit. Non-investment-grade volume growth rises by 1.6 to 2.8 percentage points per one standard deviation increase across the three specifications and is statistically significant throughout. Total volume growth is positive in all three columns and statistically significant with lender controls, while the investment-grade response is weaker and less stable across specifications.

The non-investment-grade volume response is robust across an extensive battery of checks, summarized in Table A14 in Appendix B. The estimate remains positive when we restrict attention to newly originated lending, allocate each facility across its lead arrangers, and exclude the 2007 to 2008 financial crisis. We also add the focal bank’s own employee inflows and outflows separately. These controls address the possibility that contemporaneous staffing changes at bank  $i$  jointly determine its measured outside options and lending capacity.

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<sup>13</sup>We credit each lead arranger with the full amount of a facility exactly once to correct for facilities that recur across a parent’s subsidiaries and loan amendments.

The main product-market concern is that coworker networks connect banks exposed to the same credit-demand cycle. For example, a leveraged-lending boom could increase both hiring at connected banks and risky lending at the focal bank. We address this concern by retaining the coworker tie weights while replacing connected-bank hiring with the level and annual growth of lending at those banks. The controls cover total lending and three definitions of leveraged lending based on spreads, borrower ratings, and loan purpose. Panel D adds these exposure-weighted lending measures jointly. The coefficient on inflows at ties remains stable, which indicates that lending cycles observed at the same connected banks do not account for the risky-volume response. The table displays the coefficients on total lending at ties in levels and growth so that their direct relationship with focal-bank lending remains visible.

As a separate robustness check, we construct a hiring-exposure measure using prior co-syndication relationships in place of coworker ties (Inflows at Co-syndicators, IAC). Co-syndication links capture transactional proximity, shared borrower markets, and channels of interbank information. Including this measure absorbs hiring conditions at banks connected to the focal lender through its lending network. The coefficient on inflows at coworker ties remains stable, suggesting that coworker ties do not merely proxy co-syndication relationships or common lending conditions among financially connected banks.

Recomputing the outside options measure separately from medium-senior and high-seniority job-to-job hires yields the same pattern, with no systematic difference across seniority groups (Appendix Table A16).

**Financial Risk.** Next, we examine whether outside options predict growth in financial risk at the lending bank and in the systemic risk associated with that bank, measured by VaR and CoVaR. These measures allow us to test whether the change in lending risk passes through to an established measure of the bank's tail loss and an established measure of sectoral risk conditional on that bank being in distress. We estimate Equation (15) under the same three specification choices used for lending volume.

The results are reported in Table 2. Across outcomes, the coefficients are positive, and the strongest estimates arise in specifications that absorb time-varying lender heterogeneity through lender-by-year fixed effects. At the 95% confidence level, a one standard deviation improvement

Table 1: Effect of Outside Options on Lending Volumes

|                            | Total            |                |                   | Non-Inv. Grade     |                    |                   | Inv. Grade       |                |                  |
|----------------------------|------------------|----------------|-------------------|--------------------|--------------------|-------------------|------------------|----------------|------------------|
|                            | (1)              | (2)            | (3)               | (4)                | (5)                | (6)               | (7)              | (8)            | (9)              |
| IAT                        | 0.395<br>(0.690) | 1.23<br>(1.22) | 1.62**<br>(0.807) | 1.55***<br>(0.387) | 2.81***<br>(0.997) | 1.65**<br>(0.728) | 0.344<br>(0.736) | 1.54<br>(1.31) | 1.64*<br>(0.919) |
| Lender fixed effects       | X                |                | X                 | X                  |                    | X                 | X                |                | X                |
| Year-Quarter fixed effects | X                | X              | X                 | X                  | X                  | X                 | X                | X              | X                |
| Lender-year fixed effects  |                  | X              |                   |                    | X                  |                   |                  | X              |                  |
| Lender Controls            |                  |                | X                 |                    |                    | X                 |                  |                | X                |
| Observations               | 25,378           | 25,199         | 6,296             | 25,378             | 25,199             | 6,296             | 25,378           | 25,199         | 6,296            |

Notes: The table reports results from regressing the outcome of interest on standardized  $InflowsAtTies_{it}$  and fixed effects at the lender and year-quarter level. The dependent variable is the symmetric (midpoint) growth rate,  $2(x_t - x_{t-1})/(x_t + x_{t-1})$ , transformed into percentage points, of a bank's quarterly lead-arranged lending volume. Each facility is attributed to its lead arranger exactly once at its full amount. Columns (1)–(3) use total lending volume, Columns (4)–(6) its non-investment-grade component, and Columns (7)–(9) its investment-grade component. Within each block, the first column includes lender and year-quarter fixed effects, the second replaces the lender fixed effects with saturated lender-by-year fixed effects, and the third adds time-varying lender controls. The saturated lender-by-year fixed effects preserve the sample size while still controlling for time trends at the lender with reasonable frequency. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. The data span 1994 to 2020. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 2: Effect of Outside Options on Changes in Financial Risk (in Percentage Points)

|                            | VaR 95%          |                   |                  | CoVaR 95%        |                    |                  | VaR 99%            |                     |                     | CoVaR 99%           |                     |                     |
|----------------------------|------------------|-------------------|------------------|------------------|--------------------|------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                            | (1)              | (2)               | (3)              | (4)              | (5)                | (6)              | (7)                | (8)                 | (9)                 | (10)                | (11)                | (12)                |
| IAT                        | 0.167<br>(0.194) | 0.479*<br>(0.263) | 0.226<br>(0.253) | 0.173<br>(0.150) | 0.439**<br>(0.215) | 0.179<br>(0.192) | 0.314**<br>(0.145) | 0.783***<br>(0.235) | 0.422***<br>(0.160) | 0.314***<br>(0.115) | 0.741***<br>(0.211) | 0.385***<br>(0.128) |
| Lender fixed effects       | X                |                   | X                | X                |                    | X                | X                  |                     | X                   | X                   |                     | X                   |
| Year-Quarter fixed effects | X                | X                 | X                | X                | X                  | X                | X                  | X                   | X                   | X                   | X                   | X                   |
| Lender×Year fixed effects  |                  | X                 |                  |                  | X                  |                  |                    | X                   |                     |                     | X                   |                     |
| Lender Controls            |                  |                   | X                |                  |                    | X                |                    |                     | X                   |                     |                     | X                   |
| Observations               | 15,823           | 15,755            | 11,905           | 15,823           | 15,755             | 11,905           | 15,823             | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |

Notes: The table reports coefficients from OLS regressions of symmetric growth rates in VaR and CoVaR (in percentage points) on standardized  $InflowsAtTies_{it}$ . The outcomes are VaR at the 95% confidence level (Columns 1–3), CoVaR at the 95% confidence level (Columns 4–6), VaR at the 99% confidence level (Columns 7–9), and CoVaR at the 99% confidence level (Columns 10–12). Within each outcome block, the first column includes lender and year-quarter fixed effects, the second replaces lender fixed effects with saturated lender-by-year fixed effects, and the third includes lender and year-quarter fixed effects plus time-varying lender controls. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. The data are at quarterly frequency and span 1994 to 2009. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

in outside options is associated with increases of 0.48 percentage points in VaR and 0.44 percentage points in CoVaR in the lender-by-year specification. At the 99% confidence level, the corresponding estimates rise to 0.78 and 0.74 percentage points, respectively, and remain positive and statistically significant when we instead use the lender-control specification. The dependent variables are quarterly symmetric midpoint growth rates multiplied by 100, so the estimates represent percentage-point changes in quarterly risk growth. For changes of this size, the midpoint

growth scale is approximately equal to conventional percentage growth. On the estimation samples, the lender-by-year coefficients at the 99th percentile equal 3.9% of the standard deviation of VaR growth and 3.8% of the standard deviation of CoVaR growth. The corresponding 95th-percentile effects equal 2.2% for both measures. The comparable standardized magnitudes show that the shift in risky lending remains economically visible in bank tail-risk and sectoral-distress measures.

The appendix shows that these estimates remain stable when we add the focal bank's employee inflows or outflows and hiring through co-syndication links, exclude the 2007 to 2008 financial crisis, winsorize the outside options measure, and vary the duration for which ties remain active (Tables A17 and A18). The same results obtain when outside options are recomputed separately from medium- and high-seniority job-to-job hires, with no systematic difference across seniority groups (Appendix Table A19).

Taken together, the VaR and CoVaR responses suggest a pass-through from riskier lending to the risk of the originating bank and the financial system. The similar standardized VaR and CoVaR responses further indicate that the association is equally visible in bank-specific and systemic tail risk. The concentration of the response at the 99th percentile connects the risky credit expansion to the part of the return distribution most relevant for financial stability. The lending effect therefore survives aggregation into standard, externally validated measures of bank and systemic risk.

**Effect Dynamics.** Figure 1 reports cumulative responses for risky-volume growth, VaR99, and CoVaR99 under lender plus year-quarter and lender-year plus year-quarter fixed effects. The estimates suggest immediate responses for all variables and specifications, with a statistically significant coefficient in the first quarter following the shock. At later horizons, the coefficients remain elevated but are estimated less precisely.

## 5.2 Borrower Risk

We now turn to the loan level to ask whether better outside options predict lending to riskier borrowers. Consider a bank  $i$  issuing a loan  $l$  to borrower  $b$  in year-quarter  $t$ . The loan-level

specification is given by:

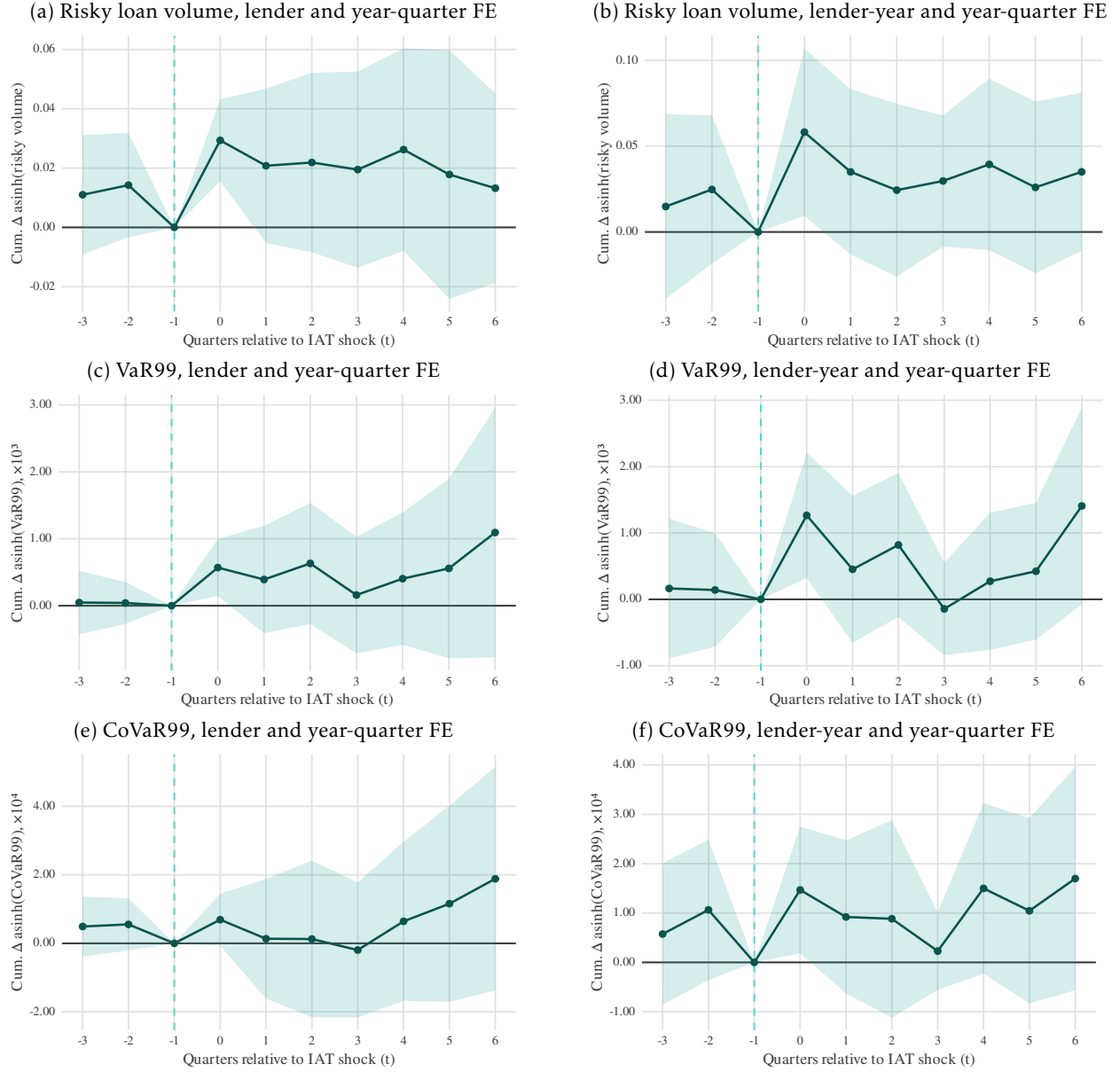
$$y_{bilit} = \beta \text{InflowsAtTies}_{it} + \Gamma V_{it} + \Theta W_{bt} + \Pi Z_{ilt} + \alpha_i + \lambda_t + \epsilon_{bilit}, \quad (16)$$

where  $y_{bilit}$  is a measure of credit risk, proxied by the borrower's credit rating. As before, we regress the outcome on inflows at ties, with lender-level controls and fixed effects at the lender and year-quarter level. The loan-level specification additionally allows us to control for the same balance sheet variables on the borrower side,  $W_{bt}$ , and for loan-level covariates  $Z_{ilt}$ , comprising the inverse hyperbolic sine of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached.

**Ex Ante Risk.** The results from estimating linear probability models are presented in Table 3. The outcome is a borrower-level indicator equal to one if the borrower is rated below investment grade by S&P. A positive coefficient thus signals a shift toward nominally riskier borrowers.  $\text{InflowsAtTies}_{it}$  is standardized, and all columns include the full set of borrower, lender, and loan controls. Column (1) is the baseline specification of Equation (16) with additive lender and year-quarter fixed effects. A one standard deviation increase in inflows at ties raises the probability that a borrower is below investment grade by about half a percentage point. Columns (2) to (4) replace the additive lender fixed effect with lender-by-year fixed effects and successively add borrower-sector-by-year-quarter and borrower-state-by-year-quarter fixed effects, which absorb time-varying demand-side factors (Huang et al., 2024, 2025). The coefficient is stable throughout and increases in magnitude under the fully saturated specification in Column (4), where it reaches 1.3 percentage points per standard deviation.

This effect is robust to alternative measures of borrower risk. Appendix Table A20 re-estimates the four specifications using the median contemporaneous Mergent FISD bond rating across Moody's, S&P, and Fitch. The result for the continuous FISD risk score is positive and significant at the 10% level or below in all four specifications. The indicator for borrowers below investment grade is positive in the first three specifications, although the estimate attenuates and becomes insignificant as the demand-side fixed effects are added. The shift toward riskier borrowers also appears in a balance sheet measure of risk. When we re-estimate the set of fixed-effects specifi-

Figure 1: Cumulative Event Studies of Risky Loan Volume Growth and Financial Risk



Notes: Each point is the coefficient from a separate regression of the cumulative inverse-hyperbolic-sine change from quarter  $t - 1$  to the indicated horizon on standardized inflows at ties. The reference quarter  $t - 1$  is normalized to zero. Shaded bands are 95% confidence intervals based on two-way clustered standard errors at the lender and year-quarter levels.

cations with the borrower’s book leverage as the dependent variable, the coefficient on inflows at ties is positive and significant across all columns (Table A24). Better outside options are therefore associated with lending to more highly levered borrowers.

For the loan-level analysis, the effect of outside options among very senior bankers is almost twice as large as the corresponding effect for less senior employees, consistent with the notion that informal networks are more important for workers at very high seniority levels (Appendix Table A21).

We subject this result to a number of robustness checks, reported in Table A22. The coefficient of interest remains stable and significant when we additionally control for the lending bank’s total employee inflows (Column 1) or outflows (Column 2), the number of lenders in the syndicate (Column 3), or hiring through co-syndication ties ( $InflowsAtCosignatories_{it}$ , Column 4, discussed below). It is similarly stable when we restrict the sample to newly originated loans (Column 5), exclude the 2007–2008 financial crisis (Column 6), or winsorize the outside options measure at the 1st and 99th percentiles (Column 7). The result is also robust to alternative assumptions about the duration for which ties remain active (Table A23).

We further probe the  $\alpha_i \varepsilon_t$  confounder by testing whether the lender’s network composition loads it onto industry-specific hiring cycles. We implement this test in two ways. First, we identify the industry segment that receives the largest share of each lender’s network weight and absorb fixed effects for that dominant segment interacted with year. This specification allows lenders whose networks point toward different industries to follow different annual paths. We estimate it with two- and three-digit NAICS segments. Second, we calculate total hiring in each connected bank’s industry segment and weight it by the lender’s network shares. We include this exposure-weighted hiring index as a single control, which removes the segment-level hiring cycle reaching the lender through its full network. We construct the index at the two- and one-digit NAICS levels. The coefficient on inflows at ties is stable across these specifications (Table A26).

Although our tie weights are predetermined relative to current hiring, they still change over time. We therefore estimate the specification in first differences and then replace the change in the baseline measure with a fixed-exposure first-difference shift-share measure,  $\sum_j w_{ij,t-1} \Delta g_{jt}$  (Borusyak et al., 2022). This measure applies the prior quarter’s network weights to changes in

Table 3: Effect of Outside Options on Borrower Risk

|                                           | Non-Prime Borrower (ex ante) |                     |                     |                     |
|-------------------------------------------|------------------------------|---------------------|---------------------|---------------------|
|                                           | (1)                          | (2)                 | (3)                 | (4)                 |
| IAT                                       | 0.005***<br>(0.001)          | 0.006***<br>(0.002) | 0.006***<br>(0.002) | 0.013***<br>(0.004) |
| Lender fixed effects                      | X                            |                     |                     |                     |
| YQ fixed effects                          | X                            | X                   |                     |                     |
| Lender $\times$ Year fixed effects        |                              | X                   | X                   | X                   |
| Borrower Sector $\times$ YQ fixed effects |                              |                     | X                   | X                   |
| Borrower State $\times$ YQ fixed effects  |                              |                     |                     | X                   |
| Borrower+Lender+Loan Controls             | X                            | X                   | X                   | X                   |
| Observations                              | 55,327                       | 55,278              | 55,223              | 50,045              |

*Notes:* The table reports the loan-level borrower-risk regression of Equation (16) across a set of increasingly demanding fixed effects. The dependent variable is an indicator equal to one if the borrower is rated below investment grade by S&P, estimated by a linear probability model, so a positive coefficient indicates a shift toward riskier borrowers.  $InflowsAtTies_{it}$  is standardized and all columns include the full set of borrower, lender, and loan controls. Column (1) includes additive lender and year-quarter fixed effects; Column (2) replaces the lender fixed effect with lender-by-year fixed effects; Column (3) adds borrower-sector-by-year-quarter fixed effects; and Column (4) further adds borrower-state-by-year-quarter fixed effects. Borrower and lender controls are the inverse hyperbolic sine of total assets, number of employees, capital expenditure, total liabilities, EBIT, and (for borrowers) fixed assets; loan controls are the inverse hyperbolic sine of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. The data span 1994 to 2020. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

connected-bank hiring, removing variation generated by quarterly changes in exposure weights. The shift toward riskier borrowers remains positive in both specifications (Table A27).

The connected-bank lending controls described in the loan-volume results produce the same conclusion for borrower risk. The coefficient on inflows at ties is stable whether we control for total or leveraged lending at connected banks, in levels or growth, individually or jointly (Table A28). We also include hiring measured over prior co-syndication links in Column (4) of Table A22. The coefficient on inflows at coworker ties remains stable. These results extend the bank-level checks to borrower composition without repeating their construction.

We also investigate whether the response of borrower ratings depends on the ease of job-to-job transitions. Non-compete agreements, NCAs, restrict work for direct competitors and reduce mobility and wages (Johnson et al., 2025). State court decisions create changes in enforceability over time. When enforceability decreases in the bank's headquarters state, bankers face fewer constraints on exercising outside options. We interact our measure with the state-level enforceability index of Mueller (2023) and include state-year fixed effects. Table A29 shows a positive

and statistically significant interaction with past decreases in enforceability in Column (3), so the borrower-risk response is concentrated in states that moved toward weaker enforceability. Interactions for increases and for the combined change measure are imprecise. Overall, the evidence is consistent with the notion that the borrower-risk response is conditional on the ability of bankers to exercise their outside options.

An important question is whether this shift reflects a genuine increase in risk-taking or yield-seeking behavior. Under yield seeking, the decline in ratings should be compensated by higher spreads. We test this by regressing the log all-in spread drawn, the standard measure of loan pricing, on inflows at ties, conditional on the same controls and fixed effects as in the baseline. As reported in Table A25, the estimated coefficients are small and statistically insignificant, allowing us to rule out spread increases larger than roughly four basis points per standard-deviation shock. This amount equals about 1% of the outcome mean at the 95% confidence level. Banks thus take on riskier borrowers without demanding significantly higher compensation, which is difficult to reconcile with yield seeking.

The estimated coefficients reported in this section are moderate in magnitude but can be considered lower-bound estimates for several reasons. First, by estimating effects at the bank level, we average across all investment bankers, including those who are unresponsive to outside options because of low observed productivity, niche specialization, or strong firm-specific attachments (Caldwell et al., 2025). Second, our identification strategy removes cyclical variation in outside options, which represents a substantial share of the overall variation and is likely the component to which bankers respond most.<sup>14</sup> Third, our measure does not capture news about labor demand at other firms that is transmitted outside personal networks, such as public job postings, positions at non-connected banks, or opportunities outside the financial industry. Taken together, our results reflect the average response to variation in one plausibly exogenous component of outside options.

**Ex Post Risk.** So far, we have measured borrower risk at origination. An important follow-up question is whether the shift toward riskier borrowers reflects genuine risk or merely a change in

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<sup>14</sup>Year-quarter fixed effects explain around 39% of the overall variation in inflows at ties, increasing to 47% after controlling for bank fixed effects.

the composition of ratings assigned when loans are made. We track two realized borrower outcomes over the three years following each loan event. The first is subsequent downgrades of the borrower's bonds from Mergent FISD records of Moody's, S&P, and Fitch rating actions. The second is the spread on the borrower's next syndicated loan, conditional on the spread of the current loan. Table 4 reports results for these outcomes in our baseline panel specification, with lender and year-quarter fixed effects and the full set of controls. Loans originated when the lending bank has better outside options are followed by more downgrades, with the effect concentrated in the tail of the outcome distribution. The likelihood of any downgrade rises only marginally in Column (1), while the likelihoods of two or more and three or more downgrade actions rise by about 1.3 percentage points per standard deviation and are statistically significant in Columns (2) and (3). The borrower's next loan is priced roughly 1% higher in Column (4), consistent with borrower credit deterioration between origination and refinancing. Because these estimates hold within the origination quarter, they are not an artifact of the credit cycle. When absorbing only lender fixed effects, however, the associations with any downgrade and subsequent default are several times larger and highly significant, in line with the procyclical, aggregate character of the mechanism (Appendix Table A31).

Appendix Table A30 further compares the outside options response of borrower risk by lender-borrower relationship status. Since screening effort plays a greater role in first-time relationships, a stronger response for new lender-borrower relationships would isolate screening as the mechanism after holding borrower composition at origination fixed. We therefore interact our proxy with an indicator equal to one if the borrower and lender did not have an ongoing lending relationship before the loan in question. This exercise allows us to include highly demanding lender-by-year-quarter fixed effects at the level of our proxy. The outside options interaction is positive for below-investment-grade origination and statistically significant for two-year downgrade risk, while the remaining realized-risk interactions are smaller and less precise. We therefore find some evidence that outside options lead to greater borrower risk when screening plays a greater role.

Together with the lender-level evidence presented above, these results show that improved outside options coincide with both a risky credit expansion and a deterioration in the borrower pool.

Table 4: Ex Post Borrower Deterioration

| Ex-post outcome            | ≥1 Downgrade                    | ≥2 Downgrades | ≥3 Downgrades | Next-Loan Spread |
|----------------------------|---------------------------------|---------------|---------------|------------------|
|                            | Subsequent bond downgrades (3y) |               |               | Next loan (3y)   |
|                            | (1)                             | (2)           | (3)           | (4)              |
| IAT                        | 0.008*                          | 0.013**       | 0.013**       | 0.009**          |
|                            | (0.004)                         | (0.005)       | (0.006)       | (0.003)          |
| Current Loan Spread        |                                 |               |               | 0.531***         |
|                            |                                 |               |               | (0.022)          |
| Lender fixed effects       | X                               | X             | X             | X                |
| Year-Quarter fixed effects | X                               | X             | X             | X                |
| Borrower Controls          | X                               | X             | X             | X                |
| Lender Controls            | X                               | X             | X             | X                |
| Loan Controls              | X                               | X             | X             | X                |
| Outcome Mean               | 0.366                           | 0.311         | 0.273         | 5.90             |
| Observations               | 29,287                          | 29,287        | 29,287        | 40,822           |

*Notes:* The table relates ex post borrower deterioration to  $InflowsAtTies_{it}$  at loan origination, estimating Equation (16) at the loan level with lender and year-quarter fixed effects and the full set of borrower, lender, and loan controls. In Columns (1)–(3), the outcome equals one if the borrower’s outstanding bonds receive at least one, two, or three rating downgrades, respectively, from Moody’s, S&P, or Fitch within three years of origination (Mergent FISD). The sample includes loans for which the borrower has a rated bond outstanding at origination. In Column (4), the outcome is the inverse hyperbolic sine of the all-in spread drawn, in basis points, on the borrower’s next syndicated loan within three years, controlling for the current loan’s spread.  $InflowsAtTies_{it}$  is standardized to unit variance. The downgrade estimates are unchanged when controlling for the number of the borrower’s outstanding bonds. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 6 Mechanism

This section examines the discipline dilution mechanism at the individual level. We use data on bankers and individually matched loans to answer two questions from the conceptual framework. We first ask whether the risk response is concentrated among younger bankers, whose mobility makes outside opportunities more likely to be acted upon. We then ask whether outside options weaken the separation penalty following poor loan performance, as predicted by Equation (10). The final subsection turns to financial advisers and provides complementary evidence from a different setting with consistent results on the insurance mechanism.

### 6.1 Evidence on Discipline Dilution

We match a subset of loans to the individual bankers who signed them and follow these bankers through the employment histories underlying our measure. For each matched loan we observe the borrower’s subsequent rating trajectory, i.e., downgrades of the borrower’s S&P rating and of

its outstanding bonds within two years of origination, as well as the signing banker's subsequent career at the lending bank.

We first document age heterogeneity in unconditional job mobility among signing bankers. Appendix Table A32 shows that each additional year of age lowers the probability of separating within eight quarters by 1.3 percentage points and within twelve quarters by 1.8 percentage points, against base rates of 26 and 36 percent. Job-to-job moves show the same steep slope. Over the fifteen-year age gap between a vice president and a senior managing director, this amounts to a difference in mobility of twenty to twenty-seven percentage points. Labor-market transitions are concentrated among young bankers, consistent with the on-the-job search dynamics introduced in the conceptual framework.

The age coefficient on mobility confirms that improvements in outside options are more relevant for younger bankers, who are more likely to exercise their options. The framework therefore predicts a stronger risk response among younger signers. Table 5 tests this prediction by regressing indicators of ex post deterioration on inflows at ties, measured over the two years following origination, interacted with the signing banker's age. Columns (1) to (3) use a downgrade of the borrower's bonds by Moody's, S&P, or Fitch within two years, while Columns (4) to (9) use downgrades and severe two-grade downgrades of the borrower's S&P rating. Where the standalone effect is identified, inflows at ties enter positively and loans signed under strong outside options are more likely to deteriorate. The interaction with age is negative and significant in all nine specifications, including those with lender-year-quarter fixed effects, which absorb the standalone coefficient on outside options and identify the age gradient from loans signed at the same bank in the same quarter by bankers of different ages. Evaluated at the sample mean age, the estimated effect on deterioration is close to zero, while for a banker in the early thirties a one standard deviation improvement in outside options raises the two-year probability of a bond rating downgrade by roughly 6 percentage points against a base level of 26 percent. The estimated response is therefore concentrated among bankers young enough to move. This age pattern weighs against senior poaching and team lift-outs as the primary explanation, since those accounts predict a stronger response among senior bankers.

The same negative age interaction is present when we measure the borrower's risk at origination rather than through its realized deterioration, though it is estimated less precisely there.

Table 5: Ex Post Loan Deterioration, Outside Options, and Banker Age

|                                     | Bond Downgrade (2 Years, pp) |                      |                     | Downgrade (2 Years, pp) |                      |                     | Severe Downgrade (2 Years, pp) |                    |                      |
|-------------------------------------|------------------------------|----------------------|---------------------|-------------------------|----------------------|---------------------|--------------------------------|--------------------|----------------------|
|                                     | (1)                          | (2)                  | (3)                 | (4)                     | (5)                  | (6)                 | (7)                            | (8)                | (9)                  |
| IAT                                 | 20.990***<br>(6.570)         | 30.703***<br>(5.508) |                     | 25.900***<br>(8.172)    | 27.575***<br>(7.316) |                     | 9.734**<br>(4.569)             | 16.235*<br>(8.186) |                      |
| Age                                 | 0.261<br>(0.166)             | 0.113<br>(0.243)     | 0.507*<br>(0.273)   | 0.192<br>(0.144)        | -0.063<br>(0.167)    | 0.348***<br>(0.108) | 0.235<br>(0.142)               | 0.021<br>(0.132)   | 0.256*<br>(0.140)    |
| IAT × Age                           | -0.531***<br>(0.157)         | -0.622***<br>(0.187) | -0.503**<br>(0.189) | -0.582**<br>(0.257)     | -0.535*<br>(0.262)   | -0.604**<br>(0.272) | -0.224*<br>(0.130)             | -0.363*<br>(0.199) | -0.368***<br>(0.109) |
| Lender fixed effects                | X                            | X                    |                     | X                       | X                    |                     | X                              | X                  |                      |
| Year-Quarter fixed effects          | X                            | X                    |                     | X                       | X                    |                     | X                              | X                  |                      |
| Lender × Year-Quarter fixed effects |                              |                      | X                   |                         |                      | X                   |                                |                    | X                    |
| Lender Controls                     |                              | X                    |                     |                         | X                    |                     |                                | X                  |                      |
| Outcome Mean                        | 26.282                       | 25.474               | 25.709              | 14.930                  | 16.585               | 15.976              | 6.620                          | 7.073              | 7.298                |
| Observations                        | 780                          | 475                  | 564                 | 710                     | 410                  | 507                 | 710                            | 410                | 507                  |

Notes: The table relates ex post loan deterioration to  $InflowsAtTies_{it}$  interacted with the signing banker's age in the sample of loans matched to individual bankers. Each deterioration indicator is multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. The indicators identify whether the borrower's outstanding bonds were downgraded by Moody's, S&P, or Fitch within two years of origination (Columns 1–3), whether the borrower's S&P rating was downgraded within two years (Columns 4–6), or whether it was downgraded by two or more notches (Columns 7–9).  $InflowsAtTies_{it}$  is standardized and measured during the second year after origination. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, EBIT, capital expenditure, total liabilities, short- and long-term debt, equity, and number of employees. Columns (3), (6), and (9) replace the separate lender and year-quarter fixed effects with lender-year-quarter fixed effects, which absorb the standalone coefficient on inflows at ties. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Appendix Table A33 reports it across the set of fixed-effects specifications for both an investment-grade indicator and a continuous risk score.

We next test the implication of Equation (10), which predicts that the separation penalty following poor performance falls as outside options improve. The employment histories record separations, without identifying whether each separation was initiated by the bank or the banker. The exercise therefore tests state dependence in the observed career penalty, while the interpretation in terms of lower employer dismissal strictness relies on the framework.

For signer  $s$  and loan  $\ell$ , we estimate

$$Separation_{s\ell} = \beta_1 Bad_\ell + \beta_2 (Bad_\ell \times InflowsAtTies_{it}) + \beta_3 InflowsAtTies_{it} + \Gamma V_{it} + \alpha_i + \lambda_t + \varepsilon_{s\ell}, \quad (17)$$

where  $Bad_\ell$  indicates that the borrower's bonds were downgraded within two years of origination,  $V_{it}$  are the lender controls, and inflows at ties are measured over the window in which the outcome materializes and the retention decision is made.

The results are reported in Table 6. Consistent with Gao et al. (2020), poor loan performance predicts an adverse career outcome. After a bad outcome on a signed loan, bankers are 7 to 11 percentage points more likely to separate from the bank within two to three years, with statistical significance in the specifications that include lender controls. The interaction  $\beta_2$  is negative in all columns and statistically significant throughout. Focusing on the point estimates, a one standard deviation improvement in outside options offsets the estimated separation penalty.

The positive baseline coefficient on inflows at ties for separations, especially job-to-job moves, confirms that the measure predicts mobility in this smaller sample. Together, the coefficients show that stronger outside options predict more mobility and a smaller observed separation penalty following poor loan outcomes, as implied by the discipline dilution channel.

## 6.2 Evidence on Adviser Misconduct

We provide complementary evidence in a related setting with worker agency and labor market outside options, namely the market for financial advisers. It differs from our main analysis because it covers a different occupation over the shorter period from 2007 to 2015. The setting adds two features to the previous analysis. Conceptually, advisers' misconduct identifies behavior that harms clients and employers more directly than an externally observed increase in credit risk. Moreover, the information on approximately 140,000 individually linked advisers provides enough variation to include rich sets of fixed effects, allowing us to examine how outside options shape the probabilities of job separation, industry exit, and reemployment. With this analysis, we contribute to a growing literature studying the determinants of adviser misconduct (Egan et al., 2019; Dimmock et al., 2021; Clifford and Gerken, 2021).

We first scrutinize how variation in outside options influences the frequency of adviser misconduct, building on Equation (16). We aggregate inflows at ties to annual frequency by taking averages and focus on the first lag of the measure, as misconduct typically takes time to be discovered and disclosed. Our analysis considers three outcome variables, which are the number of misconduct disclosures an adviser receives in a given year, the number of customer complaints filed against the adviser, and the number of complaints that result in a settlement. Given the within-firm spillovers in misconduct (Egan et al., 2019), we control for the first lag of the firm-level average of the outcome variable.

Table 6: Outside Options and the Separation Penalty for Poor Loan Outcomes

|                            | Job Separation (2 Years, pp) |                     | Job Separation (3 Years, pp) |                    | Job to Job Move (2 Years, pp) |                     |
|----------------------------|------------------------------|---------------------|------------------------------|--------------------|-------------------------------|---------------------|
|                            | (1)                          | (2)                 | (3)                          | (4)                | (5)                           | (6)                 |
| Bad Outcome                | 4.156<br>(4.343)             | 10.307**<br>(4.223) | 2.244<br>(3.546)             | 7.326**<br>(3.234) | 3.410<br>(4.406)              | 10.624**<br>(4.300) |
| IAT                        | 3.459<br>(2.798)             | 6.026<br>(4.941)    | 8.160*<br>(4.514)            | 11.416*<br>(6.496) | 8.097***<br>(2.916)           | 8.819*<br>(4.663)   |
| Bad Outcome $\times$ IAT   | -7.391***<br>(2.673)         | -8.228**<br>(3.910) | -4.730**<br>(1.625)          | -6.311*<br>(3.394) | -5.236***<br>(1.737)          | -4.396*<br>(2.526)  |
| Lender fixed effects       | X                            | X                   | X                            | X                  | X                             | X                   |
| Year-Quarter fixed effects | X                            | X                   | X                            | X                  | X                             | X                   |
| Lender Controls            |                              | X                   |                              | X                  |                               | X                   |
| Outcome Mean               | 26.303                       | 26.453              | 36.228                       | 36.473             | 21.712                        | 21.443              |
| Observations               | 806                          | 499                 | 806                          | 499                | 806                           | 499                 |

Notes: The table estimates Equation (17) on loans matched to their signing banker. *Bad Outcome* equals one if the borrower's outstanding bonds were downgraded by Moody's, S&P, or Fitch within two years of origination (Mergent FISD). Each dependent-variable indicator is multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. The indicators identify whether the banker separated from the lending bank within two years of origination (Columns 1–2), separated within three years (Columns 3–4), or moved to another employer within two years (Columns 5–6). Separations are constructed from the employment-history data and account for right-censoring. *InflowsAtTies* is standardized and measured during the second year after origination, the period in which the retention decision is made. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, EBIT, capital expenditure, total liabilities, short- and long-term debt, equity, and number of employees. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 7: Effect of Outside Options on Misconduct Disclosures

|                                       | Misconduct           |                      | Complaints          |                     | Complaints (Settled) |                      |
|---------------------------------------|----------------------|----------------------|---------------------|---------------------|----------------------|----------------------|
|                                       | (1)                  | (2)                  | (3)                 | (4)                 | (5)                  | (6)                  |
| $IAT_{t-1}$                           | 0.003***<br>(0.0009) | 0.003***<br>(0.0008) | 0.005***<br>(0.002) | 0.006***<br>(0.001) | 0.004***<br>(0.001)  | 0.004***<br>(0.0010) |
| Lender fixed effects                  | X                    |                      | X                   |                     | X                    |                      |
| Year fixed effects                    | X                    | X                    | X                   | X                   | X                    | X                    |
| Lender $\times$ Adviser fixed effects |                      | X                    |                     | X                   |                      | X                    |
| Lender Controls                       | X                    | X                    | X                   | X                   | X                    | X                    |
| Outcome Mean                          | 0.010                | 0.009                | 0.018               | 0.017               | 0.008                | 0.008                |
| Observations                          | 582,568              | 551,079              | 582,568             | 551,079             | 582,568              | 551,079              |

Notes: The table reports results from estimating Equation (16) at annual frequency at the adviser level. The dependent variables are the number of misconduct disclosures an adviser had in a given year (Columns 1 and 2), the number of complaints filed against the adviser (Columns 3 and 4), and the number of settled customer complaints (Columns 5 and 6). These outcomes remain as counts because a percentage-point transformation does not apply. The main independent variable is  $InflowsAtTies_{it-1}$ . Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity, together with the first lag of the firm-level average of the outcome variable. All models are estimated with year fixed effects and lender controls, with the latter replaced by lender-adviser fixed effects in Columns (2), (4), and (6). Standard errors are clustered at the lender level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The results are shown in Table 7. Better outside options correspond to statistically significant increases in misconduct disclosures, customer complaints, and settled complaints under both the baseline specification and a specification with adviser-firm fixed effects. The coefficients remain positive across robustness checks that add state-year fixed effects, use alternative timing assumptions, control for lagged outflows, and use binary outcomes (Tables A34, A35, and A36).

Egan et al. (2019) show that employment separations become more likely following adviser misconduct, while rehiring elsewhere partly offsets this penalty. Our mechanism posits that outside options reduce both the probability and expected cost of separation for the employee, weakening the deterrent effect of job loss. Evidence that the employment penalty itself varies with outside options is difficult to explain through overoptimism alone, although it does not exclude behavioral channels operating alongside incentive effects.

We test this prediction by estimating the differential effect of misconduct on three distinct outcomes, comparing individuals with better and worse outside options. Specifically, for an adviser  $a$  employed at bank  $i$  in year  $t$ , we define three indicators for transitions between year  $t$  and  $t + 1$ . The first is  $\text{Separation}_{at}$ , which equals one if the adviser is no longer employed at  $i$ . The second is  $\text{IndustryExit}_{at}$ , which equals one if the adviser is no longer recorded in the sample and thus exits the industry under our definition. The third is  $\text{Reemployment}_{at}$ , which equals one if the adviser is employed at another firm in the sample. We use two empirical specifications. First, we estimate the unconditional effects of misconduct, outside options, and their interaction on  $\text{Separation}_{at}$  and  $\text{IndustryExit}_{at}$ . Second, conditional on experiencing a job separation, we estimate the differential effects on  $\text{IndustryExit}_{at}$  and  $\text{Reemployment}_{at}$ .

For the unconditional effects, consider the following specification:

$$y_{at} = \beta_1 \text{Misconduct}_{at} + \beta_2 \text{Misconduct}_{at} \times \text{InflowsAtTies}_{it} + \beta_3 \text{Misconduct}_{at} \times \text{InflowsAtTies}_{it-1} + \Gamma V_{it} + \alpha_{ai} + \lambda_{it} + \epsilon_{ait}, \quad (18)$$

that is, we regress an outcome  $y_{at}$  on an indicator for misconduct, alone and interacted with contemporaneous values and the first lag of inflows at ties. We include fixed effects at the firm-year level, removing the baseline coefficient on outside options, and at the adviser or firm-adviser level.

For the conditional effects, we further augment this setup by including all potential interactions with  $\text{Separation}_{at}$ , focusing on  $\text{IndustryExit}_{at}$  and  $\text{Reemployment}_{at}$  as outcome variables.

The results are reported in Table 8. While we are particularly interested in the interactions with lagged outside options, we control for contemporaneous inflows at ties and all potential interactions in all regressions. In Columns (1) and (2), we estimate the unconditional specification for the probability of a job separation, including adviser and firm-adviser fixed effects, respectively. Misconduct enters with a large, positive, and highly significant coefficient, corroborating the results of Egan et al. (2019). The estimated coefficient on the interaction with lagged inflows at ties is negative, highly significant, and sizeable. The probability of job separation following misconduct is reduced by around 22% when inflows at ties are one standard deviation above the mean. Better outside options therefore lower the probability of experiencing a job separation following a misconduct disclosure, providing evidence of employers' reduced disciplining power.

In Columns (3) and (4), we repeat the estimation using  $\text{IndustryExit}_{at}$  as the outcome. Again, better outside options counteract the main effect, reducing the probability of leaving the industry following a misconduct disclosure. This pattern could be driven by changes in job separations and hiring at other firms. To isolate the hiring margin, we re-estimate Equation (18) and interact the variables of interest with  $\text{Separation}_{at}$ . The results in Columns (5) and (6) point to an important role for hiring. As before, misconduct-related separations raise the probability of exiting the industry, but this effect is muted for individuals with above-average outside options. Lastly, Columns (7) and (8) report similar results for  $\text{Reemployment}_{at}$ , an indicator for working at a different employer within the financial sector in the following year.

Taken together, these results suggest that outside options reduce the probability of separation following misconduct and, conditional on separation, predict continued employment within the sector. This pattern is consistent with weaker labor-market discipline when outside options improve.

## 7 Conclusion

This paper documents that better labor market outside options for bankers predict riskier lending behavior and faster growth in bank and systemic risk. Banks exposed to improved outside

Table 8: Effect of Outside Options on Labor Market Disciplining

|                                                                | Job Separation (pp) |                    |                    | Leave Industry (pp) |                     | Reemployment Next Year (pp) |                   |                   |
|----------------------------------------------------------------|---------------------|--------------------|--------------------|---------------------|---------------------|-----------------------------|-------------------|-------------------|
|                                                                | (1)                 | (2)                | (3)                | (4)                 | (5)                 | (6)                         | (7)               | (8)               |
| Misconduct                                                     | 42.5***<br>(5.25)   | 44.3***<br>(4.97)  | 23.7***<br>(2.92)  | 23.9***<br>(2.89)   | 0.395<br>(0.869)    | 0.314<br>(0.734)            | -0.598<br>(1.12)  | -0.393<br>(0.994) |
| Misconduct $\times$ IAT <sub>t</sub>                           | 1.46<br>(2.48)      | 1.81<br>(2.16)     | 1.81<br>(1.71)     | 1.77<br>(1.61)      | -0.013<br>(0.385)   | -0.063<br>(0.307)           | 0.191<br>(0.381)  | 0.285<br>(0.312)  |
| Misconduct $\times$ IAT <sub>t-1</sub>                         | -7.54***<br>(2.27)  | -8.63***<br>(1.95) | -5.39***<br>(1.31) | -5.51***<br>(1.30)  | 0.134<br>(0.434)    | 0.157<br>(0.389)            | -0.245<br>(0.508) | -0.333<br>(0.463) |
| Job Separation $\times$ Misconduct                             |                     |                    |                    |                     | 13.3**<br>(5.75)    | 12.7*<br>(6.33)             | -14.0**<br>(6.84) | -12.7*<br>(7.21)  |
| Job Separation $\times$ IAT <sub>t</sub>                       |                     |                    |                    |                     | -1.51<br>(1.63)     | 0.222<br>(1.66)             | 2.75*<br>(1.57)   | 1.41<br>(1.50)    |
| Job Separation $\times$ IAT <sub>t-1</sub>                     |                     |                    |                    |                     | -4.04***<br>(0.840) | -3.29***<br>(0.664)         | 3.19***<br>(1.05) | 2.24**<br>(0.864) |
| Job Separation $\times$ Misconduct $\times$ IAT <sub>t</sub>   |                     |                    |                    |                     | 5.29*<br>(2.72)     | 4.28<br>(2.92)              | -5.66<br>(3.51)   | -5.15<br>(3.52)   |
| Job Separation $\times$ Misconduct $\times$ IAT <sub>t-1</sub> |                     |                    |                    |                     | -6.78***<br>(1.53)  | -6.13***<br>(2.15)          | 7.69***<br>(2.00) | 7.25***<br>(2.46) |
| Adviser fixed effects                                          | X                   |                    | X                  |                     | X                   |                             | X                 |                   |
| Firm $\times$ Year fixed effects                               | X                   | X                  | X                  | X                   | X                   | X                           | X                 | X                 |
| Lender $\times$ Adviser fixed effects                          |                     | X                  |                    | X                   |                     | X                           |                   | X                 |
| Outcome Mean                                                   | 11.884              | 10.870             | 3.818              | 3.685               | 3.818               | 3.685                       | 7.943             | 7.118             |
| Observations                                                   | 598,839             | 589,763            | 598,839            | 589,763             | 598,839             | 589,763                     | 598,839           | 589,763           |

*Notes:* The table reports results from estimating versions of Equation (18) at annual frequency at the adviser level. The dependent variables are multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. The indicators identify whether the adviser is no longer employed at  $i$  in year  $t + 1$  (Columns 1 and 2), is no longer recorded in the sample (Columns 3 to 6), or is employed at another firm in the sample in  $t + 1$  (Columns 7 and 8). The main independent variables are  $InflowsAtTies_{it}$ ,  $InflowsAtTies_{it-1}$ , and an indicator for having a misconduct disclosure in year  $t$ . The specification is given by Equation (18) in Columns (1) to (4), with all variables interacted with the indicator for job separation in Columns (5) to (8). All models include fixed effects at the firm-year and at the adviser (Columns 1, 3, 5, and 7) or firm-adviser level (Columns 2, 4, 6, and 8). Standard errors are clustered at the firm level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

options expand non-investment-grade lending, while their individual loans shift toward riskier borrowers and experience worse subsequent outcomes. The evidence is consistent with stronger outside options providing bankers with implicit insurance against adverse outcomes and weakening the disciplinary role of job loss. In the individually matched sample, the risk response is concentrated among younger, more mobile bankers, and the separation penalty following poor loan performance falls when outside options improve. Additional evidence from financial advisers shows more misconduct and weaker employment penalties in stronger external labor markets.

These findings have implications for market participants and policymakers. Our results suggest that variation in sectoral labor market conditions affects employers' disciplining power, with consequences for operational risk and settlement costs. During tight labor markets, banks may benefit from strengthening incentive alignment to limit adverse on-the-job behavior, for instance by increasing the use of deferred compensation and clawback provisions and by setting stricter

risk limits. Yet an individual bank's ability to do so may be limited. The disciplinary value of these tools erodes as outside options improve, since better options shorten expected tenure, and a bank that defers compensation more aggressively becomes less attractive to talent when rivals are hiring.

For policymakers, our findings support incorporating insights from labor markets into macroprudential frameworks as a complement to the leverage-based tools already in use. The externality we document has no obvious remedy. Non-compete agreements shut down the implicit insurance channel at the cost of undercutting competition for talent and the efficient matching of workers to firms. Deferred compensation, malus, and clawback provisions induce more prudent behavior while a banker remains in place, yet their disciplinary value ends once the banker moves. Making loan performance more observable across firms could narrow this information friction in principle. A hiring bank has little stake in the fate of a new hire's legacy portfolio, and performance realized under another institution's monitoring is a noisy signal of the new hire's own performance. As a starting point, policymakers might focus on making the current state of labor-market tightness transparent and equally available to supervisors and market participants, allowing the erosion of internal discipline to be recognized as it builds even where it cannot easily be reversed.

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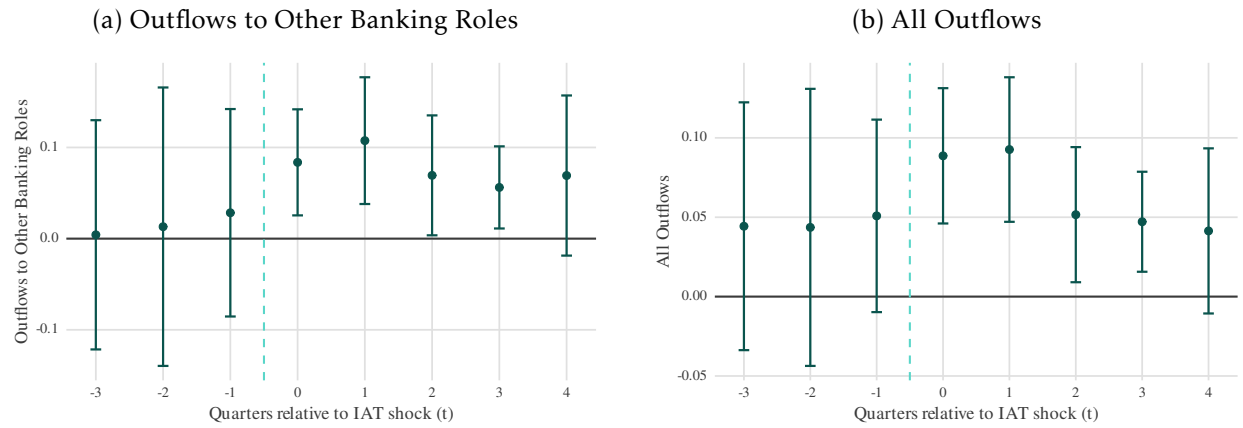
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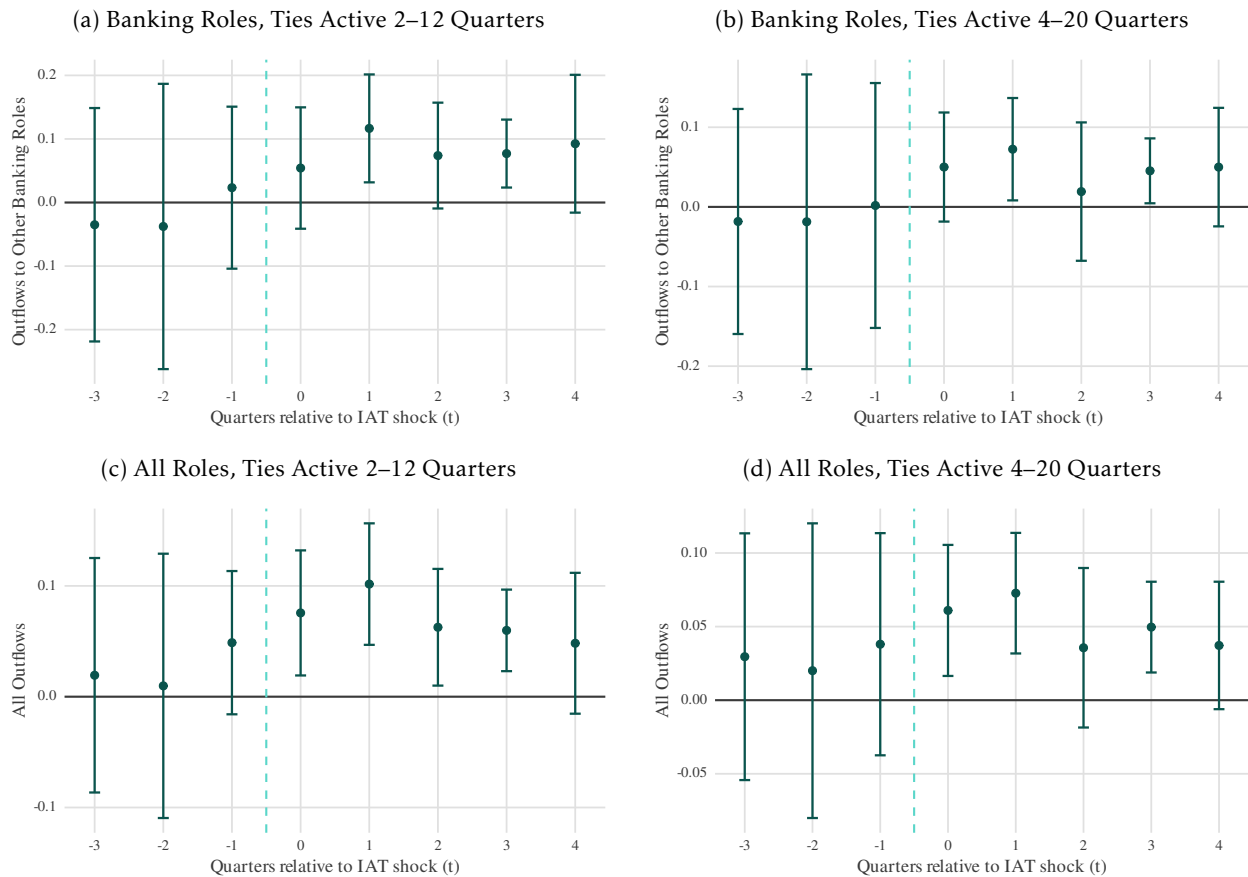
## A Additional Figures

Figure A1: Effect of Outside Options on Employee Outflows



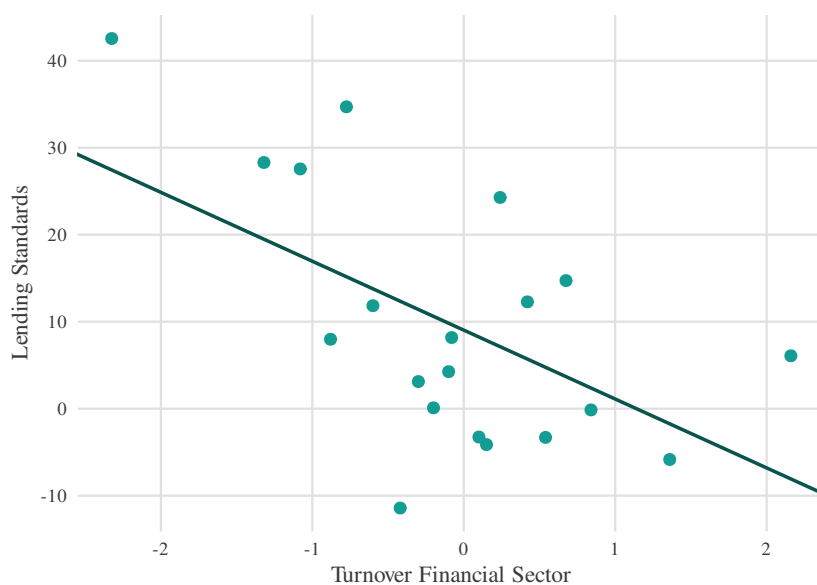
Notes: Points show coefficients from separate Poisson regressions of employee outflows on inflows at ties at horizons from three quarters before through four quarters after the shock. Panel a restricts outflows to roles at other banks in our sample, while Panel b includes all destinations. Whiskers show 95% confidence intervals based on two-way clustered standard errors at the lender and year-quarter levels. All regressions include lender and year-quarter fixed effects. Inflows at ties are standardized to have mean zero and standard deviation one.

Figure A2: Effect of Outside Options on Outflows, Robustness to Alternative Tie Lengths



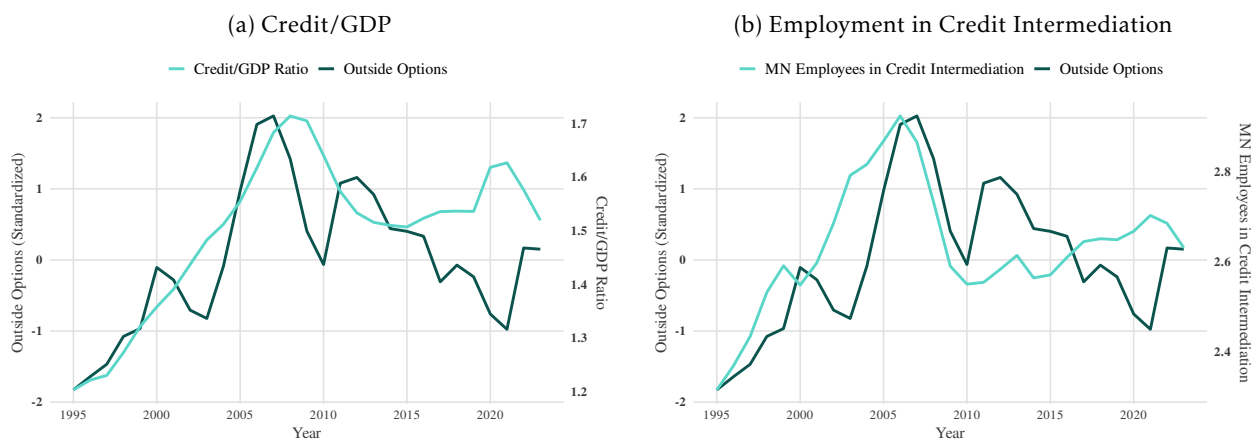
*Notes:* Points show coefficients from separate Poisson regressions of employee outflows on inflows at ties at horizons from three quarters before through four quarters after the shock. Panels a and c use ties active 2 to 12 quarters after a coworker transition, while Panels b and d use ties active 4 to 20 quarters after a transition. Whiskers show 95% confidence intervals based on two-way clustered standard errors at the lender and year-quarter levels. All regressions include lender and year-quarter fixed effects. Inflows at ties are standardized to have mean zero and standard deviation one.

Figure A3: Labor Market Turnover in the Financial Sector and Variation in Credit Standards



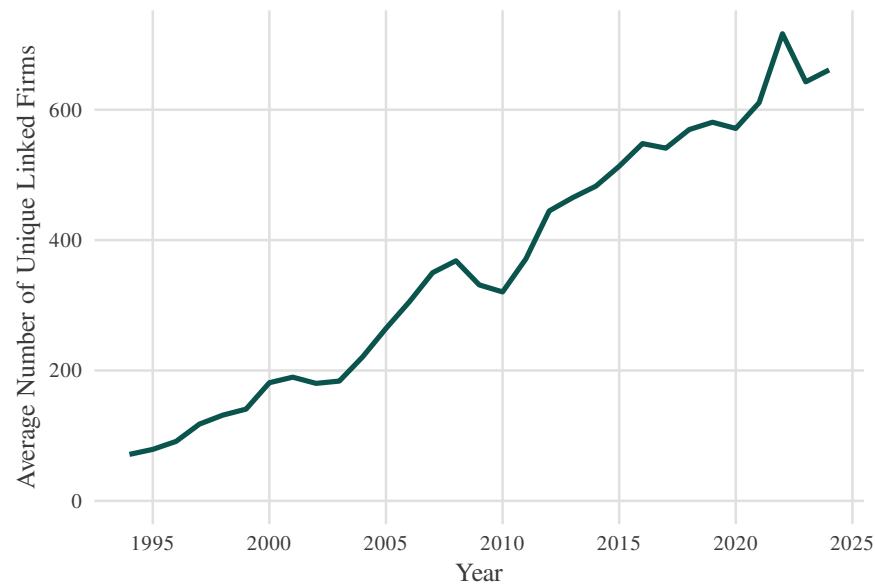
*Notes:* The figure bins voluntary labor market turnover in the financial sector against changes in lending standards. Labor market turnover is the sum of hiring and quit rates in financial activities (FRED series JTU510099HIL and JTU510099QUR), transformed into first differences. Lending standards are measured by the net percentage of domestic banks tightening their standards across loan categories. We calculate this measure using data from the Senior Loan Officer Opinion Survey on Bank Lending Practices (SLOOS) and weight it by banks' outstanding loan balances (FRED series SUBLPDMOSXWBNQ).

Figure A4: Procyclicality of Inflows at Ties



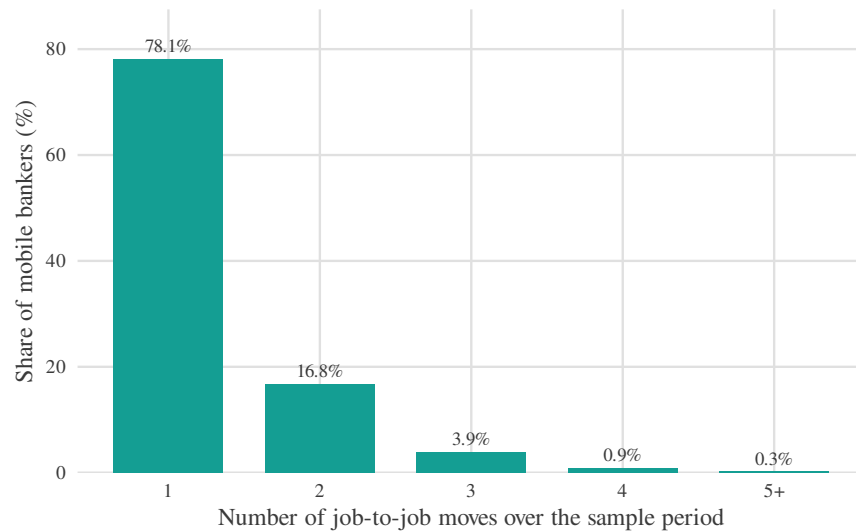
*Notes:* The plot shows average inflows at ties over time against the credit-to-GDP ratio (Panel a) and the number of employees in credit intermediation (Panel b). We obtain the aggregate data from FRED, using series QUSPAM770A and CEU5552200001.

Figure A5: Number of Connected Banks over Time



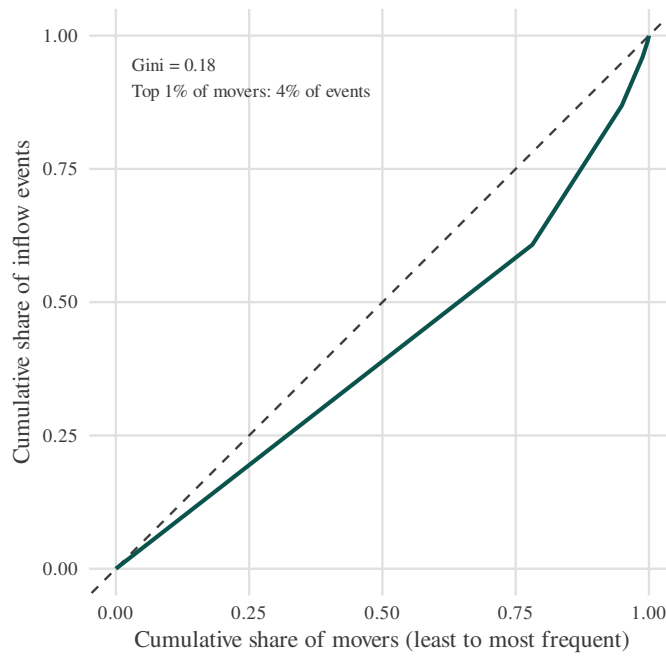
Notes: The figure plots, over time, the number of distinct banks with non-zero hiring and at least one active tie to a bank in our main estimation sample.

Figure A6: Job-to-Job Moves per Mobile Banker



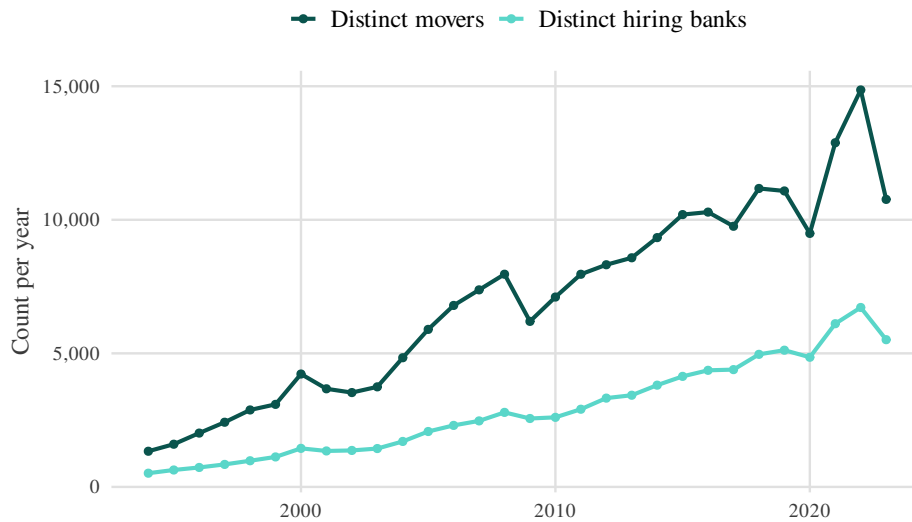
Notes: The figure shows the distribution of the number of job-to-job transitions per banker, among senior U.S. bankers who make at least one such move over the sample period (full sample, not restricted to the matched estimation sample). Moves of five or more are grouped. The overwhelming majority of movers transition only once, and serial switching is rare, indicating that the measure is not driven by a small set of highly mobile individuals.

Figure A7: Concentration of Inflow Events Across Movers



*Notes:* The figure plots the Lorenz curve of job-to-job inflow events across movers, ranked from least to most frequently mobile (full sample). The dashed line is the line of equality. The Gini coefficient is 0.18, and the most frequently mobile 1% of movers account for only about 4% of all inflow events, so no small group of serial switchers drives the identifying variation in the measure.

Figure A8: Number of Distinct Movers and Hiring Banks



*Notes:* The figure plots, per year, the number of distinct senior bankers making job-to-job moves into connected banks and the number of distinct hiring (connected) banks that generate the inflow shocks (full sample). Both series grow steadily over the sample period, reflecting a broad and expanding base of movers and institutions.

## B Additional Tables

Table A1: Summary Statistics: Dealscan Loan-Level Data

| Variable        | N     | Mean | SD   | P25  | P50 | P75  |
|-----------------|-------|------|------|------|-----|------|
| Borrower Prime  | 55327 | 0.3  | 0.46 | 0    | 0   | 1    |
| Borrower Rating | 55327 | 0.46 | 0.2  | 0.25 | 0.5 | 0.62 |
| Tranche Amount  | 55327 | 905  | 1623 | 150  | 400 | 1000 |
| Maturity        | 55327 | 50   | 23   | 36   | 58  | 60   |
| Secured         | 55327 | 0.51 | 0.5  | 0    | 1   | 1    |
| Covenants       | 55327 | 0.48 | 0.5  | 0    | 0   | 1    |

*Notes:* The table reports loan-tranche-level summary statistics for our sample of Dealscan loans. The sample includes loan observations from the 1,500 largest lead-arranging banks. *Borrower Prime* is an indicator for investment-grade borrowers. *Borrower Rating* is an ordinal borrower rating normalized to range from 0 to 1. *Tranche Amount* is measured in millions of USD. *Maturity* is measured in months. *Secured* is an indicator for whether the tranche is secured. *Covenants* is an indicator for whether financial covenants exist. The data span 1994 to 2020.

Table A2: Summary Statistics: FISD Borrower Ratings

| Variable                        | N      | Mean  | SD    | P25   | P50   | P75   |
|---------------------------------|--------|-------|-------|-------|-------|-------|
| Below-Investment-Grade Borrower | 30,146 | 0.56  | 0.50  | 0.00  | 1.00  | 1.00  |
| Borrower Risk Score, 0–100      | 30,146 | 50.35 | 18.25 | 38.10 | 47.62 | 66.67 |
| Median FISD Rating, 1–22        | 30,146 | 11.43 | 3.83  | 8.00  | 12.00 | 14.00 |
| Active Bond–Agency Ratings      | 30,146 | 28.82 | 53.42 | 4.00  | 12.00 | 33.00 |

*Notes:* The table reports loan-tranche-level summary statistics for the subsample of Dealscan loans with a contemporaneous issuer credit rating in Mergent FISD. For each tranche, we retain the most recent rating in effect on or before origination for every active bond–agency pair associated with the borrower and take the median across these ratings. Ratings from S&P, Moody’s, and Fitch are mapped to a 22-point scale, from 1 (D) to 22 (AAA/Aaa). *Below-Investment-Grade Borrower* equals one if the median rating is below 13 (BBB-/Baa3). *Borrower Risk Score* maps the median rating to a 0–100 scale, where AAA/Aaa equals zero and D equals 100. *Active Bond–Agency Ratings* is the number of ratings used to construct the median issuer rating. The sample spans 1994 to 2020.

Table A3: Summary Statistics: Employment History Data

|                           | N      | Mean   | Weighted Mean | SD    | P25    | P50    | P75    |
|---------------------------|--------|--------|---------------|-------|--------|--------|--------|
| Male                      | 801049 | 0.69   | 0.68          | 0.46  | 0.00   | 1.00   | 1.00   |
| White                     | 802908 | 0.78   | 0.78          | 0.42  | 1.00   | 1.00   | 1.00   |
| Bachelor                  | 634528 | 0.50   | 0.50          | 0.50  | 0.00   | 1.00   | 1.00   |
| More than Bachelor        | 634528 | 0.48   | 0.48          | 0.50  | 0.00   | 0.00   | 1.00   |
| Salary (in 1000)          | 803076 | 150.06 | 149.55        | 69.22 | 102.56 | 135.27 | 181.38 |
| Age at Start              | 771951 | 33.55  | 33.53         | 8.18  | 28.20  | 32.60  | 38.40  |
| Age at Middle             | 771951 | 36.69  | 36.66         | 8.28  | 30.85  | 35.68  | 41.71  |
| Age at End                | 771951 | 39.83  | 39.80         | 9.69  | 32.91  | 38.31  | 45.36  |
| Seniority                 | 803096 | 4.75   | 4.75          | 0.74  | 4.00   | 5.00   | 5.00   |
| Nr. of Previous Positions | 803096 | 3.02   | 3.03          | 2.62  | 1.00   | 3.00   | 4.00   |

*Notes:* The table reports spell-level summary statistics on demographic variables for the sample of senior employees working in credit origination described in Section 3. The full sample in Panel A includes all employment spells reported in the United States, in the respective job roles, and at seniority level 4 or above. *Male*, *White*, *Bachelor*, and *More Than Bachelor* are indicator variables equal to one for the corresponding categories. *Salary* is an imputed variable based on observable characteristics of an employment spell and measured in thousands of USD. The age variables refer to the beginning, middle, and end of an employment spell. *Seniority* is the seniority ranking, with values between 4 (lowest) and 7 (highest). *Nr. of Previous Positions* counts the number of positions held before the current spell. Panel B reports demographic summary statistics for the sample of bankers matched to individual loans at the loan-signature date. *Age* and *Job Tenure* are measured in years. *Seniority* is the seniority ranking, with values between 0 (lowest) and 7 (highest). *Salary* is an imputed variable based on observable characteristics of an employment spell and measured in thousands of USD. *Weighted Mean* uses sampling weights constructed by Revelio Labs (see here for details).

Table A4: Summary Statistics: Employment History Data vs. Administrative Data

|                        | Estimation Sample | Full Sample | CPS   | ACS    |
|------------------------|-------------------|-------------|-------|--------|
| Male                   | 0.68              | 0.63        | 0.57  | 0.57   |
| White                  | 0.78              | 0.74        | 0.84  | 0.82   |
| Bachelor               | 0.50              | 0.56        | 0.49  | 0.46   |
| More than Bachelor     | 0.48              | 0.40        | 0.17  | 0.17   |
| Salary (in 1000)       | 149.55            | 102.10      | 98.31 | 100.91 |
| Age                    | 36.66             | 32.15       | 40.84 | 40.13  |
| Observations (in 1000) | 607.60            | 1657.54     | 20.23 | 304.14 |

*Notes:* The table reports spell-level weighted averages of demographic variables, comparing the employment-biography data with demographic sources. *Estimation Sample* includes all employment spells reported in the United States, in the respective job roles, and at seniority level 4 or above, as described in Section 3. *Full Sample* relaxes the latter restriction and includes all seniority levels in the employment-spell data. *CPS* and *ACS* use administrative data from the *Current Population Survey* and the *American Community Survey*, respectively. For the CPS and ACS, we retain all observations between 1994 and 2023 for individuals who are in the labor force, are at work and employed full time, are younger than 66, and have hourly income of at least USD 10, equivalent to USD 21,600 annually. We restrict the sample to employment in the financial industry, using 1990 industry classification codes from 700 to 710, and occupations related to loan origination, using 2010 occupation codes 0120, 0830, 0840, 0850, 0910, and 4820. *Male*, *White*, *Bachelor*, and *More Than Bachelor* are indicator variables equal to one for the corresponding categories. *Salary* is measured in thousands of USD. For the employment-history data, it is an imputed measure based on observable characteristics of an employment spell. *Age* is age at the midpoint of an employment spell for the spell-level data. *Observations* is the number of observations that are nonmissing for all listed variables.

Table A5: Summary Statistics: Borrower- and Lender-Level Outcomes and Controls

|                                                              | N      | Mean         | SD         | P25        | P50          | P75          |
|--------------------------------------------------------------|--------|--------------|------------|------------|--------------|--------------|
| <i>Panel A: Borrower Characteristics</i>                     |        |              |            |            |              |              |
| Total Assets                                                 | 55,327 | 29,702.32    | 126,413.95 | 1,906.80   | 6,370.07     | 18,754.20    |
| Capex                                                        | 55,327 | 929.33       | 2,594.73   | 32.23      | 138.41       | 597.25       |
| Employees                                                    | 55,327 | 31.12        | 96.96      | 2.33       | 8.41         | 25.70        |
| Liabilities                                                  | 55,327 | 23,160.37    | 111,742.11 | 1,200.10   | 4,217.55     | 13,194.50    |
| EBIT                                                         | 55,327 | 1,308.43     | 3,782.17   | 79.63      | 349.00       | 1,097.00     |
| Net Income                                                   | 55,327 | 608.39       | 2,833.28   | -1.61      | 111.60       | 507.10       |
| Current Liabilities                                          | 55,327 | 2,651.48     | 19,409.86  | 10.00      | 74.20        | 503.00       |
| Long Term Liabilities                                        | 55,327 | 8,157.54     | 21,938.02  | 568.89     | 2,077.75     | 6,840.00     |
| Stockholder Equity                                           | 55,327 | 6,162.13     | 17,864.16  | 381.04     | 1,500.63     | 4,700.43     |
| Book Leverage (Debt/Assets, in Ppt.)                         | 55,327 | 40.73        | 22.33      | 25.33      | 38.59        | 53.56        |
| <i>Panel B: Lender Characteristics</i>                       |        |              |            |            |              |              |
| Total Assets                                                 | 55,327 | 1,325,920.98 | 809,725.31 | 687,935.00 | 1,198,942.00 | 1,930,115.00 |
| Capex                                                        | 55,327 | 1,411.17     | 2,171.85   | 73.00      | 934.11       | 1,497.20     |
| Employees                                                    | 55,327 | 112.34       | 81.66      | 51.20      | 80.00        | 163.50       |
| Liabilities                                                  | 55,327 | 1,235,089.55 | 761,271.58 | 645,607.00 | 1,108,646.00 | 1,757,212.00 |
| EBIT                                                         | 55,327 | 20,601.01    | 14,427.06  | 10,750.51  | 17,104.00    | 26,337.57    |
| Net Income                                                   | 55,327 | 7,169.74     | 7,446.01   | 2,655.00   | 5,579.94     | 10,101.96    |
| Current Liabilities                                          | 55,327 | 176,537.77   | 130,782.04 | 73,136.00  | 161,592.54   | 262,878.00   |
| Long Term Liabilities                                        | 55,327 | 120,202.79   | 84,346.79  | 32,771.00  | 127,663.64   | 178,324.00   |
| Stockholder Equity                                           | 55,327 | 88,314.10    | 62,474.38  | 44,354.00  | 75,397.31    | 110,450.56   |
| <i>Panel C: (Conditional) Value at Risk</i>                  |        |              |            |            |              |              |
| Value At Risk (95 <sup>th</sup> Pctile)                      | 10,635 | -0.08        | 23.00      | -12.00     | -0.85        | 11.00        |
| Conditional Value at Risk (95 <sup>th</sup> Pctile)          | 10,635 | -0.07        | 21.00      | -10.00     | -0.80        | 9.60         |
| Value At Risk (99 <sup>th</sup> Pctile)                      | 10,635 | -0.05        | 21.00      | -11.00     | -0.63        | 10.00        |
| Conditional Value at Risk (99 <sup>th</sup> Pctile)          | 10,635 | -0.04        | 21.00      | -9.90      | -0.45        | 9.30         |
| <i>Panel D: Compensation Data</i>                            |        |              |            |            |              |              |
| Median Salary (in \$1000)                                    | 8,457  | 784.00       | 393.00     | 506.00     | 767.00       | 1,000.00     |
| Median Bonus (in \$1000)                                     | 8,457  | 579.00       | 1,745.00   | 0.00       | 0.00         | 600.00       |
| Median Equity at Risk (in \$1000)                            | 8,333  | 9,170.00     | 19,784.00  | 2,342.00   | 5,672.00     | 11,147.00    |
| Median Salary (Growth, in Ppt.)                              | 8,462  | 3.60         | 27.00      | 0.00       | 2.70         | 9.90         |
| Median Bonus (Growth, in Ppt.)                               | 8,462  | -4.40        | 85.00      | 0.00       | 0.00         | 1.50         |
| Median Equity at Risk (Growth, in Ppt.)                      | 8,462  | 2.60         | 75.00      | -36.00     | 0.00         | 43.00        |
| <i>Panel E: Lending Outcomes</i>                             |        |              |            |            |              |              |
| Loan Volume Growth (Symmetric, in Ppt.)                      | 6,277  | 1.89         | 89.92      | -26.48     | 0.00         | 33.86        |
| Non-Investment Grade Loan Volume Growth (Symmetric, in Ppt.) | 6,277  | 1.37         | 85.60      | 0.00       | 0.00         | 0.00         |
| Loan Volume (in \$m)                                         | 6,277  | 36,293.40    | 86,011.42  | 0.00       | 695.00       | 21,305.48    |
| Non-Investment Grade Loan Volume (in \$m)                    | 6,277  | 7,340.69     | 21,812.33  | 0.00       | 0.00         | 1,788.00     |

*Notes:* This table reports summary statistics for borrower-firm (Panel A) and lender-firm (Panel B) control variables in the loan-level sample. Variables are taken from the Compustat Fundamentals Annual North America and Global databases. Total assets, total liabilities, capital expenditures, earnings before interest and taxes (EBIT), net income, current liabilities, long-term liabilities, and stockholders' equity are measured in millions of USD. Employee counts are measured in thousands. Book leverage is the ratio of short- and long-term debt to total assets, winsorized at the 1st and 99th percentiles and transformed into percentage points. Panel C reports summary statistics for (Conditional) Value at Risk, obtained from Adrian and Brunnermeier (2016). The data are quarterly and span 1994 to 2009. All variables are symmetric growth rates transformed into percentage points. Panel D reports summary statistics for compensation data from BoardEx, showing median salaries, bonuses, and equity at risk for senior non-executive bankers at the bank-year level. Growth is constructed using symmetric rates. The BoardEx data span 1997 to 2020. Panel E reports summary statistics for lending outcomes at the lender-by-year-quarter level in the estimation sample of Table 1. *Loan Volume* is the total volume of loans arranged in a given quarter, measured in millions of USD. The corresponding growth variable is the symmetric midpoint growth rate,  $2(x_t - x_{t-1})/(x_t + x_{t-1})$ , transformed into percentage points.

Table A7: Summary Statistics

| Variable                                  | N     | Mean   | SD | P25  | P50   | P75 |
|-------------------------------------------|-------|--------|----|------|-------|-----|
| Value At Risk (95 Percentile)             | 10635 | -0.079 | 23 | -12  | -0.85 | 11  |
| Conditional Value at Risk (95 Percentile) | 10635 | -0.067 | 21 | -10  | -0.8  | 9.6 |
| Value At Risk (99 Percentile)             | 10635 | -0.046 | 21 | -11  | -0.63 | 10  |
| Conditional Value at Risk (99 Percentile) | 10635 | -0.037 | 21 | -9.9 | -0.45 | 9.3 |

Notes: The table shows summary statistics of (Conditional) Value at Risk, obtained from Adrian and Brunnermeier (2016). The data are at quarterly frequency and span the period between 1994 and 2009. All variables are symmetric growth rates and are transformed into percentage points.

Table A8: Summary Statistics: Outside Options Measure

| Var                                          | N     | Mean   | SD     | P25   | P50   | P75    |
|----------------------------------------------|-------|--------|--------|-------|-------|--------|
| Nr. of Employees                             | 19378 | 316.77 | 750.68 | 19.00 | 82.00 | 258.00 |
| Nr. of Connected Bankers                     | 19378 | 40.34  | 105.46 | 2.00  | 8.00  | 29.00  |
| Nr. of Bank-Level Ties                       | 19378 | 23.58  | 45.27  | 2.00  | 7.00  | 23.00  |
| Nr. of Bank-Level Ties (Size-corrected)      | 19378 | 13.12  | 15.68  | 2.00  | 6.40  | 18.28  |
| Nr. of Bank-Level Ties With Positive Inflows | 19378 | 11.54  | 20.49  | 1.00  | 3.00  | 13.00  |
| Average Job-to-Job Hiring at Ties            | 19378 | 6.67   | 9.26   | 1.56  | 4.40  | 8.12   |
| Inflows At Ties                              | 19378 | 6.55   | 9.28   | 1.00  | 4.00  | 8.83   |

Notes: This table shows summary statistics related to the outside options measure *InflowsAtTies* by lender at the quarterly level. *Number of Employees* refers to the number of sampled individuals at the bank. *Number of Connected Bankers* is the number of individuals working in other institutions with active ties. *Number of Bank-Level Ties* is the number of distinct banks with active ties, reported equally weighted, size-corrected using the inverse Herfindahl index of Borusyak et al. (2025), and restricted to ties with positive inflows. *Average Job-to-Job Hiring at Ties* is the mean inflow at tied banks, conditional on hired individuals transitioning between jobs with a gap of one month or less. *Inflows at Ties* is the outside options proxy prior to normalization.

Table A9: Effect of Outside Options on Pay Growth (in Percentage Points)

|                            | Salary              |                     | Bonus             |                   | Equity At Risk   |                  |
|----------------------------|---------------------|---------------------|-------------------|-------------------|------------------|------------------|
|                            | (1)                 | (2)                 | (3)               | (4)               | (5)              | (6)              |
| IAT                        | 0.081***<br>(0.029) | 0.078***<br>(0.029) | -0.177<br>(0.123) | -0.182<br>(0.123) | 0.082<br>(0.094) | 0.082<br>(0.096) |
| Lender fixed effects       | X                   | X                   | X                 | X                 | X                | X                |
| Year-Quarter fixed effects | X                   | X                   | X                 | X                 | X                | X                |
| Lender Controls            |                     | X                   |                   | X                 |                  | X                |
| Observations               | 8,462               | 8,462               | 8,462             | 8,462             | 8,462            | 8,462            |

Notes: The table shows regressions of symmetric growth rates of median firm-level compensation on *InflowsAtTies<sub>it</sub>*, lender controls, and fixed effects at the lender and year-quarter level. The dependent variables are salary (Columns 1 and 2), bonus payments (Columns 3 and 4), and equity at risk (Columns 5 and 6). Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A10: Validation: Outside Options and Individual Worker Mobility

|                            | Job-to-Job (1y)<br>(1) | Job-to-Job (2y)<br>(2) | Out of Finance (2y)<br>(3) |
|----------------------------|------------------------|------------------------|----------------------------|
| IAT (+1 SD)                | 0.0010**<br>(0.0005)   | 0.0025**<br>(0.0011)   | -0.0001<br>(0.0006)        |
| Seniority                  | -0.0202***<br>(0.0009) | -0.0359***<br>(0.0018) | -0.0128***<br>(0.0009)     |
| Observations               | 906,741                | 861,156                | 861,156                    |
| Employer fixed effects     | X                      | X                      | X                          |
| Year-Quarter fixed effects | X                      | X                      | X                          |

Notes: The table reports individual-level linear probability models. Each observation is one senior U.S.-finance spell, anchored at the spell start quarter. Outcomes are indicators for a job-to-job move within one year and two years and a move out of the finance sector within two years.  $InflowsAtTies_{it}$  is standardized to unit variance. All columns include bank and year-quarter fixed effects, as well as seniority controls. Standard errors are two-way clustered at the bank and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A11: Validation: Outside Options and Bank-Level Covariates in Levels

|                            | IAT Lead 1Q<br>(1) | IAT Lead 2Q<br>(2) | IAT Lead 3Q<br>(3) | IAT Lead 4Q<br>(4) |
|----------------------------|--------------------|--------------------|--------------------|--------------------|
| Log Assets                 | -0.338<br>(0.337)  | -0.313<br>(0.332)  | -0.319<br>(0.336)  | -0.333<br>(0.345)  |
| Log EBIT                   | 0.005<br>(0.040)   | 0.005<br>(0.036)   | 0.012<br>(0.034)   | 0.028<br>(0.034)   |
| Log Current Liabilities    | 0.113*<br>(0.062)  | 0.072<br>(0.074)   | 0.052<br>(0.084)   | 0.049<br>(0.088)   |
| Log Long-Term Debt         | -0.120<br>(0.175)  | -0.144<br>(0.178)  | -0.133<br>(0.175)  | -0.131<br>(0.173)  |
| Log Liabilities            | 0.473<br>(0.369)   | 0.450<br>(0.362)   | 0.432<br>(0.354)   | 0.422<br>(0.349)   |
| Log Equity                 | 0.019<br>(0.077)   | 0.014<br>(0.076)   | 0.027<br>(0.076)   | 0.023<br>(0.072)   |
| Log Capital Expenditures   | -0.176<br>(0.112)  | -0.124<br>(0.127)  | -0.121<br>(0.131)  | -0.112<br>(0.140)  |
| Log Net Income             | -0.007<br>(0.029)  | 0.0007<br>(0.028)  | 0.001<br>(0.024)   | 0.003<br>(0.024)   |
| Lender fixed effects       | X                  | X                  | X                  | X                  |
| Year-Quarter fixed effects | X                  | X                  | X                  | X                  |
| Observations               | 11,964             | 11,964             | 11,964             | 11,964             |

Notes: The table reports results from regressing future values (one to four quarters ahead) of inflows at ties on current bank-level covariates, conditional on fixed effects at the lender and year-quarter level. Standard errors clustered at the lender level are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A12: Validation: Outside Options and Bank-Level Covariates in Growth Rates

|                             | IAT Lead 1Q<br>(1) | IAT Lead 2Q<br>(2) | IAT Lead 3Q<br>(3) | IAT Lead 4Q<br>(4) |
|-----------------------------|--------------------|--------------------|--------------------|--------------------|
| Asset Growth                | -0.217<br>(0.610)  | 0.206<br>(0.668)   | 0.201<br>(0.652)   | 0.180<br>(0.654)   |
| EBIT Growth                 | -0.008<br>(0.021)  | 0.002<br>(0.020)   | 0.022<br>(0.018)   | 0.038**<br>(0.018) |
| Current Liabilities Growth  | 0.111*<br>(0.057)  | 0.076*<br>(0.040)  | 0.044<br>(0.033)   | 0.036<br>(0.037)   |
| Long-Term Debt Growth       | 0.008<br>(0.226)   | 0.006<br>(0.230)   | -0.011<br>(0.215)  | -0.018<br>(0.198)  |
| Liabilities Growth          | 0.170<br>(0.527)   | -0.017<br>(0.476)  | -0.047<br>(0.448)  | 0.024<br>(0.421)   |
| Equity Growth               | 0.009<br>(0.044)   | 0.004<br>(0.043)   | 0.018<br>(0.048)   | 0.021<br>(0.043)   |
| Capital Expenditures Growth | -0.025<br>(0.155)  | 0.124<br>(0.101)   | 0.129<br>(0.112)   | 0.145<br>(0.144)   |
| Net Income Growth           | -0.016<br>(0.015)  | -0.009<br>(0.015)  | -0.008<br>(0.016)  | -0.004<br>(0.016)  |
| Lender fixed effects        | X                  | X                  | X                  | X                  |
| Year-Quarter fixed effects  | X                  | X                  | X                  | X                  |
| Observations                | 11,268             | 11,268             | 11,268             | 11,268             |

Notes: The table reports results from regressing future values (one to four quarters ahead) of inflows at ties on growth rates in current bank-level covariates, conditional on fixed effects at the lender and year-quarter level. Standard errors clustered at the lender level are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A13: Concentration and Co-Movement of Hiring Shocks

| <i>Panel A. Variance of hiring at connected banks explained by (<math>R^2</math>):</i> |       |
|----------------------------------------------------------------------------------------|-------|
| Common cycle (year-quarter)                                                            | 0.02  |
| Connected bank $j$                                                                     | 0.42  |
| Connected bank $j$ + year-quarter                                                      | 0.45  |
| Industry segment $\times$ year-quarter                                                 | 0.05  |
| Geographic region $\times$ year-quarter                                                | 0.06  |
| <i>Panel B. Intra-class correlation, net of the common cycle</i>                       |       |
| Connected bank $j$                                                                     | 0.43  |
| Industry segment                                                                       | 0.02  |
| Geographic region                                                                      | 0.02  |
| Industry segment $\times$ year-quarter                                                 | 0.03  |
| Geographic region $\times$ year-quarter                                                | 0.04  |
| <i>Panel C. Dispersion of identifying variation</i>                                    |       |
| Effective number of shocks ( $1/\sum_n s_n^2$ )                                        | 252   |
| Largest exposure share ( $\max_n s_n$ )                                                | 0.027 |

*Notes:* The shock is job-to-job hiring at connected bank  $j$  in quarter  $t$  (inverse hyperbolic sine). Panel A reports the  $R^2$  from regressing the shock on the indicated fixed effects. Panel B reports intra-class correlations after removing year-quarter means, i.e. the share of residual shock variance that lies *between* clusters; the industry segment is the connected bank's three-digit NAICS sector and the region is its metropolitan area. Panel C reports the inverse Herfindahl of shock-level average exposure aggregated to the connected bank, and the single largest such exposure share, over the estimation sample (gvkey-linked lenders,  $t \geq 17$ ).

Table A14: Effect of Outside Options on Lending Volumes, Robustness

|                                                                              | Total              |                  |                    | Non-Inv. Grade     |                    |                    | Inv. Grade        |                  |                    |
|------------------------------------------------------------------------------|--------------------|------------------|--------------------|--------------------|--------------------|--------------------|-------------------|------------------|--------------------|
|                                                                              | (1)                | (2)              | (3)                | (4)                | (5)                | (6)                | (7)               | (8)              | (9)                |
| <i>Panel A. Origination-only lending</i>                                     |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | 0.528<br>(0.72)    | 1.66<br>(1.32)   | 1.5<br>(1.08)      | 1.89***<br>(0.449) | 3.57***<br>(0.963) | 2.06***<br>(0.772) | 0.388<br>(0.713)  | 1.75<br>(1.33)   | 1.55<br>(0.974)    |
| Observations                                                                 | 25,378             | 25,199           | 6,296              | 25,378             | 25,199             | 6,296              | 25,378            | 25,199           | 6,296              |
| <i>Panel B. Per-lead-share allocation</i>                                    |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | -0.0447<br>(0.576) | 0.0735<br>(1.09) | 0.505<br>(0.79)    | 1.35***<br>(0.474) | 2.5**<br>(1.14)    | 1.1<br>(0.912)     | -0.107<br>(0.629) | 0.452<br>(1.19)  | 0.571<br>(0.89)    |
| Observations                                                                 | 25,378             | 25,199           | 6,296              | 25,378             | 25,199             | 6,296              | 25,378            | 25,199           | 6,296              |
| <i>Panel C. Excluding the 2007–2008 financial crisis</i>                     |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | 0.279<br>(0.75)    | 1.1<br>(1.22)    | 1.72*<br>(0.932)   | 1.55***<br>(0.416) | 2.45**<br>(1.04)   | 1.74**<br>(0.8)    | 0.311<br>(0.807)  | 1.55<br>(1.33)   | 1.75<br>(1.06)     |
| Observations                                                                 | 23,315             | 23,147           | 5,671              | 23,315             | 23,147             | 5,671              | 23,315            | 23,147           | 5,671              |
| <i>Panel D. Controlling for connected-bank lending at ties (all columns)</i> |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | 0.823<br>(0.805)   | 1.28<br>(1.36)   | 2.24***<br>(0.713) | 1.76***<br>(0.48)  | 2.71**<br>(1.05)   | 1.88**<br>(0.838)  | 0.78<br>(0.828)   | 1.54<br>(1.41)   | 2.17***<br>(0.813) |
| Lending at ties, level                                                       | 1.15<br>(2.51)     | 0.389<br>(5.3)   | 0.0305<br>(6.89)   | -0.845<br>(1.44)   | -2.94<br>(4.32)    | 5.39<br>(7.04)     | 2.44<br>(2.46)    | 4.03<br>(5.11)   | -0.515<br>(7.03)   |
| Lending at ties, growth                                                      | 2.52***<br>(0.927) | 3.26**<br>(1.25) | 0.0941<br>(2.17)   | 0.68<br>(0.609)    | 1.32<br>(0.819)    | -1.77<br>(2.11)    | 2.5***<br>(0.908) | 2.78**<br>(1.23) | 0.8<br>(1.9)       |
| Observations                                                                 | 25,378             | 25,199           | 6,296              | 25,378             | 25,199             | 6,296              | 25,378            | 25,199           | 6,296              |
| <i>Panel E. Adding Inflows at Co-Signatories (IAC)</i>                       |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | 0.616<br>(1.4)     | 1.17<br>(1.16)   | 0.763<br>(1.49)    | 1.55**<br>(0.709)  | 3.18***<br>(1.1)   | 2.09**<br>(0.839)  | 0.967<br>(1.51)   | 1.24<br>(1.37)   | 0.791<br>(1.53)    |
| IAC                                                                          | 0.434<br>(1.67)    | 1.61<br>(2.7)    | 0.963<br>(1.86)    | 2.18<br>(2.65)     | 4.99<br>(3.88)     | 5.4**<br>(2.24)    | 0.99<br>(1.47)    | 1.61<br>(2.45)   | 0.357<br>(1.85)    |
| Observations                                                                 | 4,014              | 3,645            | 2,331              | 4,014              | 3,645              | 2,331              | 4,014             | 3,645            | 2,331              |
| <i>Panel F. Adding inflows to lender i</i>                                   |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | 0.394<br>(0.693)   | 1.29<br>(1.22)   | 1.67**<br>(0.815)  | 1.58***<br>(0.401) | 2.86***<br>(1)     | 1.7**<br>(0.725)   | 0.362<br>(0.743)  | 1.6<br>(1.32)    | 1.68*<br>(0.931)   |
| Inflows to lender <i>i</i>                                                   | 0.0172<br>(0.978)  | -2.07<br>(2.23)  | -1.68<br>(2)       | -0.573<br>(0.893)  | -1.75<br>(1.92)    | -1.73<br>(2.2)     | -0.295<br>(0.877) | -2.05<br>(2.18)  | -1.38<br>(1.79)    |
| Observations                                                                 | 25,378             | 25,199           | 6,296              | 25,378             | 25,199             | 6,296              | 25,378            | 25,199           | 6,296              |
| <i>Panel G. Adding outflows from lender i</i>                                |                    |                  |                    |                    |                    |                    |                   |                  |                    |
| IAT                                                                          | 0.415<br>(0.696)   | 1.22<br>(1.22)   | 1.78*<br>(0.939)   | 1.59***<br>(0.391) | 2.77***<br>(0.973) | 1.82**<br>(0.724)  | 0.354<br>(0.742)  | 1.54<br>(1.31)   | 1.78*<br>(0.99)    |
| Outflows from lender <i>i</i>                                                | -0.631<br>(0.888)  | -1.25<br>(1.94)  | -3.41<br>(2.45)    | -1.17<br>(0.84)    | -2.75<br>(1.72)    | -3.85**<br>(1.57)  | -0.309<br>(0.871) | 0.311<br>(1.99)  | -3.17<br>(2.42)    |
| Observations                                                                 | 25,378             | 25,199           | 6,296              | 25,378             | 25,199             | 6,296              | 25,378            | 25,199           | 6,296              |
| Lender FE                                                                    | X                  |                  | X                  | X                  |                    | X                  | X                 |                  | X                  |
| Lender × Year FE                                                             |                    | X                |                    |                    | X                  |                    |                   | X                |                    |
| Year-Quarter FE                                                              | X                  | X                | X                  | X                  | X                  | X                  | X                 | X                | X                  |
| Lender controls                                                              |                    |                  | X                  |                    |                    | X                  |                   |                  | X                  |

Notes: Each panel re-estimates the lending-volume regression of Table 1 under a different robustness modification, keeping the same nine columns. The dependent variable is the symmetric midpoint growth rate of lending volume in percentage points, decomposed into total, non-investment-grade, and investment-grade lending. Within each grade block, the first column includes lender and year-quarter fixed effects, the second replaces lender fixed effects with lender-by-year fixed effects, and the third adds time-varying lender controls. Panel A uses origination volume. Panel B allocates each facility amount equally across its lead arrangers. Panel C drops 2007 and 2008. Panel D adds exposure-weighted controls for the level and annual growth of total and leveraged lending at connected banks. It displays the coefficients on total lending at ties and includes the remaining leveraged-lending controls without displaying their coefficients. Panel E adds Inflows at Co-Syndicators, *IAC*, and is restricted to lender-quarters with co-lending ties. Panels F and G add the focal lender's employee inflows and outflows separately. *InflowsAtTies<sub>it</sub>* and *IAC* are standardized to unit variance. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A15: Effect of Outside Options on Lending Volumes, Tie Length Robustness

|                                                        | Growth Rate       |                  |                   | Growth Rate (non-prime) |                    |                    | Growth Rate (prime) |                  |                   |
|--------------------------------------------------------|-------------------|------------------|-------------------|-------------------------|--------------------|--------------------|---------------------|------------------|-------------------|
|                                                        | (1)               | (2)              | (3)               | (4)                     | (5)                | (6)                | (7)                 | (8)              | (9)               |
| <i>Panel A. Baseline: ties active 4 to 12 quarters</i> |                   |                  |                   |                         |                    |                    |                     |                  |                   |
| IAT                                                    | 0.395<br>(0.690)  | 1.23<br>(1.22)   | 1.62**<br>(0.807) | 1.55***<br>(0.387)      | 2.81***<br>(0.997) | 1.65**<br>(0.728)  | 0.344<br>(0.736)    | 1.54<br>(1.31)   | 1.64*<br>(0.919)  |
| Observations                                           | 25,378            | 25,199           | 6,296             | 25,378                  | 25,199             | 6,296              | 25,378              | 25,199           | 6,296             |
| <i>Panel B. Ties active 2 to 12 quarters</i>           |                   |                  |                   |                         |                    |                    |                     |                  |                   |
| IAT                                                    | 0.354<br>(0.750)  | 0.659<br>(1.47)  | 1.47<br>(0.946)   | 1.47***<br>(0.349)      | 2.75***<br>(0.895) | 1.99***<br>(0.694) | 0.101<br>(0.751)    | 0.736<br>(1.48)  | 1.26<br>(1.13)    |
| Observations                                           | 25,378            | 25,199           | 6,296             | 25,378                  | 25,199             | 6,296              | 25,378              | 25,199           | 6,296             |
| <i>Panel C. Ties active 4 to 20 quarters</i>           |                   |                  |                   |                         |                    |                    |                     |                  |                   |
| IAT                                                    | 1.23**<br>(0.598) | 2.34**<br>(1.09) | 2.03**<br>(0.774) | 1.23***<br>(0.346)      | 2.36***<br>(0.848) | 1.94***<br>(0.449) | 1.21*<br>(0.631)    | 2.67**<br>(1.16) | 2.01**<br>(0.897) |
| Observations                                           | 25,378            | 25,199           | 6,296             | 25,378                  | 25,199             | 6,296              | 25,378              | 25,199           | 6,296             |
| Lender fixed effects                                   | X                 |                  | X                 | X                       |                    | X                  | X                   |                  | X                 |
| Year-Quarter fixed effects                             | X                 | X                | X                 | X                       | X                  | X                  | X                   | X                | X                 |
| Lender-year fixed effects                              |                   | X                |                   |                         | X                  |                    |                     | X                |                   |
| Lender Controls                                        |                   |                  | X                 |                         |                    | X                  |                     |                  | X                 |

*Notes:* The table reports results from regressing the outcome of interest on standardized  $InflowsAtTies_{it}$  and fixed effects at the lender and year-quarter level, re-estimated under alternative definitions of how long a coworker tie remains active. The dependent variable is the symmetric (midpoint) growth rate,  $2(x_t - x_{t-1})/(x_t + x_{t-1})$ , transformed into percentage points, of a bank's quarterly lead-arranged lending volume. Columns (1)–(3) use total lending volume, Columns (4)–(6) its non-investment-grade component, and Columns (7)–(9) its investment-grade component. Within each block, the first column includes lender and year-quarter fixed effects, the second replaces the lender fixed effects with saturated lender-by-year fixed effects, and the third adds time-varying lender controls (the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity). Each panel recomputes  $InflowsAtTies_{it}$  under a different tie window: *Panel A* is the baseline (ties active 4 to 12 quarters after a coworker transition, reproducing Table 1), *Panel B* uses 2 to 12 quarters, and *Panel C* uses 4 to 20 quarters. The data span 1994 to 2020. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A16: Outside Options by Seniority of Job-to-Job Hires, Lending Volumes

|                                       | Growth Rate      |                 |                   | Growth Rate (non-prime) |                    |                   | Growth Rate (prime) |                 |                   |
|---------------------------------------|------------------|-----------------|-------------------|-------------------------|--------------------|-------------------|---------------------|-----------------|-------------------|
|                                       | (1)              | (2)             | (3)               | (4)                     | (5)                | (6)               | (7)                 | (8)             | (9)               |
| <i>Panel A. Original IAT</i>          |                  |                 |                   |                         |                    |                   |                     |                 |                   |
| IAT                                   | 0.395<br>(0.690) | 1.23<br>(1.22)  | 1.62**<br>(0.807) | 1.55***<br>(0.387)      | 2.81***<br>(0.997) | 1.65**<br>(0.728) | 0.344<br>(0.736)    | 1.54<br>(1.31)  | 1.64*<br>(0.919)  |
| Observations                          | 25,378           | 25,199          | 6,296             | 25,378                  | 25,199             | 6,296             | 25,378              | 25,199          | 6,296             |
| <i>Panel B. OO – medium seniority</i> |                  |                 |                   |                         |                    |                   |                     |                 |                   |
| IAT                                   | 0.416<br>(0.711) | 1.28<br>(1.17)  | 1.52*<br>(0.791)  | 1.54***<br>(0.399)      | 2.71***<br>(1.02)  | 1.47**<br>(0.675) | 0.366<br>(0.762)    | 1.59<br>(1.27)  | 1.53*<br>(0.911)  |
| Observations                          | 25,378           | 25,199          | 6,296             | 25,378                  | 25,199             | 6,296             | 25,378              | 25,199          | 6,296             |
| <i>Panel C. OO – high seniority</i>   |                  |                 |                   |                         |                    |                   |                     |                 |                   |
| IAT                                   | 0.209<br>(0.599) | 0.639<br>(1.32) | 1.83**<br>(0.888) | 1.30***<br>(0.424)      | 2.48***<br>(0.921) | 2.34**<br>(1.00)  | 0.165<br>(0.609)    | 0.831<br>(1.35) | 1.92**<br>(0.937) |
| Observations                          | 25,378           | 25,199          | 6,296             | 25,378                  | 25,199             | 6,296             | 25,378              | 25,199          | 6,296             |
| Lender fixed effects                  | X                |                 | X                 | X                       |                    | X                 | X                   |                 | X                 |
| Year-Quarter fixed effects            | X                | X               | X                 | X                       | X                  | X                 | X                   | X               | X                 |
| Lender × Year fixed effects           |                  | X               |                   |                         | X                  |                   |                     | X               |                   |
| Lender Controls                       |                  |                 | X                 |                         |                    | X                 |                     |                 | X                 |

*Notes:* The table reports regressions of the symmetric, midpoint growth rate of a bank's quarterly lead-arranged lending volume, transformed into percentage points, on standardized IAT. Each facility is credited to its lead arranger exactly once at its full amount. Columns (1)–(3) use total lending volume, Columns (4)–(6) use its non-investment-grade component, and Columns (7)–(9) use its investment-grade component. Panel A uses the original IAT. Panel B recomputes IAT from job-to-job hires with seniority levels 4–5, and Panel C recomputes IAT from job-to-job hires with seniority levels 6–7. Within each panel, IAT is standardized in the estimation sample. Within each outcome block, the first column includes lender and year-quarter fixed effects, the second replaces lender fixed effects with saturated lender-by-year fixed effects, and the third adds time-varying lender controls. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. The data span 1994 to 2020. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A17: Effect of Outside Options on Changes in Financial Risk, Robustness

|                                                            | VaR 95%           |                     |                    | CoVaR 95%           |                      |                     | VaR 99%             |                     |                     | CoVaR 99%           |                     |                     |
|------------------------------------------------------------|-------------------|---------------------|--------------------|---------------------|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                                            | (1)               | (2)                 | (3)                | (4)                 | (5)                  | (6)                 | (7)                 | (8)                 | (9)                 | (10)                | (11)                | (12)                |
| <i>Panel A. Baseline</i>                                   |                   |                     |                    |                     |                      |                     |                     |                     |                     |                     |                     |                     |
| IAT                                                        | 0.167<br>(0.194)  | 0.479*<br>(0.263)   | 0.226<br>(0.253)   | 0.173<br>(0.150)    | 0.439**<br>(0.215)   | 0.179<br>(0.192)    | 0.314**<br>(0.145)  | 0.783***<br>(0.235) | 0.422***<br>(0.160) | 0.314***<br>(0.115) | 0.741***<br>(0.211) | 0.385***<br>(0.128) |
| Observations                                               | 15,823            | 15,755              | 11,905             | 15,823              | 15,755               | 11,905              | 15,823              | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel B. Adding inflows to lender <math>i</math></i>    |                   |                     |                    |                     |                      |                     |                     |                     |                     |                     |                     |                     |
| IAT                                                        | 0.201<br>(0.192)  | 0.509*<br>(0.265)   | 0.260<br>(0.251)   | 0.233<br>(0.149)    | 0.491**<br>(0.214)   | 0.233<br>(0.191)    | 0.353**<br>(0.151)  | 0.805***<br>(0.236) | 0.472***<br>(0.164) | 0.366***<br>(0.122) | 0.775***<br>(0.211) | 0.445***<br>(0.134) |
| Inflows to lender $i$                                      | -0.213<br>(0.173) | -0.506<br>(0.339)   | -0.221<br>(0.222)  | -0.369**<br>(0.164) | -0.862***<br>(0.307) | -0.363*<br>(0.203)  | -0.240<br>(0.220)   | -0.371<br>(0.377)   | -0.329<br>(0.243)   | -0.320<br>(0.205)   | -0.580<br>(0.352)   | -0.400*<br>(0.224)  |
| Observations                                               | 15,823            | 15,755              | 11,905             | 15,823              | 15,755               | 11,905              | 15,823              | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel C. Adding outflows from lender <math>i</math></i> |                   |                     |                    |                     |                      |                     |                     |                     |                     |                     |                     |                     |
| IAT                                                        | 0.202<br>(0.195)  | 0.463*<br>(0.259)   | 0.259<br>(0.253)   | 0.222<br>(0.150)    | 0.414**<br>(0.208)   | 0.219<br>(0.192)    | 0.354**<br>(0.155)  | 0.772***<br>(0.229) | 0.456***<br>(0.169) | 0.361***<br>(0.125) | 0.723***<br>(0.204) | 0.422***<br>(0.136) |
| Outflows                                                   | -0.360<br>(0.229) | -0.608<br>(0.376)   | -0.470*<br>(0.272) | -0.501**<br>(0.230) | -1.001***<br>(0.338) | -0.590**<br>(0.268) | -0.411<br>(0.274)   | -0.442<br>(0.433)   | -0.490<br>(0.305)   | -0.487*<br>(0.272)  | -0.691*<br>(0.403)  | -0.547*<br>(0.306)  |
| Observations                                               | 15,823            | 15,755              | 11,905             | 15,823              | 15,755               | 11,905              | 15,823              | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel D. Adding IAC</i>                                 |                   |                     |                    |                     |                      |                     |                     |                     |                     |                     |                     |                     |
| IAT                                                        | 0.158<br>(0.191)  | 0.443*<br>(0.255)   | 0.212<br>(0.251)   | 0.161<br>(0.147)    | 0.403*<br>(0.209)    | 0.160<br>(0.190)    | 0.304**<br>(0.142)  | 0.748***<br>(0.228) | 0.407**<br>(0.155)  | 0.301***<br>(0.112) | 0.702***<br>(0.207) | 0.366***<br>(0.124) |
| IAC                                                        | 0.085<br>(0.080)  | 0.212*<br>(0.112)   | 0.149*<br>(0.088)  | 0.122**<br>(0.059)  | 0.215**<br>(0.090)   | 0.184**<br>(0.077)  | 0.099<br>(0.116)    | 0.214<br>(0.150)    | 0.154<br>(0.147)    | 0.127<br>(0.101)    | 0.229*<br>(0.135)   | 0.184<br>(0.135)    |
| Observations                                               | 15,823            | 15,755              | 11,905             | 15,823              | 15,755               | 11,905              | 15,823              | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel E. Excluding 2007–2008</i>                        |                   |                     |                    |                     |                      |                     |                     |                     |                     |                     |                     |                     |
| IAT                                                        | 0.288*<br>(0.153) | 0.700***<br>(0.217) | 0.436**<br>(0.181) | 0.232*<br>(0.119)   | 0.567***<br>(0.180)  | 0.328**<br>(0.143)  | 0.443***<br>(0.151) | 0.933***<br>(0.241) | 0.573***<br>(0.166) | 0.404***<br>(0.130) | 0.831***<br>(0.225) | 0.502***<br>(0.140) |
| Observations                                               | 14,092            | 13,815              | 10,396             | 14,092              | 13,815               | 10,396              | 14,092              | 13,815              | 10,396              | 14,092              | 13,815              | 10,396              |
| <i>Panel F. Winsorized IAT</i>                             |                   |                     |                    |                     |                      |                     |                     |                     |                     |                     |                     |                     |
| IAT                                                        | 0.215<br>(0.264)  | 0.681*<br>(0.379)   | 0.358<br>(0.326)   | 0.245<br>(0.216)    | 0.666**<br>(0.327)   | 0.315<br>(0.257)    | 0.410*<br>(0.216)   | 1.098***<br>(0.381) | 0.623***<br>(0.220) | 0.420**<br>(0.183)  | 1.055***<br>(0.349) | 0.580***<br>(0.182) |
| Observations                                               | 15,823            | 15,755              | 11,905             | 15,823              | 15,755               | 11,905              | 15,823              | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| Lender fixed effects                                       | X                 |                     | X                  | X                   |                      | X                   | X                   |                     | X                   | X                   |                     | X                   |
| Year-Quarter fixed effects                                 | X                 | X                   | X                  | X                   | X                    | X                   | X                   | X                   | X                   | X                   | X                   | X                   |
| Lender×Year fixed effects                                  |                   | X                   |                    |                     | X                    |                     |                     | X                   |                     |                     | X                   |                     |
| Lender Controls                                            |                   |                     | X                  |                     |                      | X                   |                     |                     | X                   |                     |                     | X                   |

Notes: The table reports robustness checks for the financial-risk regressions in Table 2. Outcomes are symmetric growth rates in VaR and CoVaR, transformed into percentage points. Columns (1)–(3) use VaR at the 95% confidence level, Columns (4)–(6) use CoVaR at the 95% confidence level, Columns (7)–(9) use VaR at the 99% confidence level, and Columns (10)–(12) use CoVaR at the 99% confidence level. Within each outcome block, specifications follow Table 2. Panel A reproduces the baseline. Panels B and C add the focal lender's employee inflows and outflows separately. Panel D adds hiring through the co-syndication network,  $InflowsAtCosignatories_{it}$ . Panel E excludes 2007 and 2008. Panel F winsorizes  $InflowsAtTies_{it}$  at the 1st and 99th percentiles. The data are quarterly and span 1994 to 2009. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A18: Effect of Outside Options on Changes in Financial Risk, Tie Length Robustness

|                                                        | VaR 95%          |                   |                   | CoVaR 95%          |                    |                   | VaR 99%            |                     |                     | CoVaR 99%           |                     |                     |
|--------------------------------------------------------|------------------|-------------------|-------------------|--------------------|--------------------|-------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                                        | (1)              | (2)               | (3)               | (4)                | (5)                | (6)               | (7)                | (8)                 | (9)                 | (10)                | (11)                | (12)                |
| <i>Panel A. Baseline: ties active 4 to 12 quarters</i> |                  |                   |                   |                    |                    |                   |                    |                     |                     |                     |                     |                     |
| IAT                                                    | 0.167<br>(0.194) | 0.479*<br>(0.263) | 0.226<br>(0.253)  | 0.173<br>(0.150)   | 0.439**<br>(0.215) | 0.179<br>(0.192)  | 0.314**<br>(0.145) | 0.783***<br>(0.235) | 0.422***<br>(0.160) | 0.314***<br>(0.115) | 0.741***<br>(0.211) | 0.385***<br>(0.128) |
| Observations                                           | 15,823           | 15,755            | 11,905            | 15,823             | 15,755             | 11,905            | 15,823             | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel B. Ties active 2 to 12 quarters</i>           |                  |                   |                   |                    |                    |                   |                    |                     |                     |                     |                     |                     |
| IAT                                                    | 0.189<br>(0.159) | 0.418*<br>(0.233) | 0.251<br>(0.216)  | 0.190<br>(0.130)   | 0.426**<br>(0.200) | 0.208<br>(0.174)  | 0.316**<br>(0.121) | 0.664***<br>(0.232) | 0.420***<br>(0.152) | 0.312***<br>(0.104) | 0.660***<br>(0.220) | 0.385***<br>(0.130) |
| Observations                                           | 15,823           | 15,755            | 11,905            | 15,823             | 15,755             | 11,905            | 15,823             | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel C. Ties active 4 to 20 quarters</i>           |                  |                   |                   |                    |                    |                   |                    |                     |                     |                     |                     |                     |
| IAT                                                    | 0.223<br>(0.185) | 0.485<br>(0.313)  | 0.399*<br>(0.220) | 0.307**<br>(0.124) | 0.500**<br>(0.247) | 0.322*<br>(0.174) | 0.208<br>(0.204)   | 0.594*<br>(0.315)   | 0.414**<br>(0.194)  | 0.249<br>(0.151)    | 0.594**<br>(0.266)  | 0.354**<br>(0.174)  |
| Observations                                           | 15,983           | 15,915            | 12,033            | 15,983             | 15,915             | 12,033            | 15,983             | 15,915              | 12,033              | 15,983              | 15,915              | 12,033              |
| Lender fixed effects                                   | X                |                   | X                 | X                  |                    | X                 | X                  |                     | X                   | X                   |                     | X                   |
| Year-Quarter fixed effects                             | X                | X                 | X                 | X                  | X                  | X                 | X                  | X                   | X                   | X                   | X                   | X                   |
| Lender×Year fixed effects                              |                  | X                 |                   |                    | X                  |                   |                    | X                   |                     |                     | X                   |                     |
| Lender Controls                                        |                  |                   | X                 |                    |                    | X                 |                    |                     | X                   |                     |                     | X                   |

Notes: The table reports coefficients from OLS regressions of symmetric growth rates in VaR and CoVaR (in percentage points) on standardized  $InflowsAtTies_{it}$ , re-estimated under alternative definitions of how long a coworker tie remains active. The outcomes are VaR at the 95% confidence level (Columns 1–3), CoVaR at the 95% confidence level (Columns 4–6), VaR at the 99% confidence level (Columns 7–9), and CoVaR at the 99% confidence level (Columns 10–12). Within each outcome block, the first column includes lender and year-quarter fixed effects, the second replaces lender fixed effects with saturated lender-by-year fixed effects, and the third includes lender and year-quarter fixed effects plus time-varying lender controls. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. Each panel recomputes  $InflowsAtTies_{it}$  under a different tie window: *Panel A* is the baseline (ties active 4 to 12 quarters after a coworker transition, reproducing Table 2), *Panel B* uses 2 to 12 quarters, and *Panel C* uses 4 to 20 quarters. The data are at quarterly frequency and span 1994 to 2009. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A19: Outside Options by Seniority of Job-to-Job Hires, Financial Risk

|                                       | VaR 95%          |                   |                  | CoVaR 95%        |                    |                  | VaR 99%            |                     |                     | CoVaR 99%           |                     |                     |
|---------------------------------------|------------------|-------------------|------------------|------------------|--------------------|------------------|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                       | (1)              | (2)               | (3)              | (4)              | (5)                | (6)              | (7)                | (8)                 | (9)                 | (10)                | (11)                | (12)                |
| <i>Panel A. Original IAT</i>          |                  |                   |                  |                  |                    |                  |                    |                     |                     |                     |                     |                     |
| IAT                                   | 0.167<br>(0.194) | 0.479*<br>(0.263) | 0.226<br>(0.253) | 0.173<br>(0.150) | 0.439**<br>(0.215) | 0.179<br>(0.192) | 0.314**<br>(0.145) | 0.783***<br>(0.235) | 0.422***<br>(0.160) | 0.314***<br>(0.115) | 0.741***<br>(0.211) | 0.385***<br>(0.128) |
| Observations                          | 15,823           | 15,755            | 11,905           | 15,823           | 15,755             | 11,905           | 15,823             | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel B. OO – medium seniority</i> |                  |                   |                  |                  |                    |                  |                    |                     |                     |                     |                     |                     |
| IAT                                   | 0.175<br>(0.195) | 0.485*<br>(0.259) | 0.231<br>(0.252) | 0.177<br>(0.149) | 0.436**<br>(0.212) | 0.177<br>(0.191) | 0.313**<br>(0.148) | 0.767***<br>(0.237) | 0.426***<br>(0.159) | 0.309**<br>(0.118)  | 0.716***<br>(0.213) | 0.384***<br>(0.126) |
| Observations                          | 15,823           | 15,755            | 11,905           | 15,823           | 15,755             | 11,905           | 15,823             | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| <i>Panel C. OO – high seniority</i>   |                  |                   |                  |                  |                    |                  |                    |                     |                     |                     |                     |                     |
| IAT                                   | 0.110<br>(0.176) | 0.363<br>(0.240)  | 0.177<br>(0.229) | 0.135<br>(0.142) | 0.371*<br>(0.201)  | 0.165<br>(0.175) | 0.282**<br>(0.126) | 0.705***<br>(0.205) | 0.353**<br>(0.158)  | 0.298***<br>(0.104) | 0.703***<br>(0.187) | 0.342**<br>(0.138)  |
| Observations                          | 15,823           | 15,755            | 11,905           | 15,823           | 15,755             | 11,905           | 15,823             | 15,755              | 11,905              | 15,823              | 15,755              | 11,905              |
| Lender fixed effects                  | X                |                   | X                | X                |                    | X                | X                  |                     | X                   | X                   |                     | X                   |
| Year-Quarter fixed effects            | X                | X                 | X                | X                | X                  | X                | X                  | X                   | X                   | X                   | X                   | X                   |
| Lender × Year fixed effects           |                  | X                 |                  |                  | X                  |                  |                    | X                   |                     |                     | X                   |                     |
| Lender Controls                       |                  |                   | X                |                  |                    | X                |                    |                     | X                   |                     |                     | X                   |

*Notes:* The table reports coefficients from OLS regressions of symmetric growth rates in VaR and CoVaR, transformed into percentage points, on standardized IAT. The outcomes are VaR at the 95% confidence level in Columns (1)–(3), CoVaR at the 95% confidence level in Columns (4)–(6), VaR at the 99% confidence level in Columns (7)–(9), and CoVaR at the 99% confidence level in Columns (10)–(12). Panel A uses the original IAT. Panel B recomputes IAT from job-to-job hires with seniority levels 4–5, and Panel C recomputes IAT from job-to-job hires with seniority levels 6–7. Within each panel, IAT is standardized in the estimation sample. Within each outcome block, the first column includes lender and year-quarter fixed effects, the second replaces lender fixed effects with saturated lender-by-year fixed effects, and the third includes lender and year-quarter fixed effects plus time-varying lender controls. Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, and equity. The data are at quarterly frequency and span 1994 to 2009. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A20: Borrower Risk (Median FISD Ratings)

|                                                                    | Borrower Risk (Median FISD Ratings) |                   |                   |                    |
|--------------------------------------------------------------------|-------------------------------------|-------------------|-------------------|--------------------|
|                                                                    | (1)                                 | (2)               | (3)               | (4)                |
| <i>Panel A. Below-Investment-Grade Borrower, percentage points</i> |                                     |                   |                   |                    |
| IAT                                                                | 1.370***<br>(0.339)                 | 0.915*<br>(0.490) | 0.886<br>(0.604)  | 0.040<br>(0.636)   |
| Observations                                                       | 30,144                              | 30,091            | 30,022            | 28,078             |
| <i>Panel B. Continuous Borrower-Risk Score, 0–100</i>              |                                     |                   |                   |                    |
| IAT                                                                | 0.481***<br>(0.161)                 | 0.334*<br>(0.166) | 0.321*<br>(0.182) | 0.389**<br>(0.182) |
| Observations                                                       | 30,144                              | 30,091            | 30,022            | 28,078             |
| Lender fixed effects                                               | X                                   |                   |                   |                    |
| YQ fixed effects                                                   | X                                   | X                 |                   |                    |
| Lender × Year fixed effects                                        |                                     | X                 | X                 | X                  |
| Borrower Sector × YQ fixed effects                                 |                                     |                   | X                 | X                  |
| Borrower State × YQ fixed effects                                  |                                     |                   |                   | X                  |
| Borrower+Lender+Loan Controls                                      | X                                   | X                 | X                 | X                  |

*Notes:* The table re-estimates the four borrower-risk specifications of Table 3, measuring borrower quality at origination from Mergent FISD bond ratings. For each loan and each bond-agency pair, we retain the most recent Moody's, S&P, or Fitch rating in effect on or before origination, exclude withdrawn and unrated bonds, map ratings to a 22-point scale from D= 1 to AAA/Aaa= 22, and take the median across active bond-agency pairs. Panel A reports a linear probability model in which the outcome equals one when the median rating is below BBB–/Baa3, expressed in percentage points. Panel B maps the median rating to a continuous 0–100 risk score, where AAA/Aaa equals zero and D equals 100. A positive coefficient indicates a shift toward riskier borrowers in both panels. All columns include the full set of borrower, lender, and loan controls. Column (1) includes lender and year-quarter fixed effects, Column (2) replaces lender fixed effects with lender-by-year fixed effects, Column (3) adds borrower-sector-by-year-quarter fixed effects, and Column (4) adds borrower-state-by-year-quarter fixed effects.  $InflowsAtTies_{it}$  is standardized to unit variance. Standard errors are two-way clustered at the lender and year-quarter levels. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A21: Outside Options by Seniority of Job-to-Job Hires, Borrower Risk

|                                       | Non-Prime Borrower (ex ante, pp) |                     |                     |                     |
|---------------------------------------|----------------------------------|---------------------|---------------------|---------------------|
|                                       | (1)                              | (2)                 | (3)                 | (4)                 |
| <i>Panel A. Original IAT</i>          |                                  |                     |                     |                     |
| IAT                                   | 0.475***<br>(0.147)              | 0.583***<br>(0.178) | 0.622***<br>(0.176) | 1.258***<br>(0.382) |
| Observations                          | 55,327                           | 55,278              | 55,223              | 50,045              |
| <i>Panel B. OO – medium seniority</i> |                                  |                     |                     |                     |
| IAT                                   | 0.439***<br>(0.140)              | 0.543***<br>(0.166) | 0.593***<br>(0.169) | 1.17***<br>(0.349)  |
| Observations                          | 55,327                           | 55,278              | 55,223              | 50,045              |
| <i>Panel C. OO – high seniority</i>   |                                  |                     |                     |                     |
| IAT                                   | 0.783**<br>(0.297)               | 0.931**<br>(0.375)  | 0.786*<br>(0.417)   | 1.77***<br>(0.634)  |
| Observations                          | 55,327                           | 55,278              | 55,223              | 50,045              |
| Lender fixed effects                  | X                                |                     |                     |                     |
| Year-Quarter fixed effects            | X                                | X                   |                     |                     |
| Lender × Year fixed effects           |                                  | X                   | X                   | X                   |
| Borrower Sector × YQ fixed effects    |                                  |                     | X                   | X                   |
| Borrower State × YQ fixed effects     |                                  |                     |                     | X                   |
| Borrower+Lender+Loan Controls         | X                                | X                   | X                   | X                   |

*Notes:* The table reports loan-level linear probability regressions in which the dependent variable is an indicator equal to one if the borrower is rated below investment grade by S&P, so a positive coefficient indicates a shift toward riskier borrowers. Panel A uses the original IAT. Panel B recomputes IAT from job-to-job hires with seniority levels 4–5, and Panel C recomputes IAT from job-to-job hires with seniority levels 6–7. IAT is standardized within each panel's estimation sample. All columns include the full set of borrower, lender, and loan controls. Column (1) includes additive lender and year-quarter fixed effects. Column (2) replaces the lender fixed effect with lender-by-year fixed effects. Column (3) adds borrower-sector-by-year-quarter fixed effects, and Column (4) further adds borrower-state-by-year-quarter fixed effects. Borrower and lender controls are the inverse hyperbolic sine of total assets, number of employees, capital expenditure, total liabilities, EBIT, and, for borrowers, fixed assets. Loan controls are the inverse hyperbolic sine of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. The data span 1994 to 2020. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A22: Effect of Outside Options on Borrower Risk, Robustness

| Robustness                                | Additional Control    |                        |                                        |                    | Sample / Variable  |                   |                  |
|-------------------------------------------|-----------------------|------------------------|----------------------------------------|--------------------|--------------------|-------------------|------------------|
|                                           | Inflows at lender $i$ | Outflows at lender $i$ | Nr. Lenders<br>Non-Prime Borrower (pp) | IAC                | Origination        | Excl. GFC         | Winsor.          |
|                                           | (1)                   | (2)                    | (3)                                    | (4)                | (5)                | (6)               | (7)              |
| IAT                                       | 1.25***<br>(0.389)    | 1.25***<br>(0.381)     | 1.26***<br>(0.407)                     | 1.47***<br>(0.366) | 1.69***<br>(0.412) | 4.01***<br>(1.48) | 3.15**<br>(1.36) |
| Inflows at lender $i$                     | 0.003<br>(0.006)      |                        |                                        |                    |                    |                   |                  |
| Outflows at lender $i$                    |                       | -0.003<br>(0.003)      |                                        |                    |                    |                   |                  |
| Number of Lenders                         |                       |                        | -0.739<br>(0.671)                      |                    |                    |                   |                  |
| IAC                                       |                       |                        |                                        | 0.321<br>(0.344)   |                    |                   |                  |
| Lender $\times$ Year fixed effects        | X                     | X                      | X                                      | X                  | X                  | X                 | X                |
| Borrower Sector $\times$ YQ fixed effects | X                     | X                      | X                                      | X                  | X                  | X                 | X                |
| Borrower State $\times$ YQ fixed effects  | X                     | X                      | X                                      | X                  | X                  | X                 | X                |
| Borrower+Lender+Loan Controls             | X                     | X                      | X                                      | X                  | X                  | X                 | X                |
| Observations                              | 50,045                | 50,045                 | 50,045                                 | 45,542             | 21,625             | 46,730            | 50,045           |

Notes: The table reports robustness exercises for the borrower-risk result in Table 3, obtained by estimating Equation (16) at the loan level. The dependent variable in all columns is an indicator for a below-investment-grade (non-prime) borrower multiplied by 100, so coefficients are in percentage points and a positive coefficient indicates a shift toward riskier borrowers. All columns are built on the fully saturated specification in Column (4) of Table 3, namely lender-by-year, borrower-sector-by-year-quarter, and borrower-state-by-year-quarter fixed effects, together with the full set of borrower, lender, and loan controls. Borrower and lender controls include the inverse hyperbolic sine transformation of total assets, number of employees, capital expenditure, total liabilities, EBIT, and, for borrowers, fixed assets. Loan controls are the inverse hyperbolic sine transformation of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. Relative to that specification, Column (1) adds total employee inflows at lender  $i$  in year-quarter  $t$ , Column (2) adds total employee outflows, Column (3) adds the inverse hyperbolic sine transformation of the number of lenders, and Column (4) adds IAC ( $InflowsAtCosignatories_{it}$ ), a variant of  $InflowsAtTies_{it}$  in which ties from coworker networks are replaced by ties resulting from co-arranging loans. Column (5) restricts the sample to newly originated loans, Column (6) excludes the 2007–2008 financial crisis, and Column (7) winsorizes  $InflowsAtTies_{it}$  at the 1st and 99th percentiles. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A23: Effect of Outside Options on Borrower Risk, Tie Length Robustness

|                                                        | Non-Prime Borrower, percentage points |                     |                     |                     |
|--------------------------------------------------------|---------------------------------------|---------------------|---------------------|---------------------|
|                                                        | (1)                                   | (2)                 | (3)                 | (4)                 |
| <i>Panel A. Baseline, ties active 4 to 12 quarters</i> |                                       |                     |                     |                     |
| IAT                                                    | 0.475***<br>(0.147)                   | 0.583***<br>(0.178) | 0.622***<br>(0.176) | 1.258***<br>(0.382) |
| Observations                                           | 55,327                                | 55,278              | 55,223              | 50,045              |
| <i>Panel B. Ties active 2 to 12 quarters</i>           |                                       |                     |                     |                     |
| IAT                                                    | 0.484***<br>(0.143)                   | 0.655***<br>(0.209) | 0.692***<br>(0.205) | 1.269***<br>(0.364) |
| Observations                                           | 55,327                                | 55,278              | 55,223              | 50,045              |
| <i>Panel C. Ties active 4 to 20 quarters</i>           |                                       |                     |                     |                     |
| IAT                                                    | 0.618***<br>(0.124)                   | 0.461**<br>(0.180)  | 0.484***<br>(0.129) | 0.943***<br>(0.327) |
| Observations                                           | 55,327                                | 55,278              | 55,223              | 50,045              |
| Lender fixed effects                                   | X                                     |                     |                     |                     |
| YQ fixed effects                                       | X                                     | X                   |                     |                     |
| Lender $\times$ Year fixed effects                     |                                       | X                   | X                   | X                   |
| Borrower Sector $\times$ YQ fixed effects              |                                       |                     | X                   | X                   |
| Borrower State $\times$ YQ fixed effects               |                                       |                     |                     | X                   |
| Borrower, lender, and loan controls                    | X                                     | X                   | X                   | X                   |

*Notes:* The table reports the loan-level borrower-risk regression of Equation (16) across a set of increasingly demanding fixed effects, re-estimated under alternative definitions of how long a coworker tie remains active. The below-investment-grade indicator is multiplied by 100 before estimation, so coefficients are in percentage points and a positive coefficient indicates a shift toward riskier borrowers.  $InflowsAtTies_{it}$  is standardized and all columns include the full set of borrower, lender, and loan controls. Column (1) includes additive lender and year-quarter fixed effects; Column (2) replaces the lender fixed effect with lender-by-year fixed effects; Column (3) adds borrower-sector-by-year-quarter fixed effects; and Column (4) further adds borrower-state-by-year-quarter fixed effects. Borrower and lender controls are the inverse hyperbolic sine of total assets, number of employees, capital expenditure, total liabilities, EBIT, and (for borrowers) fixed assets; loan controls are the inverse hyperbolic sine of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. Each panel recomputes  $InflowsAtTies_{it}$  under a different tie window: *Panel A* is the baseline (ties active 4 to 12 quarters after a coworker transition, reproducing Table 3), *Panel B* uses 2 to 12 quarters, and *Panel C* uses 4 to 20 quarters; the estimation sample is held fixed across panels. The data span 1994 to 2020. Standard errors are two-way clustered at the lender and year-quarter level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A24: Effect of Outside Options on Borrower Leverage

|                                           | Book Leverage, percentage points |          |          |          |
|-------------------------------------------|----------------------------------|----------|----------|----------|
|                                           | (1)                              | (2)      | (3)      | (4)      |
| IAT                                       | 0.143*                           | 0.270*** | 0.246*** | 0.515*** |
|                                           | (0.077)                          | (0.048)  | (0.048)  | (0.160)  |
| Lender fixed effects                      | X                                |          |          |          |
| YQ fixed effects                          | X                                | X        |          |          |
| Lender $\times$ Year fixed effects        |                                  | X        | X        | X        |
| Borrower Sector $\times$ YQ fixed effects |                                  |          | X        | X        |
| Borrower State $\times$ YQ fixed effects  |                                  |          |          | X        |
| Borrower+Lender+Loan Controls             | X                                | X        | X        | X        |
| Observations                              | 55,327                           | 55,278   | 55,223   | 50,045   |

Notes: The table re-estimates the loan-level borrower-risk regression of Table 3 with the borrower's book leverage as the dependent variable, defined as the sum of short- and long-term debt over total assets and winsorized at the 1st and 99th percentiles. Book leverage is multiplied by 100 before estimation, so coefficients are in percentage points. A positive coefficient indicates a shift toward more highly levered borrowers.  $InflowsAtTies_{it}$  is standardized and all columns include the full set of lender and loan controls together with the borrower controls, the latter excluding the three items that construct the dependent variable, namely short-term debt, long-term debt, and total assets. Column (1) includes additive lender and year-quarter fixed effects; Column (2) replaces the lender fixed effect with lender-by-year fixed effects; Column (3) adds borrower-sector-by-year-quarter fixed effects; and Column (4) further adds borrower-state-by-year-quarter fixed effects. Borrower and lender controls are the inverse hyperbolic sine of total assets, number of employees, capital expenditure, total liabilities, EBIT, and (for borrowers) fixed assets; loan controls are the inverse hyperbolic sine of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. The data span 1994 to 2020. Standard errors are two-way clustered at the lender and year-quarter level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A25: Effect of Outside Options on Loan Price

|                               | All-In-Spread-Drawn (log bps) |         |         |         |         |         |
|-------------------------------|-------------------------------|---------|---------|---------|---------|---------|
|                               | (1)                           | (2)     | (3)     | (4)     | (5)     | (6)     |
| IAT                           | -0.004                        | 0.000   | 0.002   | -0.005  | -0.002  | 0.000   |
|                               | (0.006)                       | (0.005) | (0.003) | (0.005) | (0.005) | (0.003) |
| Lender fixed effects          | X                             | X       | X       | X       | X       | X       |
| Year-Quarter fixed effects    | X                             | X       | X       | X       | X       | X       |
| Borrower Rating fixed effects |                               |         |         | X       | X       | X       |
| Lender Controls               | X                             | X       | X       | X       | X       | X       |
| Borrower Controls             |                               | X       | X       |         | X       | X       |
| Loan Controls                 |                               |         | X       |         |         | X       |
| Outcome Mean                  | 5.935                         | 5.935   | 5.935   | 5.935   | 5.935   | 5.935   |
| Observations                  | 47,760                        | 47,760  | 47,760  | 47,760  | 47,760  | 47,760  |

Notes: The table reports results from estimating Equation (16) at the loan level. The dependent variable is the log all-in-spread-drawn (in basis points). The independent variable is  $InflowsAtTies_{it}$ . Borrower and lender controls include the inverse hyperbolic sine transformation of total assets, number of employees, capital expenditure, total liabilities, EBIT, and (for borrowers) fixed assets. Loan controls are the inverse hyperbolic sine transformation of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. All regressions include fixed effects at the lender and year-quarter level, and standard errors are double-clustered at the same levels. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A26: Robustness to Segment-Time Confounders

|                                 | Non-Prime Borrower, percentage points |                          |                    |                                        |                     |
|---------------------------------|---------------------------------------|--------------------------|--------------------|----------------------------------------|---------------------|
|                                 | (1)                                   | (2)                      | (3)                | (4)                                    | (5)                 |
|                                 | Baseline                              | Segment $\times$ year FE |                    | Exposure-weighted segment-time control |                     |
|                                 |                                       | NAICS-2                  | NAICS-3            | NAICS-2                                | NAICS-1             |
| Inflows at Ties                 | 0.475***<br>(0.147)                   | 0.358**<br>(0.164)       | 0.283**<br>(0.140) | 0.386***<br>(0.135)                    | 0.375***<br>(0.140) |
| Borrower, lender, loan controls | X                                     | X                        | X                  | X                                      | X                   |
| Lender + year-quarter FE        | X                                     | X                        | X                  | X                                      | X                   |
| Segment $\times$ year FE        |                                       | X                        | X                  |                                        |                     |
| Exposure-weighted control       |                                       |                          |                    | X                                      | X                   |
| Observations                    | 55,327                                | 52,070                   | 52,069             | 52,070                                 | 52,070              |

Notes: The dependent variable is a below-investment-grade borrower indicator multiplied by 100 before estimation, so coefficients are in percentage points and a positive coefficient indicates a shift toward riskier borrowers. The specification is the baseline loan-level regression in Column (1) of Table 3, with  $InflowsAtTies_{it}$ , the full set of borrower, lender, and loan controls, and lender and year-quarter fixed effects. Columns (2)–(3) add fixed effects for the lender’s dominant connected-bank industry segment interacted with the year at the two- and three-digit NAICS levels. Columns (4)–(5) add a single exposure-weighted control  $\sum_j w_{ijt} \text{asinh}(\text{segment-year hiring})$ , capturing the segment-time hiring to which the lender’s network is exposed. Column (4) constructs the control from two-digit NAICS segments, while Column (5) uses one-digit segments. These are empirical counterparts of a measured  $\alpha_i \varepsilon_t$  confounder, with network composition loading the lender on a segment-level aggregate shock. Stability across these specifications provides evidence against the observed segment-time channels represented by these controls. The panel cannot support segment-by-year-quarter interactions because they saturate the design, so the table reports feasible year-level interactions. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A27: First-Difference Specification

|                                                    | (1)                 | (2)                |
|----------------------------------------------------|---------------------|--------------------|
|                                                    | First difference    | FD, fixed exposure |
| $\Delta$ Inflows at Ties                           | 1.396***<br>(0.426) |                    |
| $\sum_j s_{ij,t-1} \Delta g_{jt}$ , fixed exposure |                     | 2.764*<br>(1.384)  |
| Year-quarter FE                                    | X                   | X                  |
| Observations                                       | 1,956               | 1,956              |

Notes: The table reports first-difference estimates relating outside option shocks to borrower risk, as a robustness check for the panel’s time-varying shares (Borusyak et al., 2022, fn. 32). The lender-quarter below-investment-grade loan share is multiplied by 100 before first-differencing and estimation, so coefficients are in percentage points and a positive coefficient indicates a shift toward riskier borrowers. Column (1) regresses the change in the risky-loan share on the change in standardized inflows at ties. Column (2) replaces this regressor with a fixed-exposure first-difference shift-share measure,  $\sum_j s_{ij,t-1} \Delta g_{jt}$ , which applies prior-quarter exposure weights to changes in hiring at connected banks. This construction removes variation generated by quarterly changes in the network weights. Both regressors are standardized separately. All specifications absorb year-quarter fixed effects. Standard errors clustered at the lender level are reported in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A28: Robustness to Connected-Bank Lending Cycles

|                                       | Below-Investment-Grade Borrower (pp) |                      |                      |                     |                     |                     |
|---------------------------------------|--------------------------------------|----------------------|----------------------|---------------------|---------------------|---------------------|
|                                       | (1)                                  | (2)                  | (3)                  | (4)                 | (5)                 | (6)                 |
| IAT                                   | 0.475***<br>(0.147)                  | 0.611***<br>(0.127)  | 0.650***<br>(0.124)  | 0.612***<br>(0.131) | 0.605***<br>(0.136) | 0.571***<br>(0.153) |
| Lending at Ties (Level)               |                                      | -0.870***<br>(0.308) |                      |                     |                     | -0.375<br>(3.99)    |
| Lending at Ties (Growth)              |                                      | 0.526<br>(0.560)     |                      |                     |                     | 0.669<br>(0.791)    |
| Leveraged Lending at Ties (Level)     |                                      |                      | -0.800***<br>(0.283) |                     |                     | -5.66<br>(3.44)     |
| Leveraged Lending at Ties (Growth)    |                                      |                      | 0.126<br>(0.305)     |                     |                     | 0.323<br>(0.421)    |
| Lev. Lending at Ties, rated (Level)   |                                      |                      |                      | -0.554<br>(0.365)   |                     | 2.13<br>(1.95)      |
| Lev. Lending at Ties, rated (Growth)  |                                      |                      |                      | -0.054<br>(0.295)   |                     | -0.413<br>(0.340)   |
| Lev. Lending at Ties, buyout (Level)  |                                      |                      |                      |                     | -0.652**<br>(0.308) | 3.12<br>(2.04)      |
| Lev. Lending at Ties, buyout (Growth) |                                      |                      |                      |                     | 0.253<br>(0.375)    | -0.156<br>(0.524)   |
| Lender fixed effects                  | X                                    | X                    | X                    | X                   | X                   | X                   |
| Year-Quarter fixed effects            | X                                    | X                    | X                    | X                   | X                   | X                   |
| Borrower Controls                     | X                                    | X                    | X                    | X                   | X                   | X                   |
| Lender Controls                       | X                                    | X                    | X                    | X                   | X                   | X                   |
| Loan Controls                         | X                                    | X                    | X                    | X                   | X                   | X                   |
| Outcome Mean                          | 70.427                               | 70.427               | 70.427               | 70.427              | 70.427              | 70.427              |
| Observations                          | 55,327                               | 55,327               | 55,327               | 55,327              | 55,327              | 55,327              |

Notes: The dependent variable is a below-investment-grade borrower indicator multiplied by 100 before estimation, so coefficients are in percentage points and a positive coefficient indicates a shift toward riskier borrowers. The specification is the baseline loan-level regression in Column (1) of Table 3, with  $InflowsAtTies_{it}$ , the full set of borrower, lender, and loan controls, and lender and year-quarter fixed effects. Each column adds one or more exposure-weighted controls for lending at connected banks. These are constructed as a shift-share that retains the  $InflowsAtTies$  tie weights  $w_{ijt}$  but replaces the hiring shift with the lending activity of connected banks  $j$ ,  $\sum_j w_{ijt} Lending_{jt}$ , entered both in levels, using the inverse hyperbolic sine of quarterly lead-arranged volume in millions of USD, and in year-on-year growth. Column (2) uses total connected-bank lending. Columns (3)–(5) use leveraged lending, defined as loans priced at or above 200 basis points, loans to borrowers rated below investment grade, and loans whose primary purpose is a buyout, merger or acquisition, or recapitalization, respectively. Column (6) includes all of these controls jointly. Every lending control is standardized. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A29: Effect of Outside Options on Borrower Risk by NCA Enforceability

|                                   | Below-Investment-Grade Borrower (pp) |         |         |
|-----------------------------------|--------------------------------------|---------|---------|
|                                   | (1)                                  | (2)     | (3)     |
| IAT                               | 0.662*                               | 0.448   | 0.398   |
|                                   | (0.356)                              | (0.435) | (0.500) |
| IAT $\times$ Enforcement Change   | -1.05                                |         |         |
|                                   | (1.09)                               |         |         |
| IAT $\times$ Enforcement Increase |                                      | 1.32    |         |
|                                   |                                      | (1.06)  |         |
| IAT $\times$ Enforcement Decrease |                                      |         | 3.63*** |
|                                   |                                      |         | (1.11)  |
| Lender fixed effects              | X                                    | X       | X       |
| Year-Quarter fixed effects        | X                                    | X       | X       |
| HQ State-Year fixed effects       | X                                    | X       | X       |
| Borrower Controls                 | X                                    | X       | X       |
| Lender Controls                   | X                                    | X       | X       |
| Loan Controls                     | X                                    | X       | X       |
| Outcome Mean                      | 69.762                               | 69.762  | 69.762  |
| Observations                      | 50,490                               | 50,490  | 50,490  |

*Notes:* The table reports a robustness exercise for Column (1) of Table 3, estimating Equation (16) at the loan level. The below-investment-grade borrower indicator is multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. The main independent variable is  $InflowsAtTies_{it}$ , interacted with measures of changes in non-compete agreement (NCA) enforceability in the state of the bank's headquarters. *Enforcement Change* equals 1 if a state increased its NCA enforceability and -1 if it decreased it in any past or current period, and 0 otherwise. *Enforcement Increase* and *Enforcement Decrease* are separate indicators for past increases and decreases. Borrower and lender controls include the inverse hyperbolic sine transformation of total assets, number of employees, capital expenditure, total liabilities, EBIT, and, for borrowers, fixed assets. Loan controls are the inverse hyperbolic sine transformation of tranche amount and maturity and indicators for whether the loan is secured and whether covenants are attached. All columns include lender, year-quarter, and state-year fixed effects. Standard errors are double-clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A30: Outside Options and First-Time Lead-Borrower Relationships

|                                           | Below IG<br>at origination<br>(1) | Downgrade<br>within 2 years<br>(2) | Large downgrade<br>within 2 years<br>(3) | Default<br>within 3 years<br>(4) |
|-------------------------------------------|-----------------------------------|------------------------------------|------------------------------------------|----------------------------------|
| First-time lead-borrower relationship     | -0.33<br>(0.98)                   | 3.77***<br>(1.27)                  | 1.64**<br>(0.74)                         | 0.43*<br>(0.22)                  |
| Outside options × First-time relationship | 0.74*<br>(0.44)                   | 1.23**<br>(0.55)                   | 0.46<br>(0.56)                           | 0.09<br>(0.24)                   |
| Borrower and deal controls                | Yes                               | Yes                                | Yes                                      | Yes                              |
| Origination-rating controls               |                                   | Yes                                | Yes                                      | Yes                              |
| Lender × Year-Quarter fixed effects       | Yes                               | Yes                                | Yes                                      | Yes                              |
| Lender and deal clustering                | Yes                               | Yes                                | Yes                                      | Yes                              |
| Observations                              | 19,351                            | 9,611                              | 9,611                                    | 19,351                           |

Notes: The unit of observation is a lead arranger-deal pair. *First-time lead-borrower relationship* equals one when Dealscan records no strictly earlier deal between the lead arranger and borrower. All dependent variables are measured in percentage points. Column (1) measures below-investment-grade status at origination. Columns (2)–(4) measure an S&P rating downgrade, a downgrade of at least two notches, and default over the stated windows. All regressions include lender-by-year-quarter fixed effects, which absorb the standalone outside options coefficient and compare first-time with repeat relationships for the same lender in the same quarter. Borrower and deal controls follow the main loan-level specification. The realized-risk regressions also control for the origination rating. Standard errors are two-way clustered by lender and deal. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A31: Ex Post Borrower Deterioration Across Fixed-Effects Specifications

| Rating source              | Downgrade (3y, pp)<br>Mergent FISD (bonds) |                  | Default (3y, pp)<br>Mergent FISD (bonds) |                  | Downgrade (3y, pp)<br>Compustat / S&P (issuer) |                  | Default (3y, pp)<br>Compustat / S&P (issuer) |     |
|----------------------------|--------------------------------------------|------------------|------------------------------------------|------------------|------------------------------------------------|------------------|----------------------------------------------|-----|
|                            | (1)                                        | (2)              | (3)                                      | (4)              | (5)                                            | (6)              | (7)                                          | (8) |
| IAT                        | 2.88**<br>(1.11)                           | 0.707<br>(0.444) | 0.687**<br>(0.324)                       | 0.528<br>(0.363) | 1.49<br>(1.17)                                 | 0.405<br>(0.421) |                                              |     |
| Lender fixed effects       | X                                          | X                | X                                        | X                | X                                              | X                |                                              |     |
| Year-Quarter fixed effects |                                            | X                |                                          | X                |                                                |                  |                                              |     |
| Borrower Controls          | X                                          | X                | X                                        | X                | X                                              | X                |                                              |     |
| Lender Controls            | X                                          | X                | X                                        | X                | X                                              | X                |                                              |     |
| Loan Controls              | X                                          | X                | X                                        | X                | X                                              | X                |                                              |     |
| Outcome Mean               | 36.267                                     | 36.269           | 3.015                                    | 3.015            | 25.026                                         | 2.244            |                                              |     |
| Observations               | 29,288                                     | 29,287           | 29,288                                   | 29,287           | 25,462                                         | 41,586           |                                              |     |

Notes: The table relates ex post borrower deterioration to  $InflowsAtTies_{it}$  at origination, estimating Equation (16) at the loan level with the full set of borrower, lender, and loan controls. All dependent-variable indicators are multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. Columns (1)–(4) use Mergent FISD records of Moody's, S&P, and Fitch rating actions to measure bond downgrades and defaults within three years of origination. Columns (5)–(6) use Compustat records of the borrower's S&P long-term issuer rating to measure downgrades and defaults within three years. *Downgrade* equals one if the borrower's rating falls below its origination level within the window, and *Default* equals one if the rating reaches D or SD. All columns include lender fixed effects, while Columns (2) and (4) also include year-quarter fixed effects.  $InflowsAtTies_{it}$  is standardized to unit variance. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A32: Banker Age and Labor Market Transitions

|                            | Separation (8q, pp) |                     | Separation (12q, pp) |                     | Job-to-Job Move (8q, pp) |                     |
|----------------------------|---------------------|---------------------|----------------------|---------------------|--------------------------|---------------------|
|                            | (1)                 | (2)                 | (3)                  | (4)                 | (5)                      | (6)                 |
| Bad Outcome                | 6.88<br>(25.3)      | -6.60<br>(35.7)     | -9.06<br>(21.1)      | -19.7<br>(30.4)     | 18.5<br>(22.3)           | 19.6<br>(27.8)      |
| Age                        | -1.28***<br>(0.206) | -1.27***<br>(0.385) | -1.78***<br>(0.174)  | -1.77***<br>(0.484) | -1.32***<br>(0.196)      | -1.38***<br>(0.352) |
| Bad Outcome × Age          | -0.086<br>(0.501)   | 0.334<br>(0.766)    | 0.293<br>(0.450)     | 0.624<br>(0.690)    | -0.365<br>(0.446)        | -0.261<br>(0.595)   |
| Lender fixed effects       | X                   | X                   | X                    | X                   | X                        | X                   |
| Year-Quarter fixed effects | X                   | X                   | X                    | X                   | X                        | X                   |
| Lender Controls            |                     | X                   |                      | X                   |                          | X                   |
| Outcome Mean               | 26.418              | 26.539              | 36.211               | 36.306              | 21.778                   | 21.444              |
| Observations               | 776                 | 471                 | 776                  | 471                 | 776                      | 471                 |

*Notes:* The table relates the signing banker's subsequent labor market transitions to the banker's age, on the sample of loans matched to individual bankers. Each dependent-variable indicator is multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. The indicators identify whether the banker separated from the lending bank within eight quarters of loan origination (Columns 1–2), within twelve quarters (Columns 3–4), or moved to a job at another employer within eight quarters (Columns 5–6). Separations are constructed from the employment-history data and account for right-censoring. *Bad Outcome* equals one if the borrower's outstanding bonds were downgraded by Moody's, S&P, or Fitch within two years of origination (Mergent FISD). Lender controls are the inverse hyperbolic sine transformations of total assets, net income, EBIT, capital expenditure, total liabilities, short- and long-term debt, equity, and number of employees. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A33: Outside Options and Ex Ante Borrower Risk by Banker Age: Fixed-Effects Ladder

|                                   | Below Investment Grade (pp) |                     |                      |                     | Borrower Risk Score |                     |                       |                     |
|-----------------------------------|-----------------------------|---------------------|----------------------|---------------------|---------------------|---------------------|-----------------------|---------------------|
|                                   | (1)                         | (2)                 | (3)                  | (4)                 | (5)                 | (6)                 | (7)                   | (8)                 |
| IAT (+1 SD)                       | 8.067<br>(6.492)            | 15.88<br>(14.50)    | 15.85**<br>(6.459)   |                     | 0.0118<br>(0.0235)  | 0.0182<br>(0.0697)  | 0.0773***<br>(0.0258) |                     |
| IAT × Age                         | -0.1063<br>(0.1278)         | -0.0616<br>(0.1413) | -0.3143*<br>(0.1693) | -0.1117<br>(0.1618) | -0.0005<br>(0.0005) | -0.0001<br>(0.0007) | -0.0016*<br>(0.0008)  | -0.0003<br>(0.0007) |
| Lender fixed effects              | X                           |                     | X                    |                     | X                   |                     | X                     |                     |
| Year-quarter fixed effects        | X                           | X                   | X                    |                     | X                   | X                   | X                     |                     |
| Lender-Year fixed effects         |                             | X                   |                      |                     |                     | X                   |                       |                     |
| Lender-Year-quarter fixed effects |                             |                     |                      | X                   |                     |                     |                       | X                   |
| Lender controls                   |                             |                     | X                    |                     |                     |                     | X                     |                     |
| Outcome mean                      | 81.594                      | 81.054              | 81.646               | 80.940              | 0.505               | 0.500               | 0.497                 | 0.497               |
| Observations                      | 1,869                       | 1,726               | 948                  | 1,511               | 1,869               | 1,726               | 948                   | 1,511               |

*Notes:* The table relates the borrower's risk measured *at origination* to  $InflowsAtTies_{it}$  interacted with the signing banker's age, on the sample of loans matched to individual bankers and without banker fixed effects. In Columns (1)–(4) the below-investment-grade indicator is multiplied by 100 before estimation, so coefficients and outcome means are in percentage points. In Columns (5)–(8) the dependent variable is the S&P quality ranking normalized to the unit interval and oriented so that higher values denote riskier borrowers, which remains on its continuous scale. Each block runs the same fixed-effects ladder: Column (1)/(5) includes additive lender and year-quarter fixed effects; Column (2)/(6) replaces the lender fixed effect with lender-by-year fixed effects; Column (3)/(7) adds lender balance sheet controls; and Column (4)/(8) uses lender-by-year-quarter fixed effects, which absorb the standalone coefficient on inflows at ties and identify the age interaction purely from loans signed at the same bank in the same quarter. The age main effect is absorbed by the fixed effects and is not reported.  $InflowsAtTies_{it}$  is standardized to unit variance. Consistent with the realized-deterioration result of Table 5, the interaction with age is negative throughout, so the risk response to outside options declines with age; at origination the gradient is directionally robust but estimated less precisely than in realized deterioration. Standard errors are two-way clustered at the lender and year-quarter level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A34: Effect of Outside Options on Misconduct Disclosures

|                                | Number of Misconduct Disclosures |                      |                      |                      |                      | Binarized (pp)       |                     |
|--------------------------------|----------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|---------------------|
|                                | (1)                              | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                 |
| IAT <sub>t-1</sub>             | 0.003***<br>(0.0009)             | 0.003***<br>(0.0009) | 0.003***<br>(0.0008) | 0.003***<br>(0.0008) | 0.002***<br>(0.0006) | 0.004***<br>(0.0009) | 0.328***<br>(0.082) |
| IAT <sub>t</sub>               |                                  |                      |                      |                      | 0.003**<br>(0.001)   |                      |                     |
| IAT <sub>t-2</sub>             |                                  |                      |                      |                      | -0.0008<br>(0.0005)  |                      |                     |
| Outflows <sub>t-1</sub>        |                                  |                      |                      |                      |                      | 0.003<br>(0.002)     |                     |
| Lender fixed effects           | X                                | X                    |                      |                      |                      |                      |                     |
| Year fixed effects             | X                                | X                    | X                    |                      | X                    | X                    | X                   |
| Adviser fixed effects          |                                  | X                    |                      |                      |                      |                      |                     |
| Lender × Adviser fixed effects |                                  |                      | X                    | X                    | X                    | X                    | X                   |
| State × Year fixed effects     |                                  |                      |                      | X                    |                      |                      |                     |
| Lender Controls                | X                                | X                    | X                    | X                    | X                    | X                    | X                   |
| Outcome Mean                   | 0.010                            | 0.010                | 0.009                | 0.009                | 0.009                | 0.009                | 0.948               |
| Observations                   | 582,568                          | 561,018              | 551,079              | 551,079              | 551,079              | 551,079              | 551,079             |

Notes: The table reports results from estimating Equation (16) at annual frequency at the adviser level. Columns (1)–(6) use the number of misconduct disclosures on its original count scale. Column (7) uses an indicator for any misconduct disclosure multiplied by 100 before estimation, so its coefficient and outcome mean are in percentage points. The main independent variable is  $InflowsAtTies_{it-1}$ . Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, equity, and the first lag of the firm-level average of the outcome variable. Column (5) includes contemporaneous inflows at ties and its second lag. Column (6) controls for the inverse hyperbolic sine transformation of lagged outflows. All models are estimated with year fixed effects and lender controls. Standard errors are clustered at the lender level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A35: Effect of Outside Options on Customer Complaints

|                                | Number of Customer Complaints |                     |                     |                     |                     | Binarized (pp)      |                     |
|--------------------------------|-------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                | (1)                           | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 |
| IAT <sub>t-1</sub>             | 0.005***<br>(0.002)           | 0.006***<br>(0.002) | 0.006***<br>(0.001) | 0.005***<br>(0.002) | 0.003***<br>(0.001) | 0.006***<br>(0.001) | 0.417***<br>(0.108) |
| IAT <sub>t</sub>               |                               |                     |                     |                     | 0.005*<br>(0.003)   |                     |                     |
| IAT <sub>t-2</sub>             |                               |                     |                     |                     | 0.0002<br>(0.001)   |                     |                     |
| Outflows <sub>t-1</sub>        |                               |                     |                     |                     |                     | 0.003<br>(0.004)    |                     |
| Lender fixed effects           | X                             | X                   |                     |                     |                     |                     |                     |
| Year fixed effects             | X                             | X                   | X                   |                     | X                   | X                   | X                   |
| Adviser fixed effects          |                               | X                   |                     |                     |                     |                     |                     |
| Lender × Adviser fixed effects |                               |                     | X                   | X                   | X                   | X                   | X                   |
| State × Year fixed effects     |                               |                     |                     | X                   |                     |                     |                     |
| Lender Controls                | X                             | X                   | X                   | X                   | X                   | X                   | X                   |
| Outcome Mean                   | 0.018                         | 0.017               | 0.017               | 0.017               | 0.017               | 0.017               | 1.479               |
| Observations                   | 582,568                       | 561,018             | 551,079             | 551,079             | 551,079             | 551,079             | 551,079             |

Notes: The table reports results from estimating Equation (16) at annual frequency at the adviser level. Columns (1)–(6) use the number of customer complaints on its original count scale. Column (7) uses an indicator for any customer complaint multiplied by 100 before estimation, so its coefficient and outcome mean are in percentage points. The main independent variable is  $InflowsAtTies_{it-1}$ . Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, equity, and the first lag of the firm-level average of the outcome variable. Column (5) includes contemporaneous inflows at ties and its second lag. Column (6) controls for the inverse hyperbolic sine transformation of lagged outflows. All models are estimated with year fixed effects and lender controls. Standard errors are clustered at the lender level. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A36: Effect of Outside Options on Settled Customer Complaints

|                                       | Number of Settled Customer Complaints |                     |                      |                      |                      | Binarized (pp)      |                     |
|---------------------------------------|---------------------------------------|---------------------|----------------------|----------------------|----------------------|---------------------|---------------------|
|                                       | (1)                                   | (2)                 | (3)                  | (4)                  | (5)                  | (6)                 | (7)                 |
| $IAT_{t-1}$                           | 0.004***<br>(0.001)                   | 0.004***<br>(0.001) | 0.004***<br>(0.0010) | 0.004***<br>(0.0009) | 0.003***<br>(0.0007) | 0.004***<br>(0.001) | 0.326***<br>(0.068) |
| $IAT_t$                               |                                       |                     |                      |                      | 0.004*<br>(0.002)    |                     |                     |
| $IAT_{t-2}$                           |                                       |                     |                      |                      | -0.0005<br>(0.0008)  |                     |                     |
| $Outflows_{t-1}$                      |                                       |                     |                      |                      |                      | 0.003<br>(0.003)    |                     |
| Lender fixed effects                  | X                                     | X                   |                      |                      |                      |                     |                     |
| Year fixed effects                    | X                                     | X                   | X                    |                      | X                    | X                   | X                   |
| Adviser fixed effects                 |                                       | X                   |                      |                      |                      |                     |                     |
| Lender $\times$ Adviser fixed effects |                                       |                     | X                    | X                    | X                    | X                   | X                   |
| State $\times$ Year fixed effects     |                                       |                     |                      | X                    |                      |                     |                     |
| Lender Controls                       | X                                     | X                   | X                    | X                    | X                    | X                   | X                   |
| Outcome Mean                          | 0.008                                 | 0.008               | 0.008                | 0.008                | 0.008                | 0.008               | 0.672               |
| Observations                          | 582,568                               | 561,018             | 551,079              | 551,079              | 551,079              | 551,079             | 551,079             |

Notes: The table reports results from estimating Equation (16) at annual frequency at the adviser level. Columns (1)–(6) use the number of settled customer complaints on its original count scale. Column (7) uses an indicator for any settled customer complaint multiplied by 100 before estimation, so its coefficient and outcome mean are in percentage points. The main independent variable is  $InflowsAtTies_{it-1}$ . Lender controls are the inverse hyperbolic sine transformations of total assets, net income, number of employees, EBIT, capital expenditure, total liabilities, short- and long-term debt, equity, and the first lag of the firm-level average of the outcome variable. Column (5) includes contemporaneous inflows at ties and its second lag. Column (6) controls for the inverse hyperbolic sine transformation of lagged outflows. All models are estimated with year fixed effects and lender controls. Standard errors are clustered at the lender level.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .